

Development of the Musculoskeletal and Integumentary System

- ✓ Development of the skeletal system
- ✓ Development of the muscular system
- ✓ Development of limbs
- ✓ Development of the Integumentary system

Development of the Musculoskeletal system

- Development of bone and muscle occurs within mesenchymal regions of the embryo.
- Muscle development occurs in the embryo through the formation of **myoblasts**, which undergo extensive proliferation to form terminally differentiated, postmitotic **myocytes**.
- Bones development occurs in the embryo through the process called **Ossification**, or *osteogenesis*
- **Myocytes express** Actin, Myosin, and other contractile proteins and fuse to form contractile myofibrils.

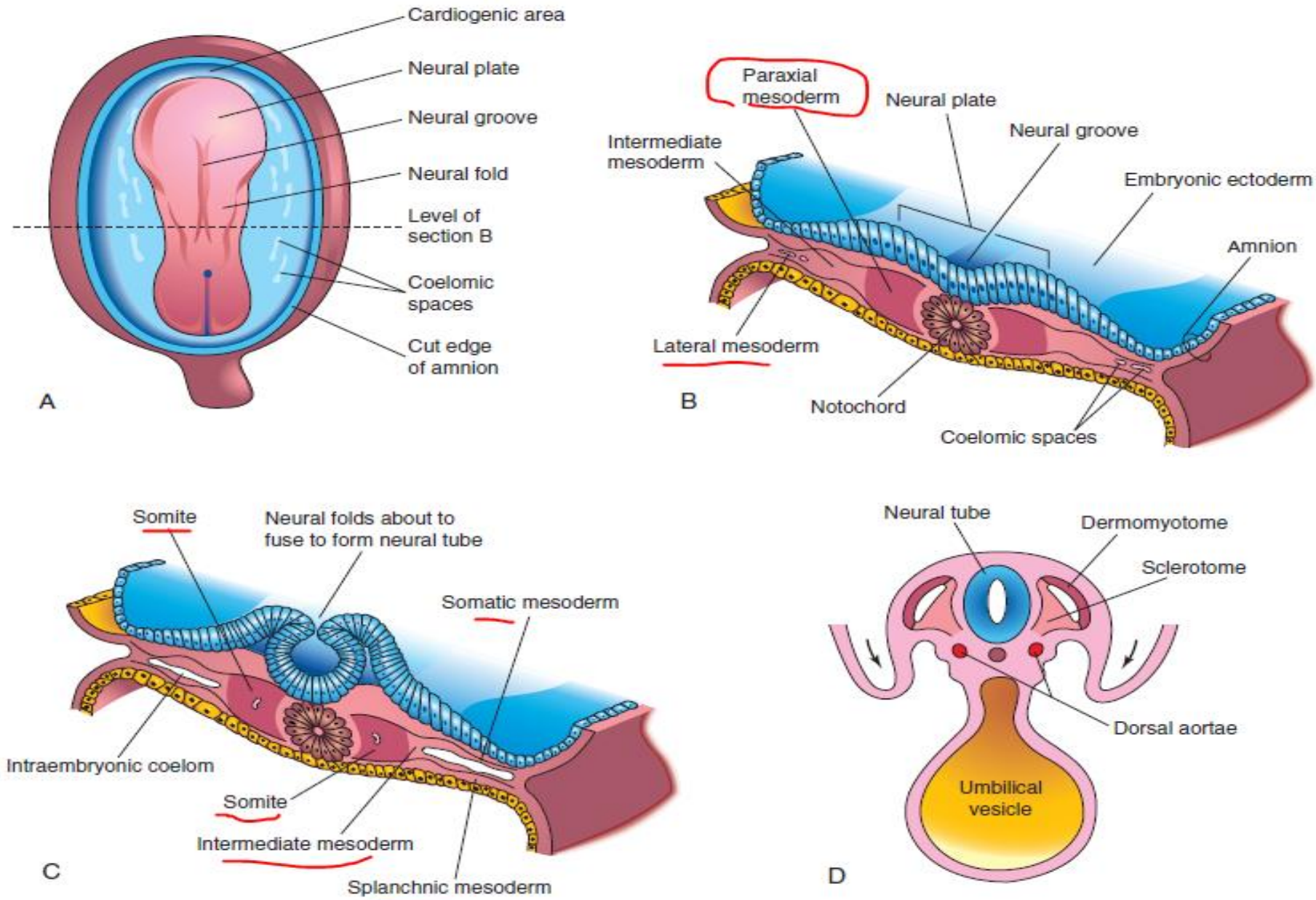
Development of the Musculoskeletal system,,

- **Three types of muscles form in the embryo:** skeletal, smooth, and cardiac.
 - Skeletal (striated or **voluntary**), smooth (**involuntary**), and **cardiac** (structurally striated but functionally smooth muscle).
 - Skeletal, or voluntary, muscle develops in association with bone.
- I. The **skeletal muscles** mostly develop from somites (paraxial mesoderm).
 - II. The **smooth muscles** mostly develop from splanchnic mesoderm.
 - III. The **cardiac muscle** develops from splanchnic mesoderm surrounding developing heart tube.

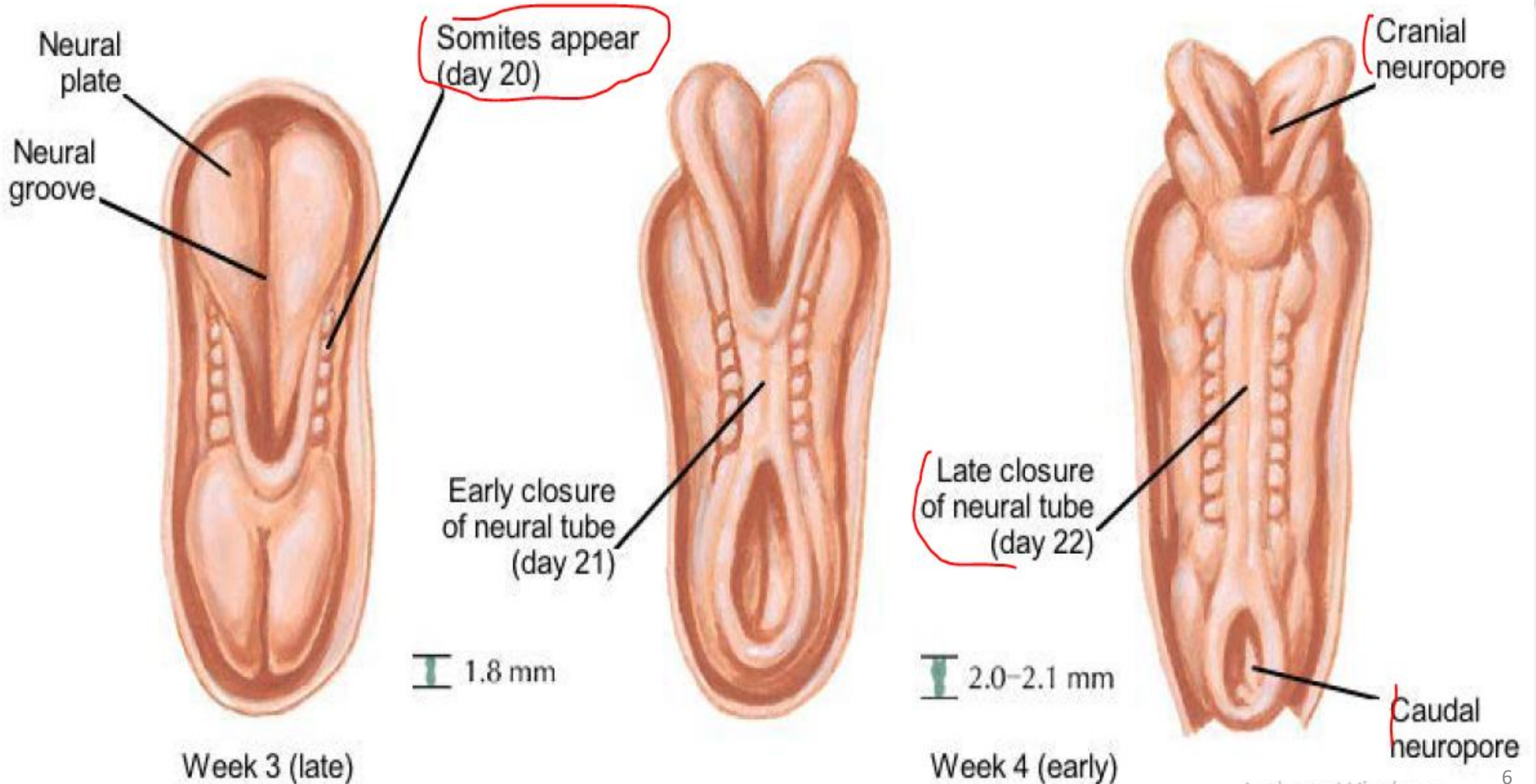
Development of the Musculoskeletal system,,

- **As the notochord and neural tube** form in the **third week**, the intraembryonic mesoderm lateral to these structures thickens to form **two** longitudinal columns of **paraxial mesoderm**.
- The skeletal system develops from **paraxial** and **lateral plate (somatic layer) mesoderm** and from **neural crest**.
- **Paraxial mesoderm** forms a segmented series of tissue blocks on each side of the neural tube, known as **somitomeres** in the head region and **somites** from the occipital region caudally.
- **Toward the end of the third week**, these dorsolateral columns, located in the trunk, become segmented into blocks of mesoderm—**somites**.

Development of the Musculoskeletal system,,

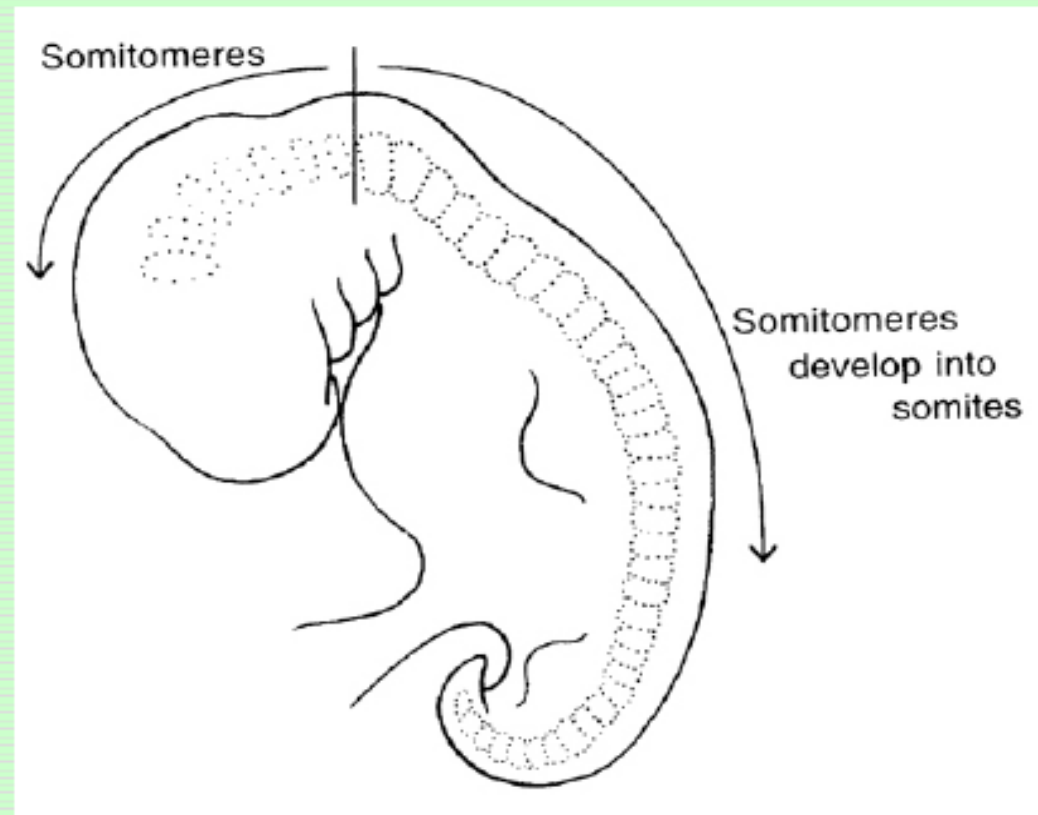


Development of the Musculoskeletal system,,,



Development of the Musculoskeletal system,,

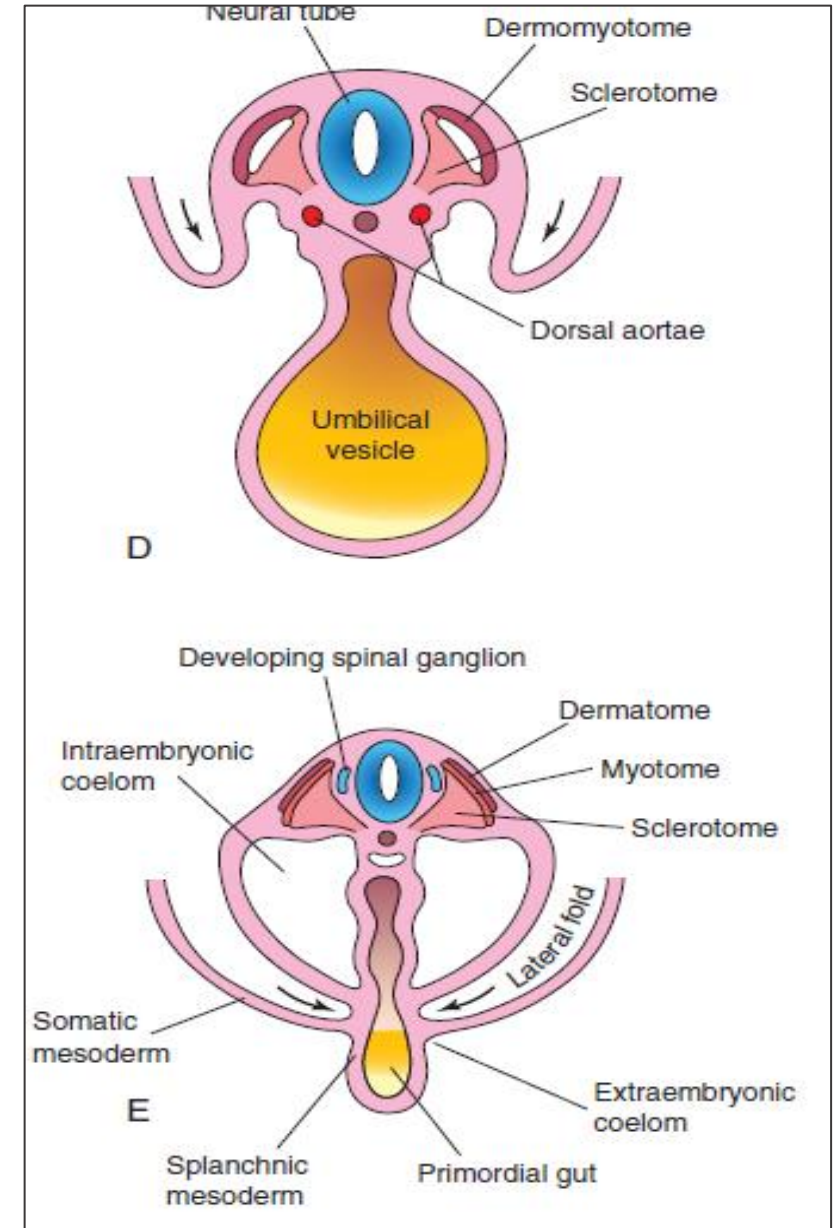
At the end of the fifth week, 42 to 44 pairs somites are present.



Lateral view

Development of the Musculoskeletal system,,,

- Each somite differentiates into two parts:
 - ✓ The ventromedial part is the **sclerotome**; its cells form the vertebrae and ribs.
 - ✓ The dorsolateral part is the **dermomyotome**; its cells from the myotome region form myoblasts (primordial muscle cells), and those from its dermatome region form the dermis (fibroblasts).



Development of the Musculoskeletal system,,

- The muscles and bones of the trunk derive from the somites.
- Each somite forms two distinct zones: a **sclerotome** and a **dermomyotome**.
- ❖ The **sclerotome** gives rise to the bones of the **axial skeleton**.
- ❖ The **dermomyotome** gives rise to the dermatome—which forms the dermis of the back skin of the trunk and neck (with the remainder of the dermis of these regions forming from **lateral plate mesoderm**)&the **myotome**—which forms the **muscles of the trunk**.
- The **myotome** also gives rise to the musculature of **the limbs and tongue**.
- ❖ The **dermis & bones of the limbs** form from **lateral plate mesoderm**.

Development of the Musculoskeletal system,,

❖ **Anatomically the skeletal system** consists of bones and cartilages.

- Both bones and cartilages develop from mesenchyme (loose tissue of mesoderm).

❑ **The mesenchyme forming bones and cartilages is derived from following three sources:**

I. Paraxial mesoderm (main source)

II. Parietal layer of lateral plate mesoderm

III. Neural crest cells

❖ Whatever may be the source of origin of mesenchyme, the general process of formation of cartilages and bones is same.

Formation of Cartilage

- Cartilage development begins from mesenchyme during the **fifth** week.
- In areas where cartilage is to develop, the mesenchyme condenses to form **chondrification centers** where the cartilage is to be formed.
- The mesenchymal cells differentiate 1st **into precondrocytes &** then into **chondroblasts**, which secrete collagenous fibrils and ground substance (extracellular matrix).
- The chondroblasts form the collagen fibers and ground substance of the matrix (intercellular substance).
- **Some chondroblasts** get imprisoned within the intercellular substance of developing cartilage and are called **chondrocytes**.
- ❖ **Depending upon the presence and predominance of collagen or elastic fibers the cartilages are classified into 3 types: *fibrocartilage, hyaline cartilage, and elastic cartilage***

Formation of Cartilage,,

❑ **The hyaline cartilage** has a homogeneous, glassy, bluish, and opalescent appearance.

- The collagen fibers present in the matrix **are fine and not seen easily.**

❑ **The fibrocartilage** is a dense, fasciculated, opaque white tissue.

- The collagen fibers present in fibrocartilage **are numerous and very obvious.**

❑ **The elastic cartilage** is pliable and imparts yellowish tinge.

- The **intercellular substance of elastic cartilage**, as the name indicates, is permeated by elastic fibers.
- The mesenchymal cells surrounding the surface of developing cartilage form the **perichondrium.**

Bone Development

❑ All the bones are mesodermal in origin.

- **Ossification**, or *osteogenesis*, refers to the formation and development of bone connective tissue.
- **Ossification** begins in the **embryo** and continues as the skeleton grows during childhood and adolescence.
- Even after the adult bones have formed, ossification continues.
- The bone continues to grow, and new **osteoblasts** are trapped in the expanding bone.
- **Additional osteoblasts** are continually produced as mesenchymal cells grow and develop.
- Newly formed blood vessels also branch throughout this region.

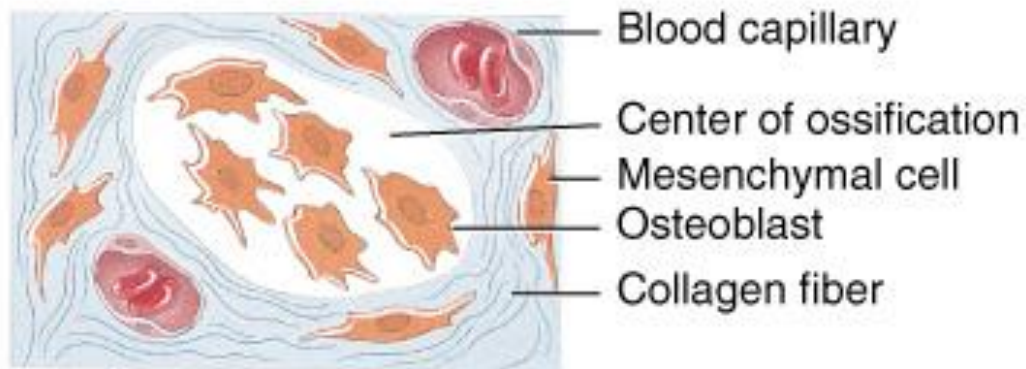
Bone Development,,

- Bone primarily develops in two types of connective tissue, **mesenchyme** and **cartilage**, but it can also develop in other connective tissues (e.g., the patella develops in a **tendon**).
- All embryonic connective tissue begins as **mesenchyme**.
- ❖ Two types of *osteogenesis or ossification* occur.
 - ❑ *Intramembranous ossification* is the formation of bone directly from or within **mesenchymal cells**.
 - It is called **dermal ossification**
 - e.g.: flat bone particularly part of the skull, mandible, clavicle
 - ❑ *Endochondrial ossification* is the formation of bone from **hyaline cartilage models**.
 - e.g. long bones

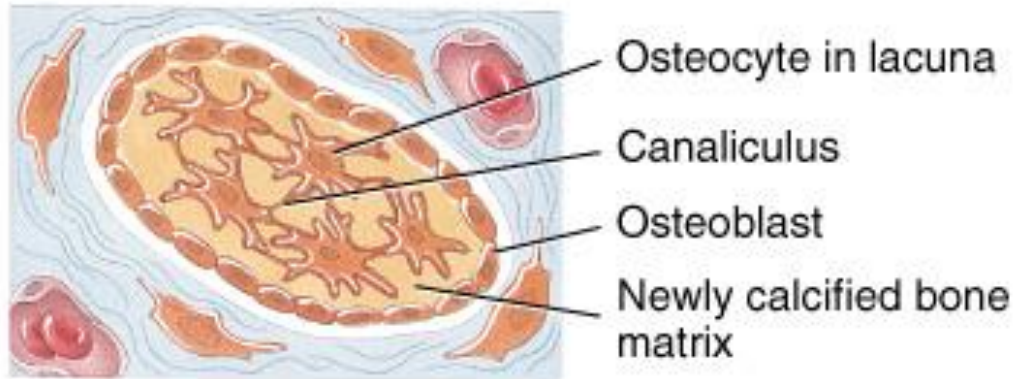
Membranous Ossification

- At the site where the **membrane bone** is to be formed, the mesenchyme condenses and becomes highly vascular (**mesenchymal model**).
- **Some cells of mesenchymal** model differentiate into **osteoblasts**.
- The osteoblasts lay down the matrix of bone called **osteoid**.
- The osteoid tissue consists of **collagen fibers and ground substance**.
- **The osteoid tissue under** the influence of osteoblasts becomes **mineralized** with calcium salts and becomes **the bone**.
- The osteoblasts are trapped in the bone that is formed around it and are converted into **osteocytes**.
- **The bone formed by membranous** ossification is in the form of **spicules** or **trabeculae** that are arranged irregularly with spaces between them. This type of bone is called **spongy bone**.

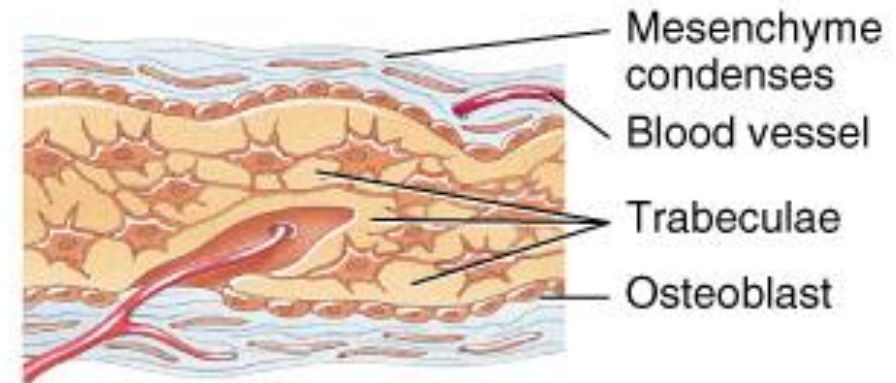
Process of intramembranous Ossification



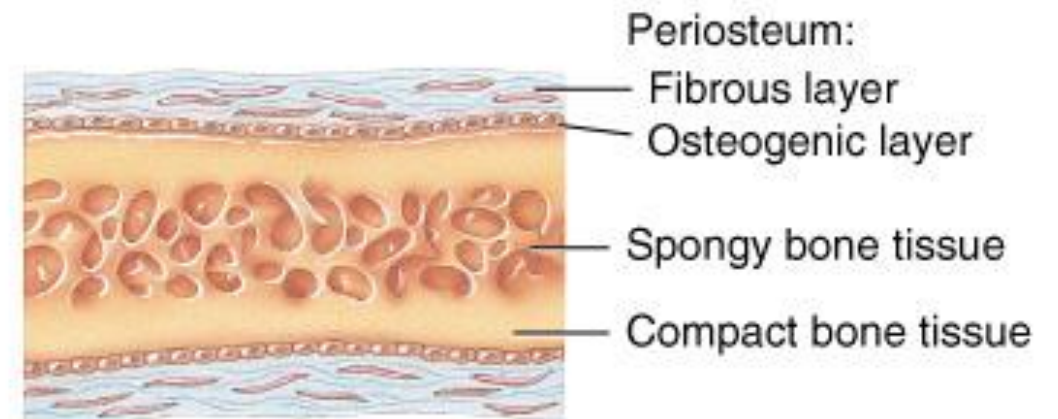
1 Development of center of ossification



2 Osteocytes deposit mineral salts (calcification)

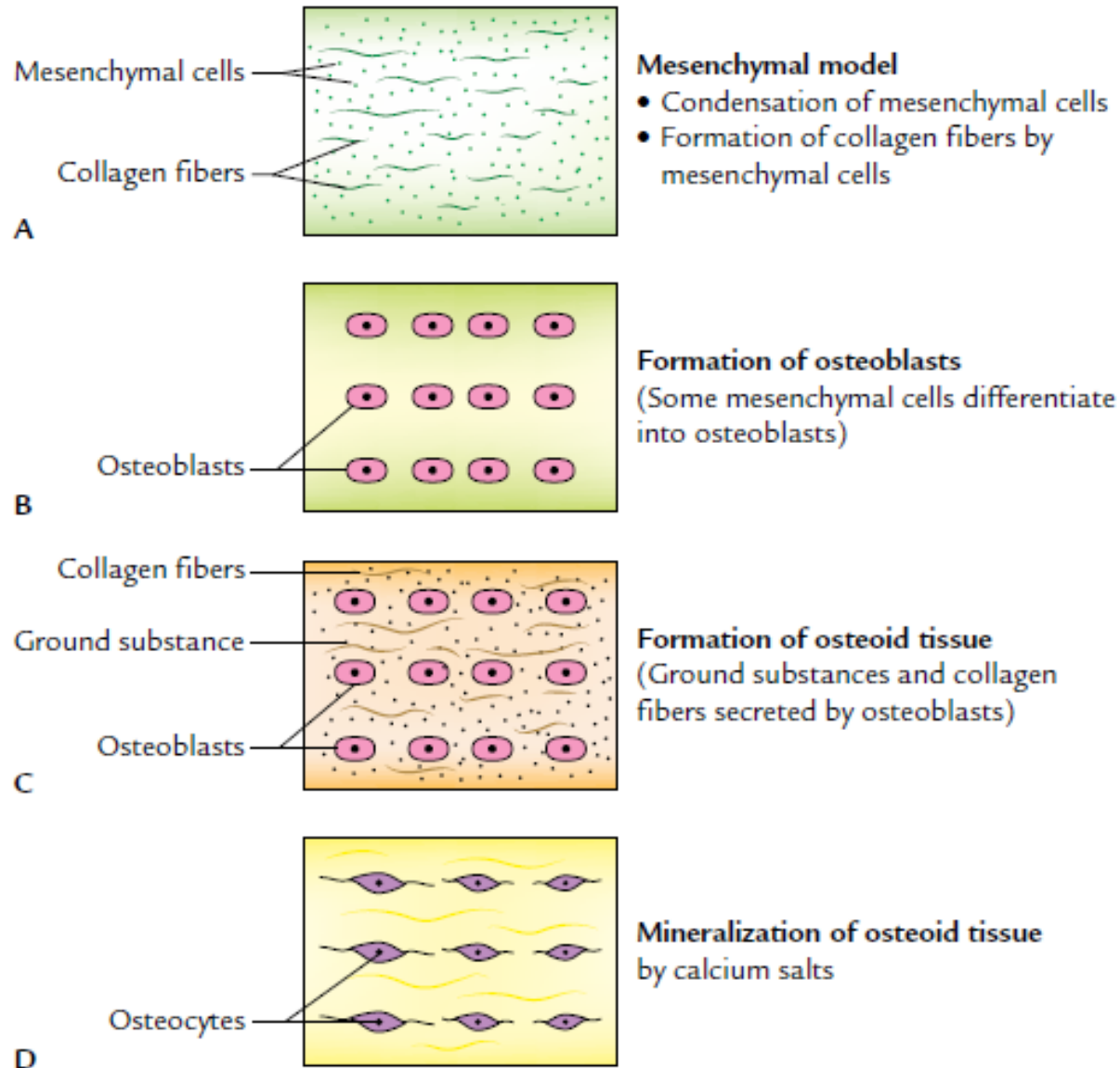


3 Formation of trabeculae

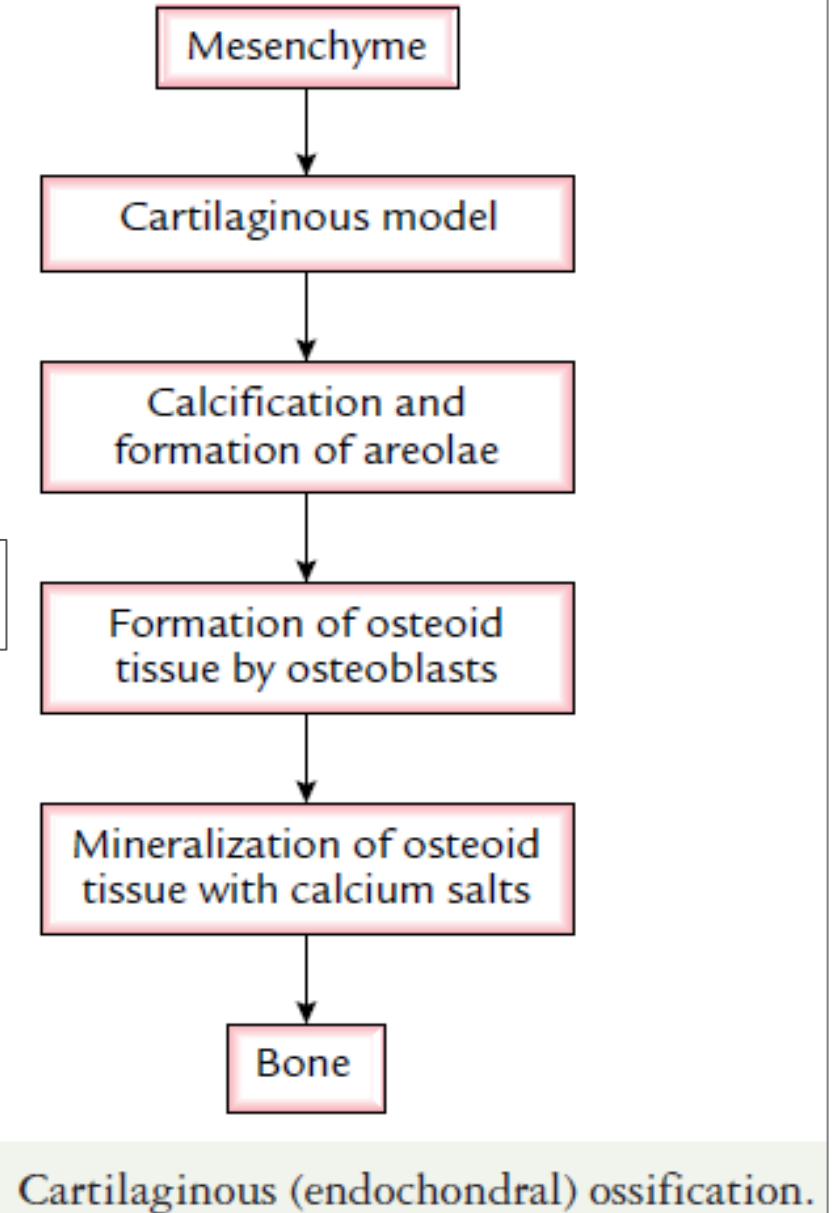


4 Development of periosteum, spongy bone, and compact bone tissue

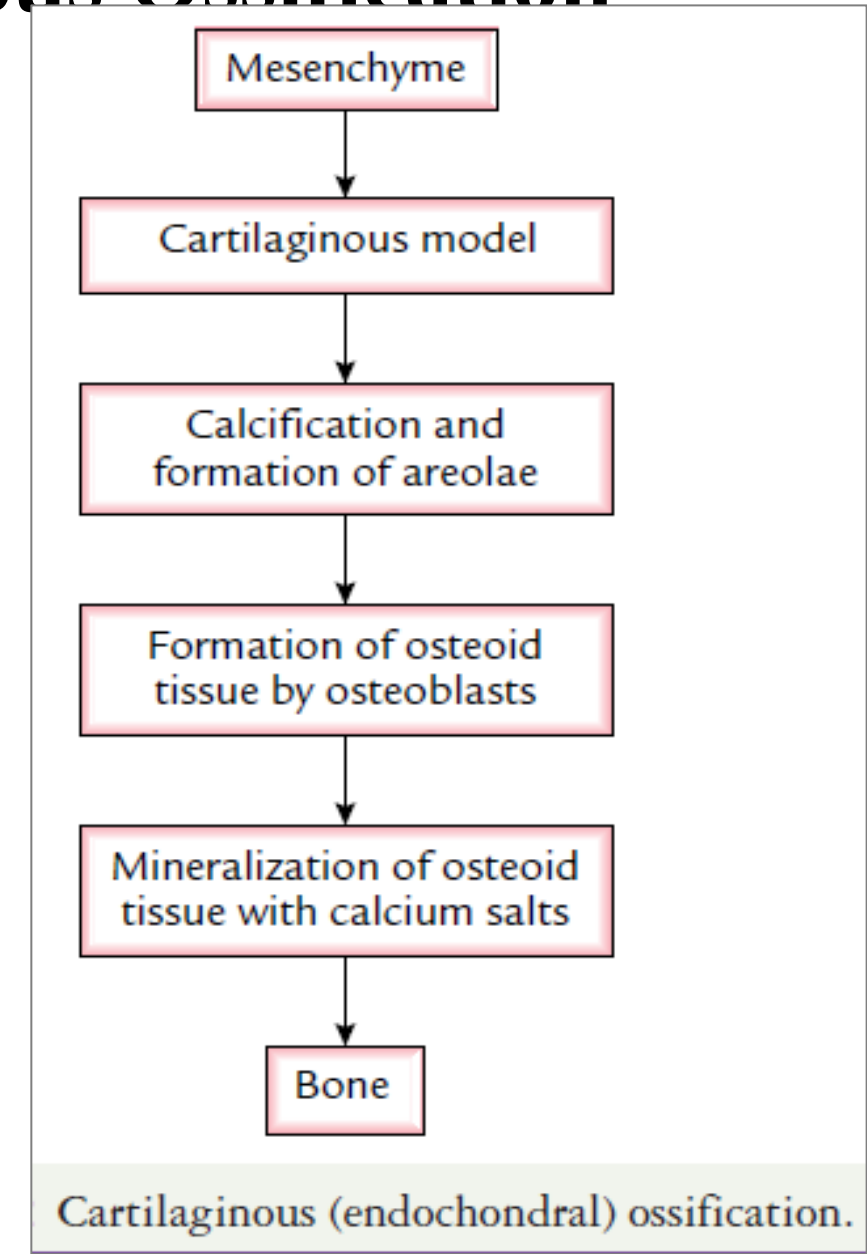
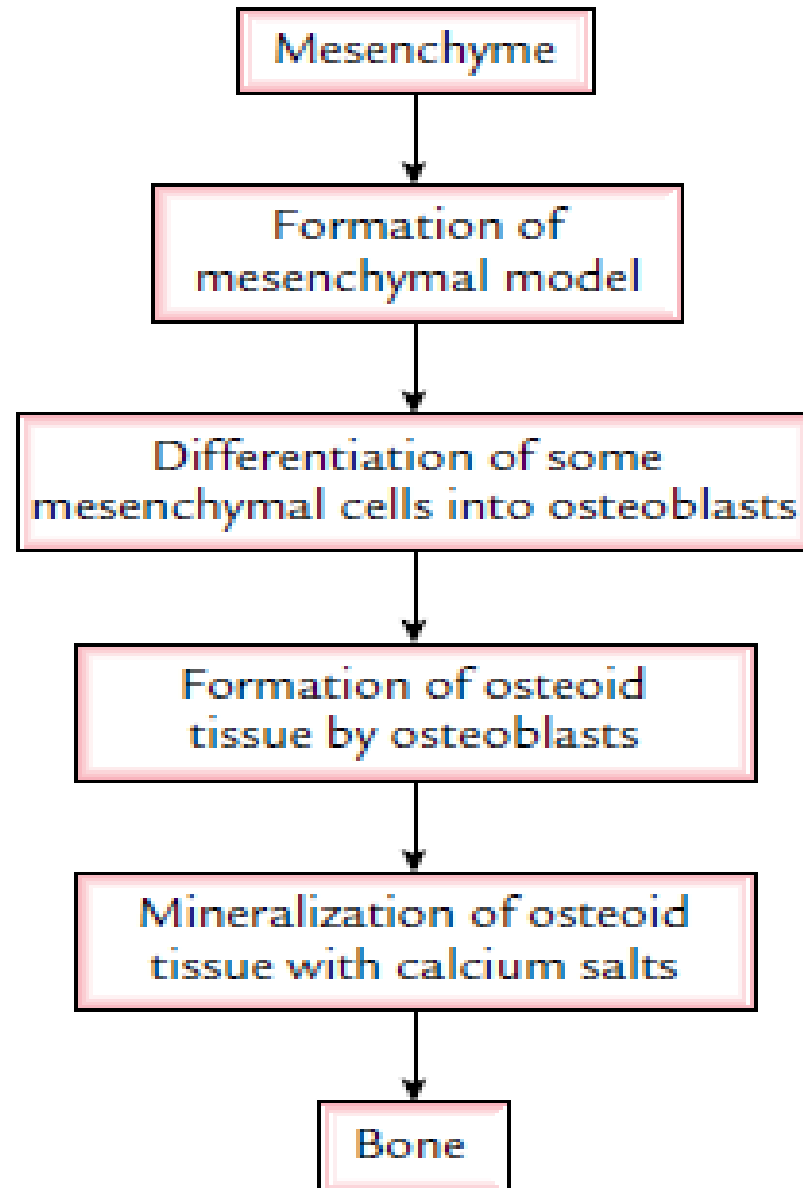
Process of intramembranous Ossification



VS



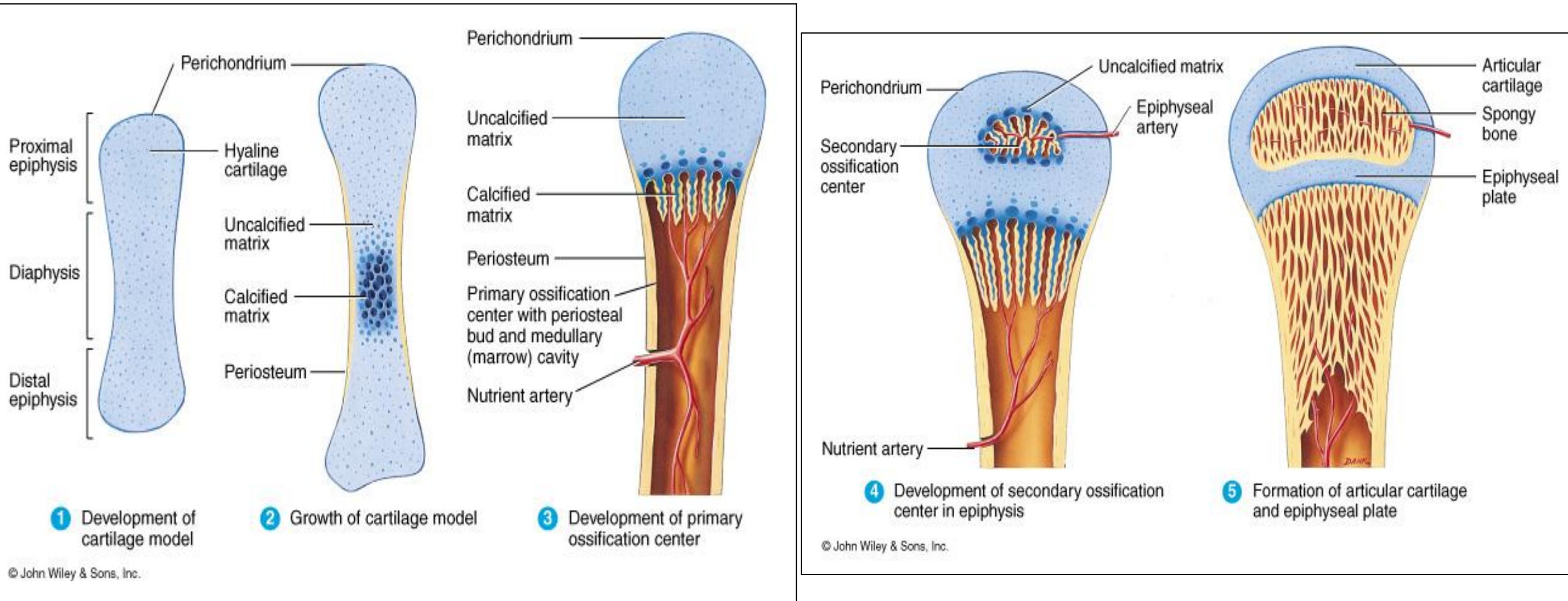
Process of intramembranous Ossification



Cartilagenous (Endochondral) Ossification

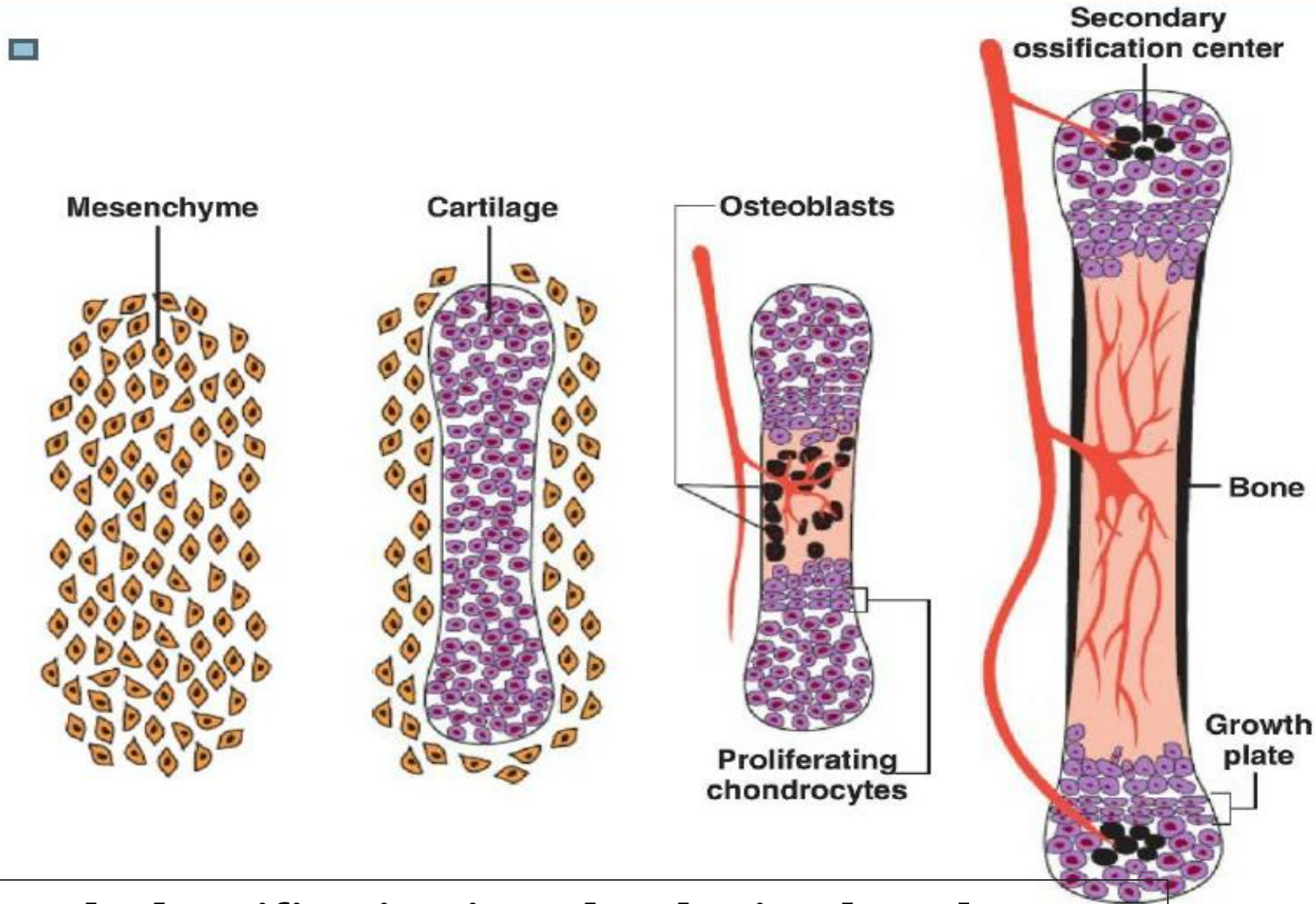
- ❖ **During endochondral ossification**, a cartilage model first forms and is eventually replaced with bone.
- The cartilage cells enlarge and matrix surrounding them gets calcified under the influence of an enzyme (**alkaline phosphatase**) secreted by cartilage cells.
- **The cartilage cells die and disappear**, leaving behind the empty spaces (**primary areolae**).
- The cells on the surface of cartilage, i.e., cells of perichondrium (primodium of periosteum) differentiate into **osteoblasts**.

Cartilagenous (Endochondral) Ossification,,



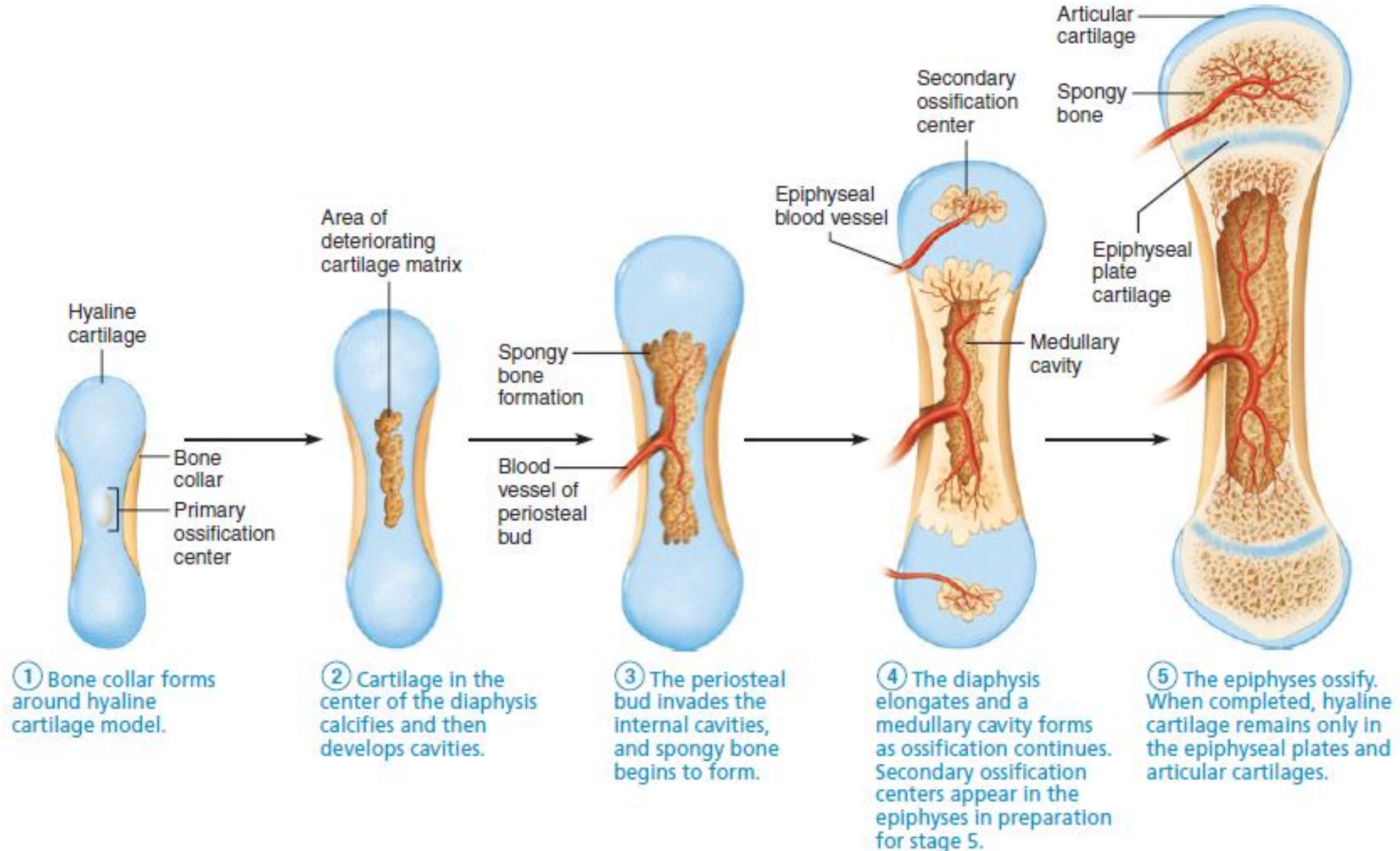
Endochondral ossification in a developing long bone.

Cartilagenous (Endochondral) Ossification,,,



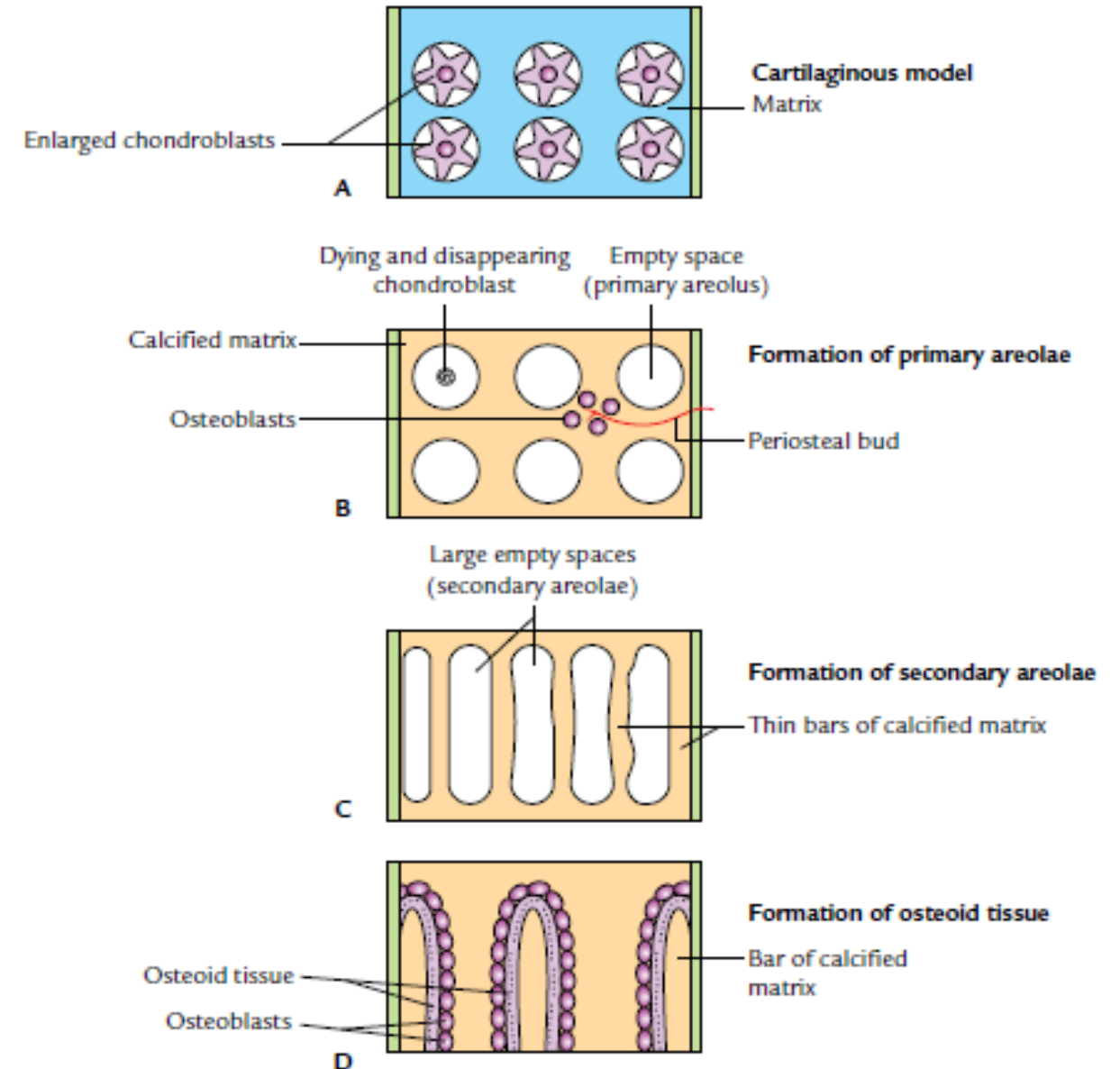
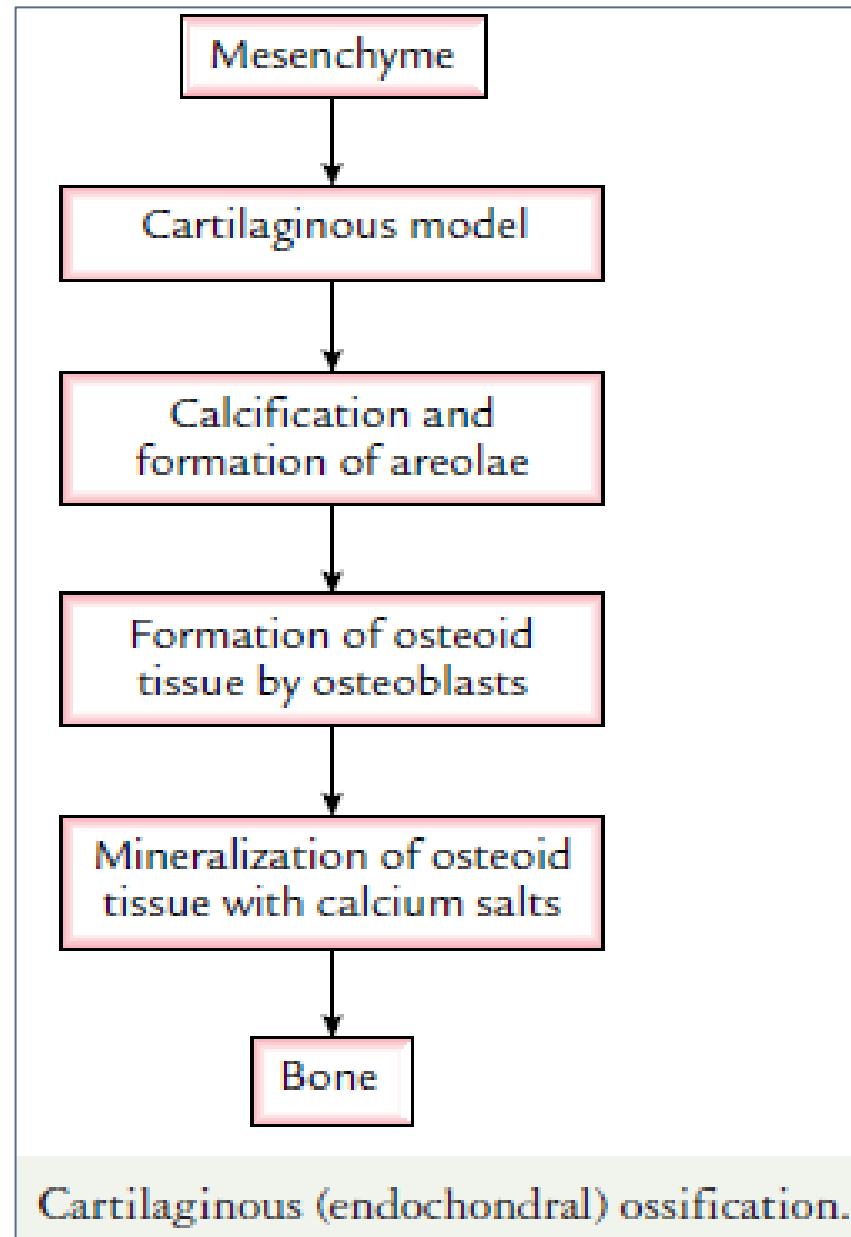
Endochondral ossification in a developing long bone.

Cartilagenous (Endochondral) Ossification,,



Endochondral ossification in a developing long bone.

Cartilagenous (Endochondral) Ossification,,



Important stages of cartilaginous ossification.

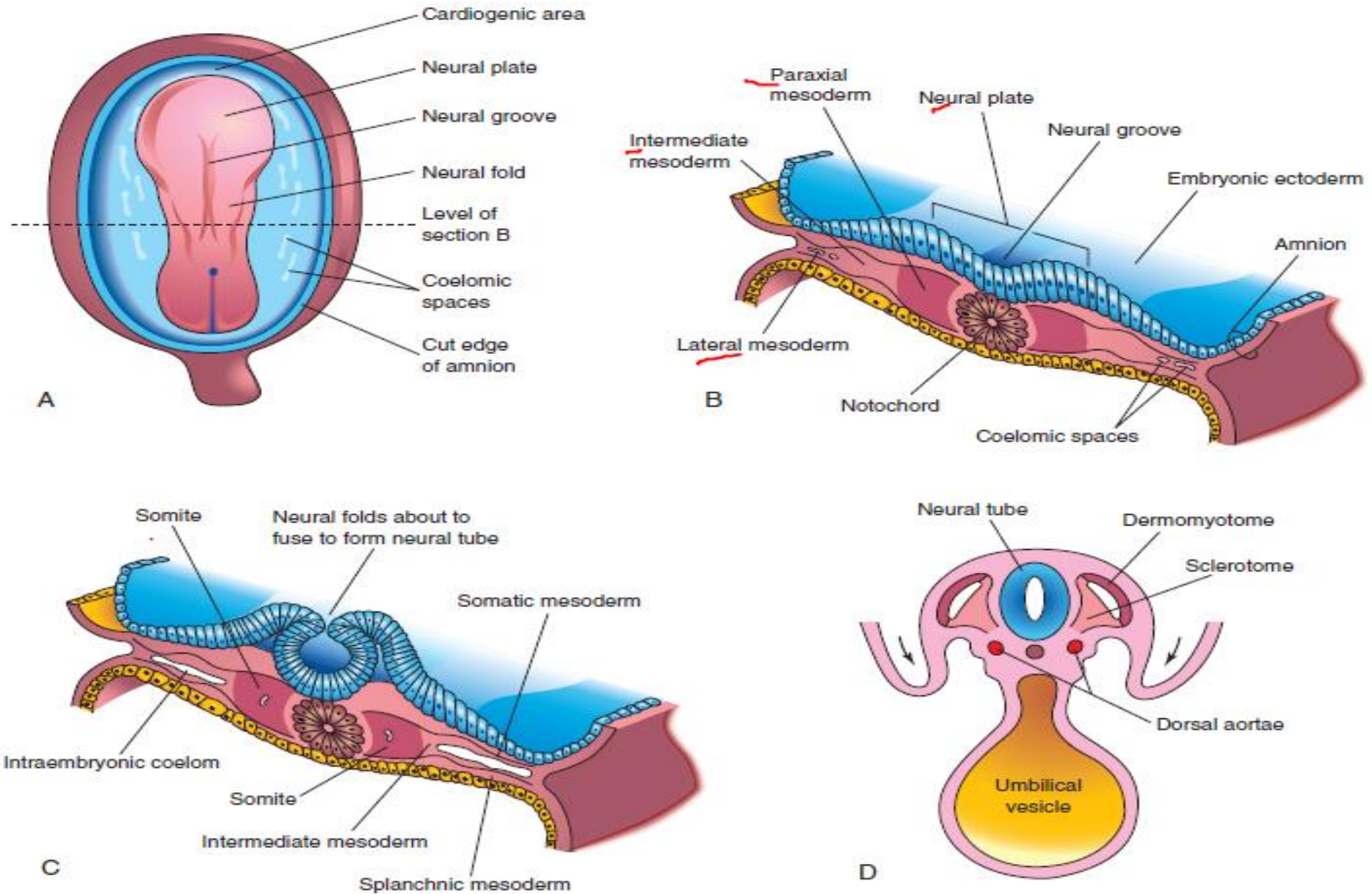
Cartilagenous (Endochondral) Ossification,,

- Endochondral bones are formed by three cell types:
- **chondrocytes** (cartilage cells), **osteoblasts** (boneforming cells), and **osteoclasts** (bone-resorbing cells).

❖ Chondrocytes have three tissue origins:

- ❑ The paraxial mesoderm forms the axial skeleton, including the occipital portion of the cranial base;
 - ❑ The lateral plate mesoderm forms the appendicular skeleton and sternum; and
 - ❑ Neural crest cells (i.e., ectodermal cells) give rise to the cartilaginous elements in the face & neck.
- **The origin of osteoblasts and osteoclasts is less diverse:** osteoblasts arise from mesenchymal stem cells and osteoclasts arise from the hematopoietic system.

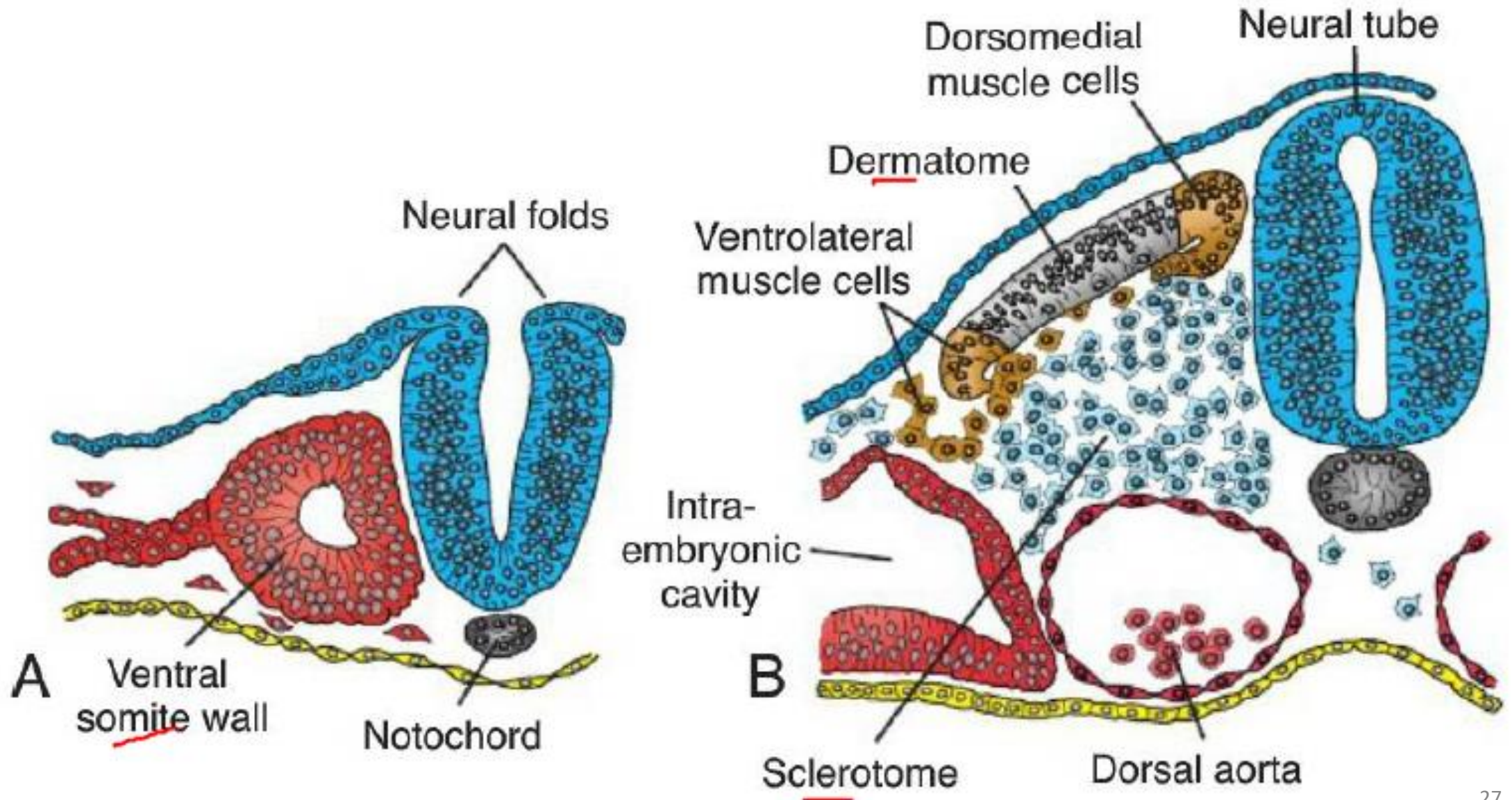
Cartilagenous (Endochondral) Ossification,,



Development of Axial Skeleton

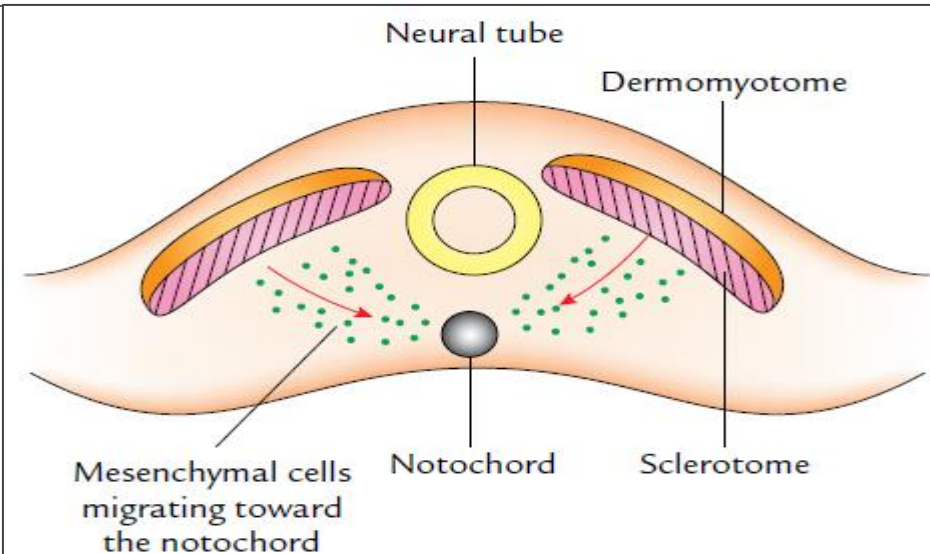
- In general, the skeletal system develops from **paraxial** and **lateral plate mesoderm** and from **neural crest**.
 - **Dermal bones develop from neural crest** cells (facial bones and the frontal bone of the skull) or unsegmented paraxial (head) mesoderm (e.g., parietal bone of the skull).
 - **In dermal bones, the osteoblasts directly differentiate within the mesenchyme.**
- ❑ **Paraxial mesoderm** forms a **segmented** series of tissue blocks on each side of the neural tube, known as **somitomeres** in the head region & **somites** from the occipital region caudally.
- **Somites differentiate** into a ventromedial part, the **sclerotome**, and a dorsolateral part, the **dermomyotome**.

Development of Axial Skeleton,,



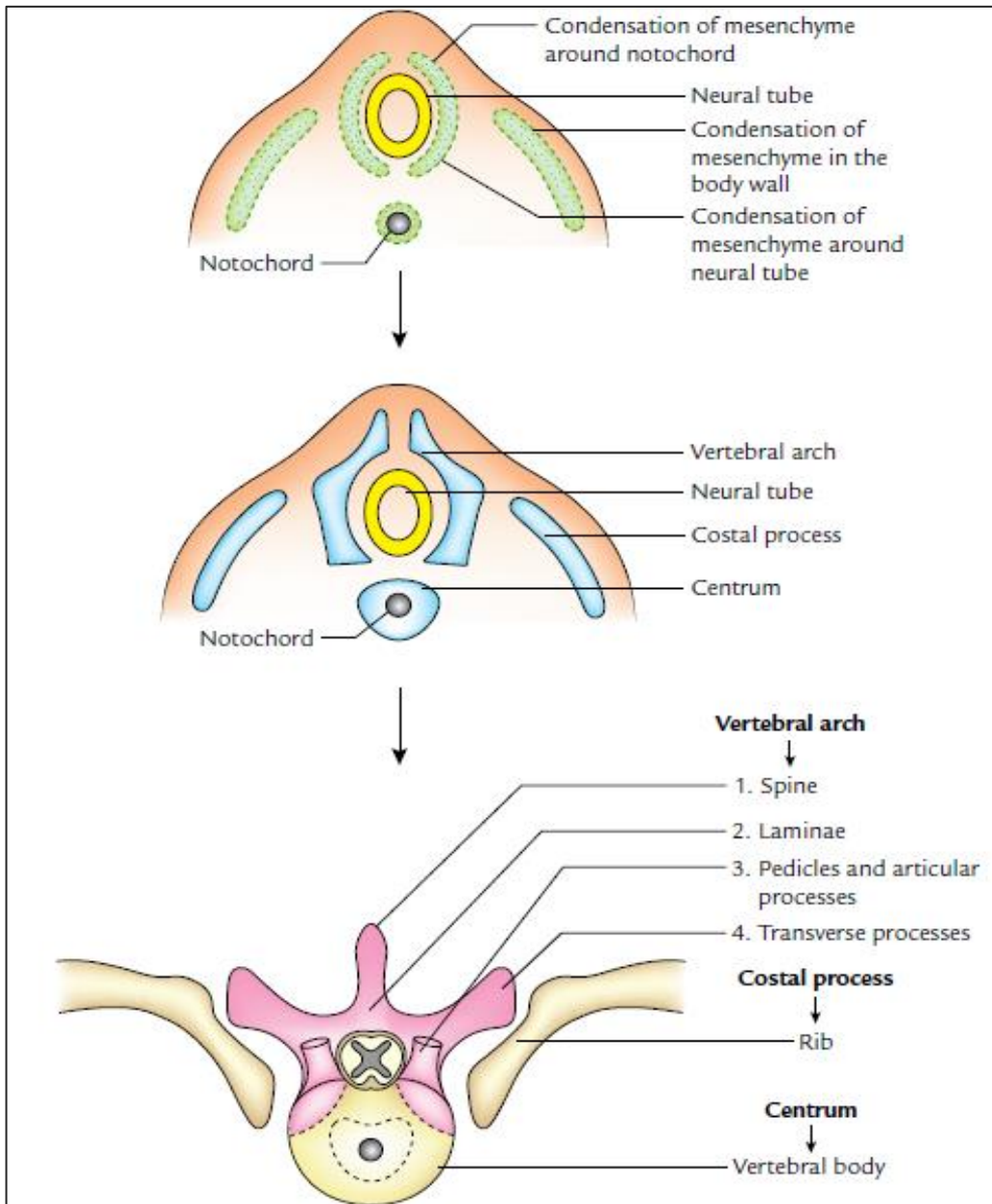
Development of Vertebral Column

- Vertebrae form from the sclerotome portions of the somites, which are derived from paraxial Mesoderm
- **The cells of sclerotome become** large, polymorphic, and get loosely arranged to form embryonic tissue called **mesenchyme**.
- **The mesenchymal** cells have potential to differentiate into **fibroblasts, chondroblasts, and osteoblasts**.



➤ Note the migration of mesenchymal cells towards the notochord (red arrows)

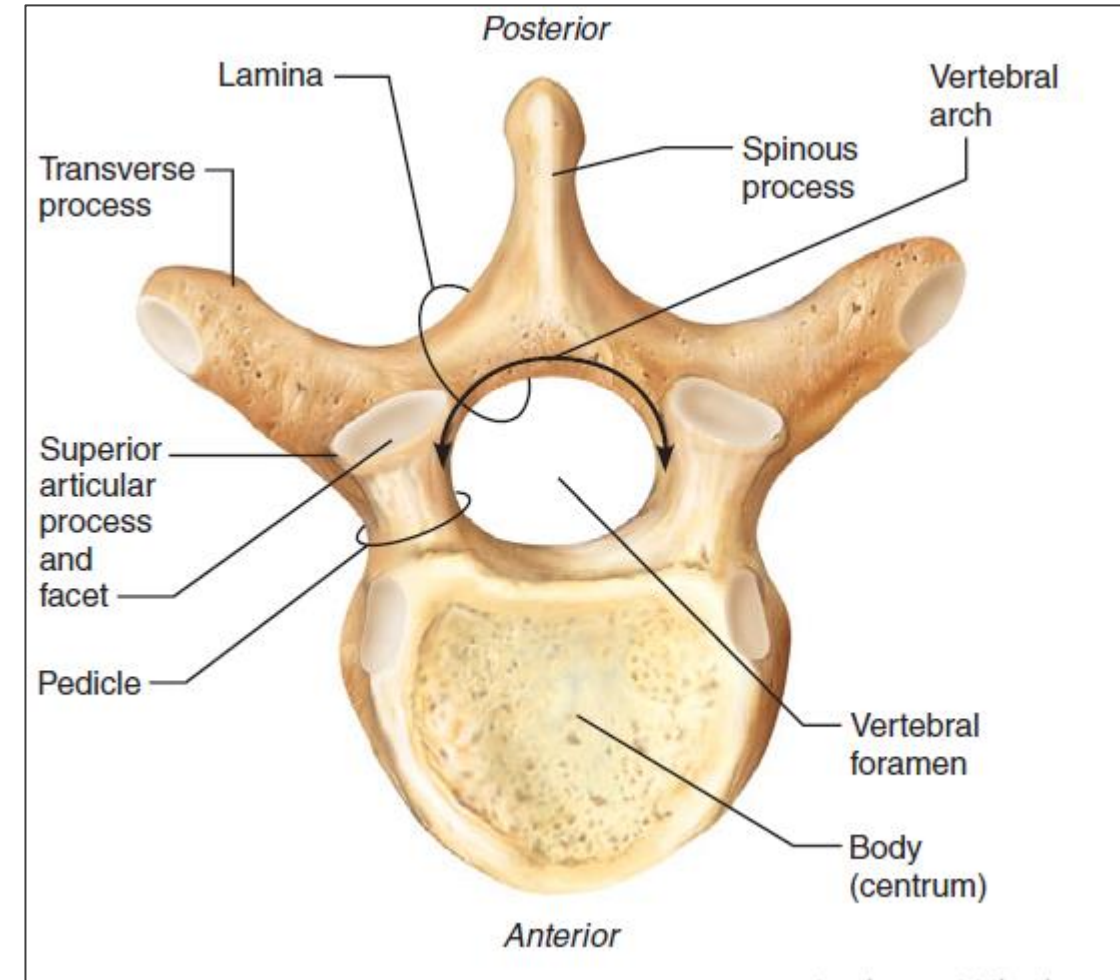
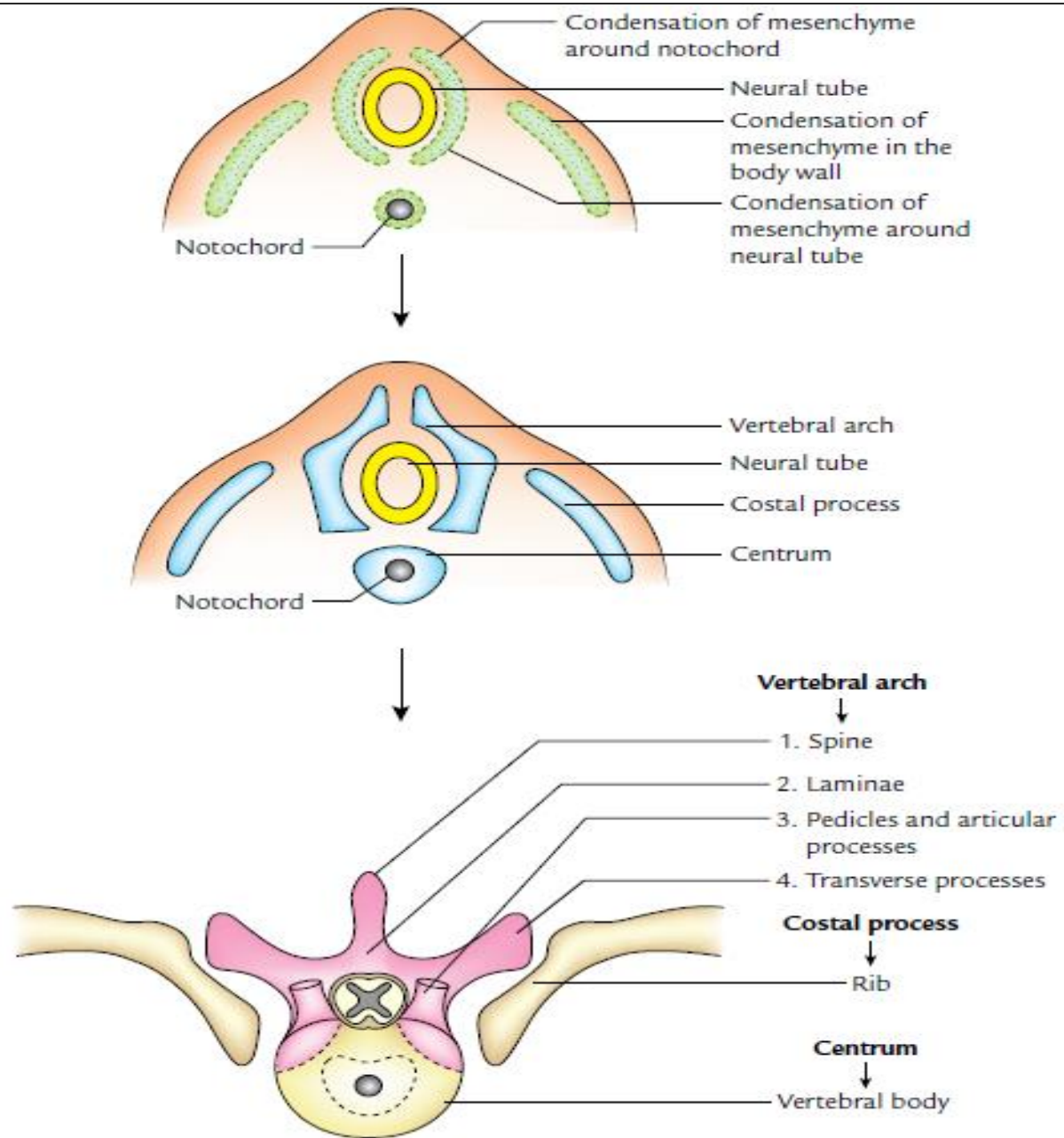
Development of Vertebral Column,,



- ❑ The mesenchymal cells from the sclerotomes migrate medially towards notochord to form the **vertebral column**.
- ❑ The mesenchymal cells condense (a) around the notochord to form the **centrum**, (b) around the neural tube to form right and left **neural (vertebral) arches**, and (c) in the body wall adjacent to the proximal part of neural arches to form **costal processes**.
- ❖ The adult derivatives of these structures are as follows:
 - The **centrum** forms the **vertebral body**.
 - The **vertebral arches** form the **pedicles, laminae, spine, articular processes, and transverse processes**.
 - The **costal processes** form the **costal elements of the transverse processes**.

Development of Vertebral Column,,

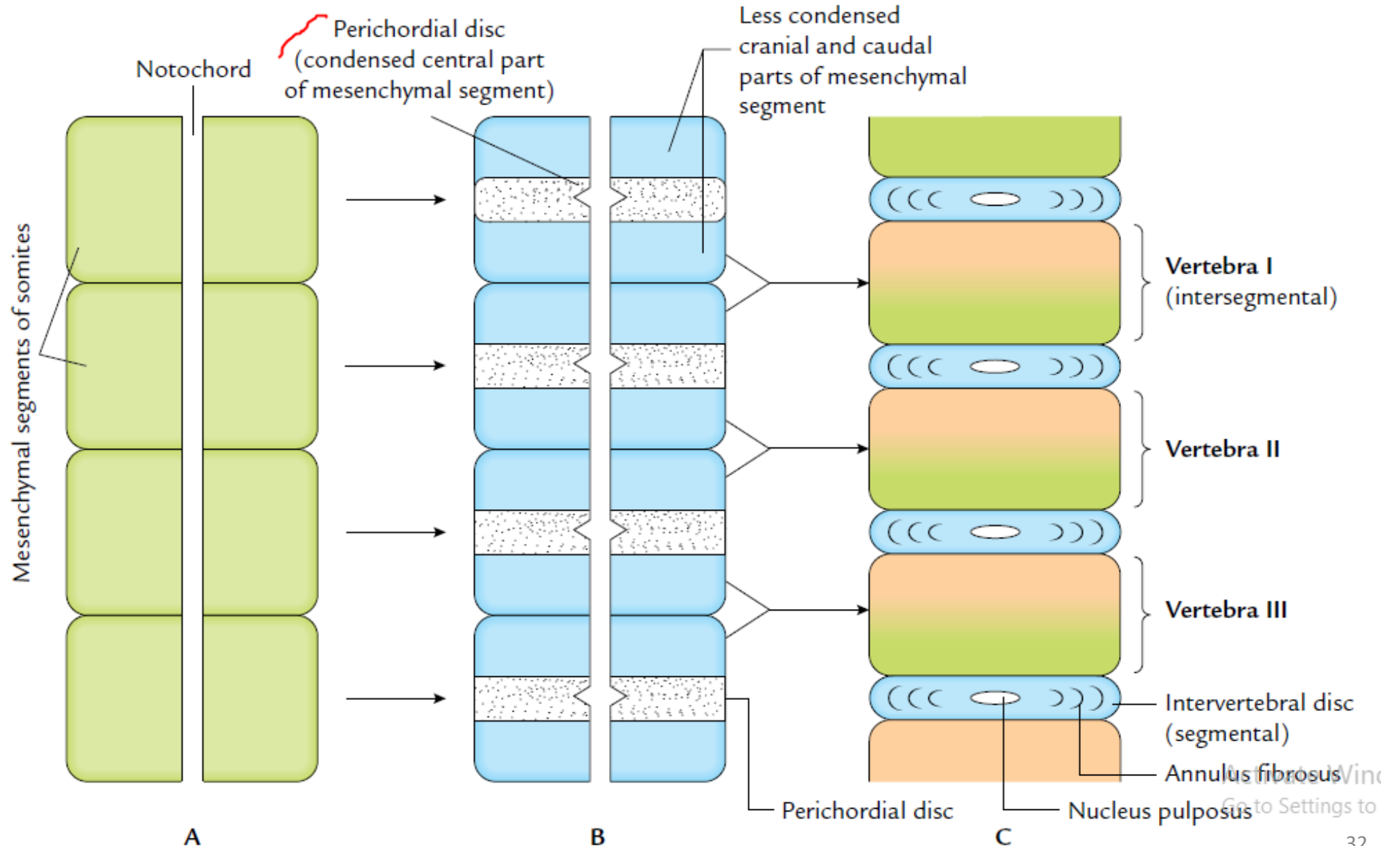
The adult derivatives of vertebral structures



Development of Vertebral Column,,

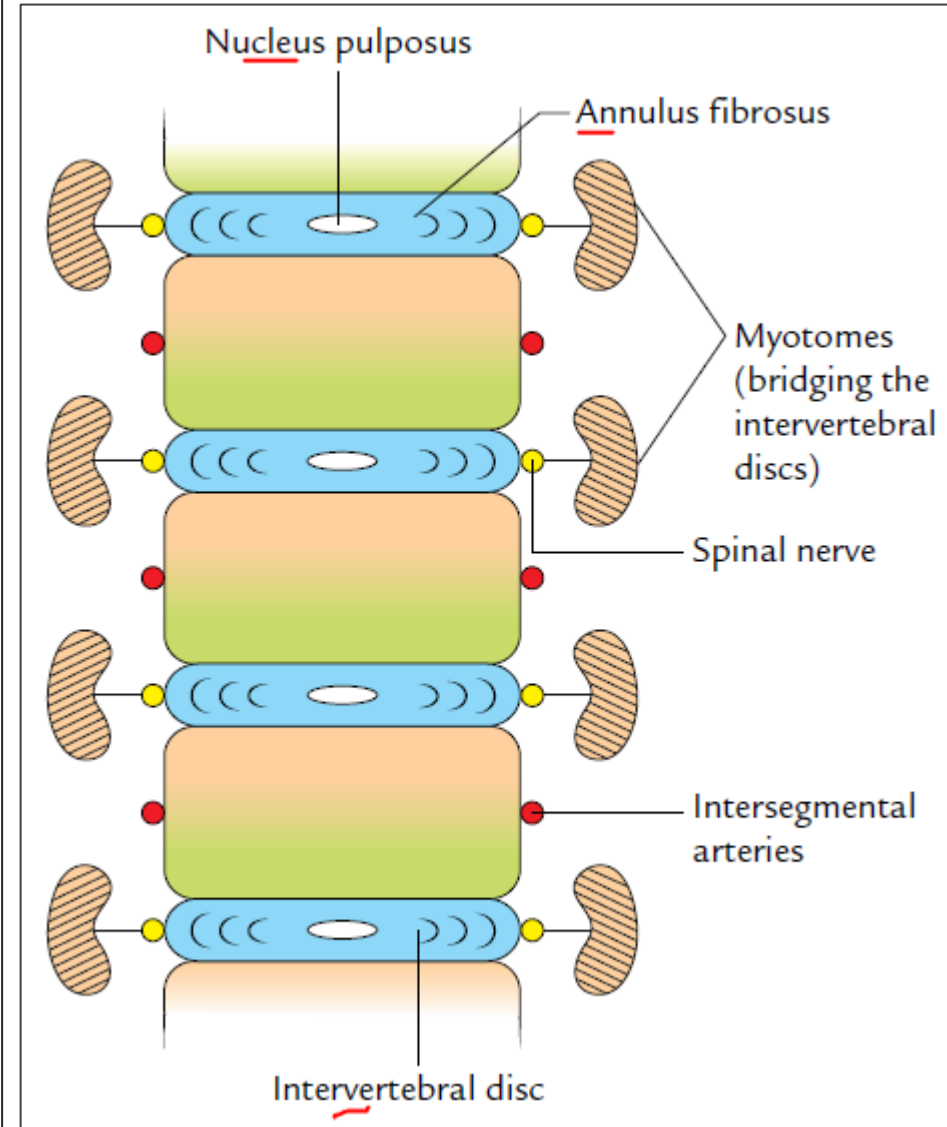
- **Lateral extension** of the mesenchyme forms the transverse process of the vertebra while its **ventral extension** forms the rib.
- **There is condensation of cells in the middle of the segment.** This condensed part is called *perichordal disk*.
- The perichordal disk intervenes between the two vertebral bodies and forms the *intervertebral disk*.
- **The notochord gives rise nucleus pulposus part of the *intervertebral disk*. while the rest of the notochord disappears.**
- ❖ Remember that the vertebra is an example of intersegmental structure which is formed by the parts of the two somites.

Development of Vertebral Column,,

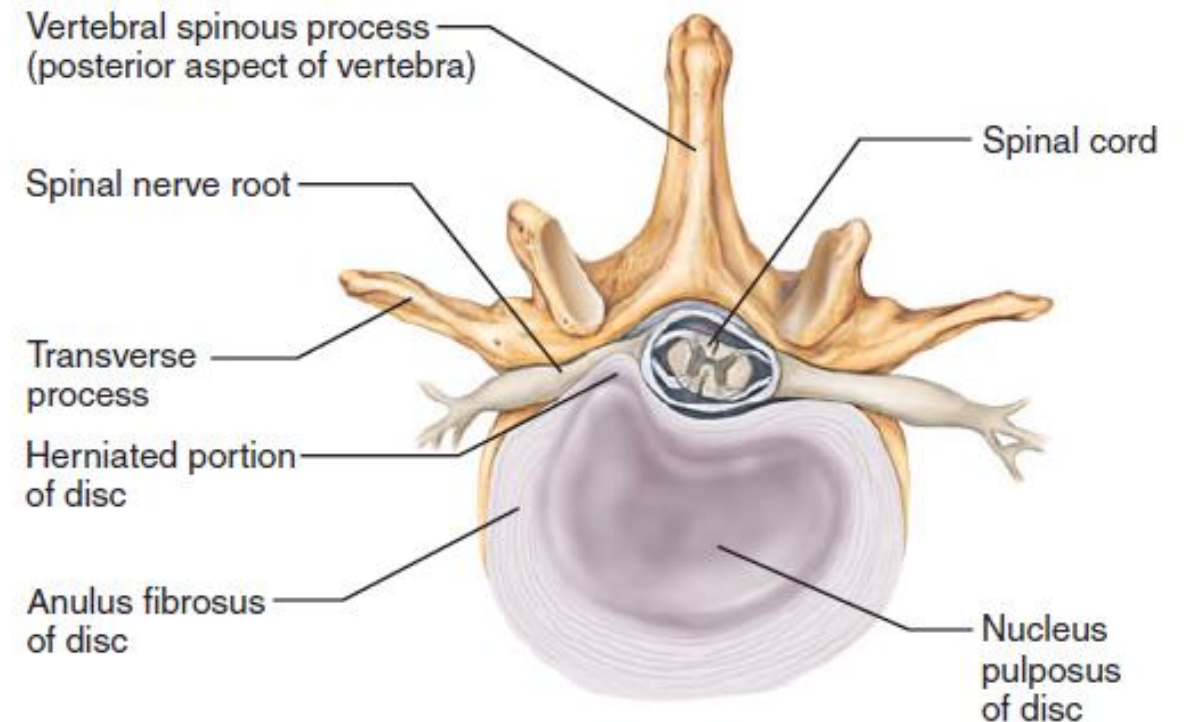
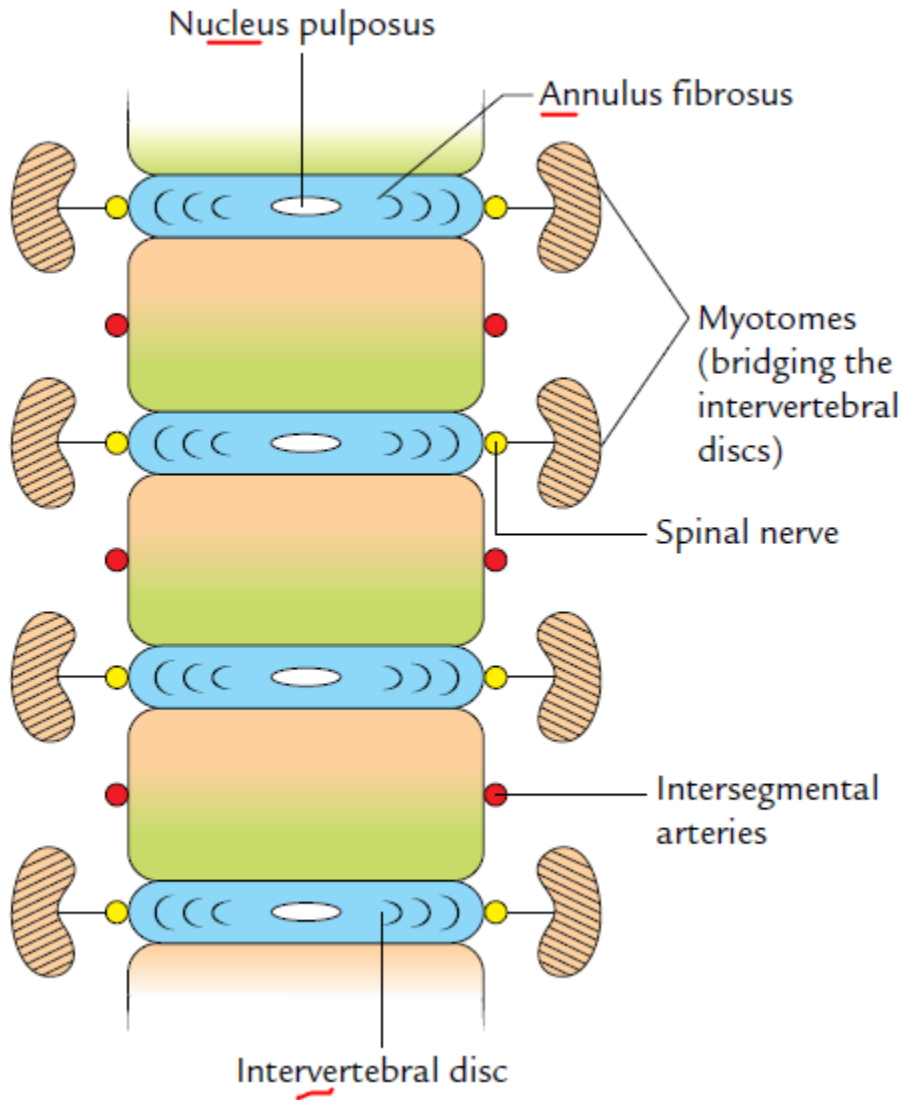


Development of Vertebral Column,,

- The notochord degenerates and disappears where it is surrounded by the developing vertebral bodies.
- **Between the vertebrae**, the **notochord** expands to form the gelatinous center of the IV disc—the **nucleus pulposus**.
- This nucleus is later surrounded by circularly arranged fibers that form the **annulus fibrosus**.
- **The nucleus pulposus and annulus fibrosus together form the IV disc.**
- The mesenchymal cells, surrounding the neural tube, form the **neural arch**, that is, the **primordium of vertebral arch**.
- The mesenchymal cells in the body wall form **costal processes**, which form the ribs in the thoracic region.



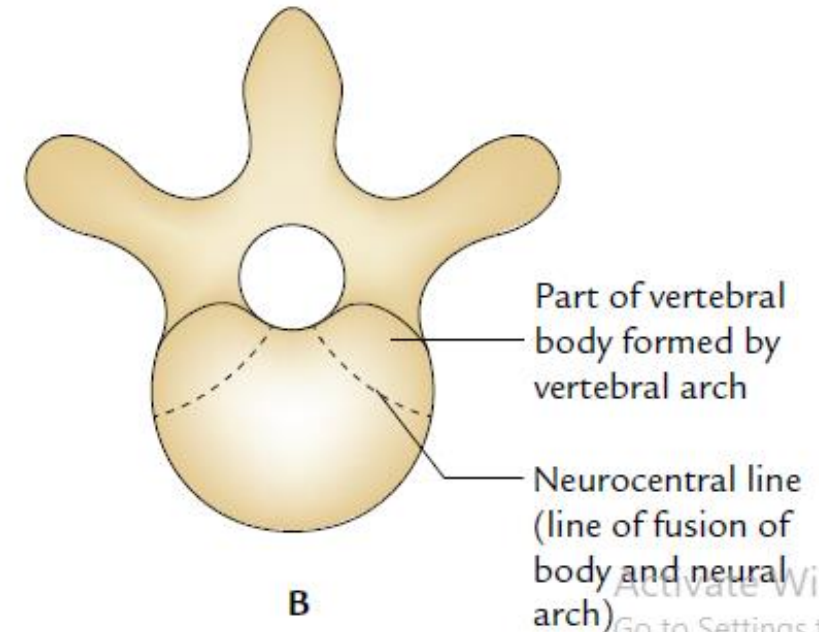
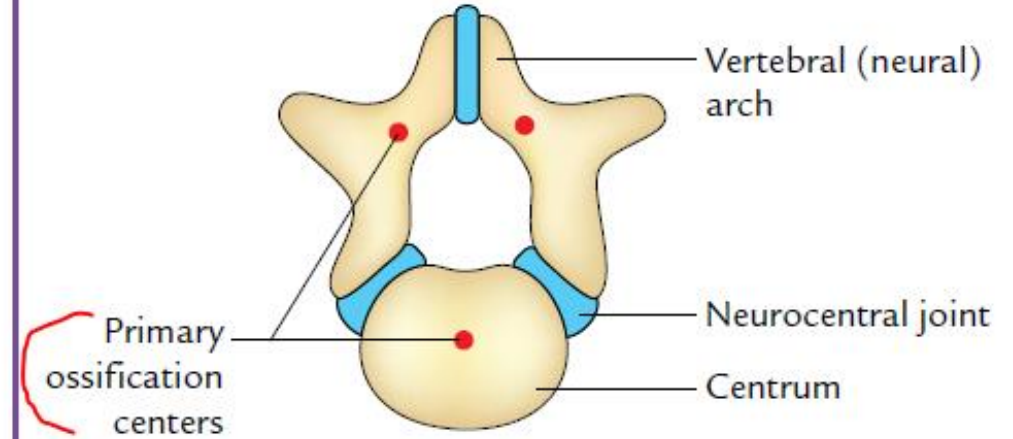
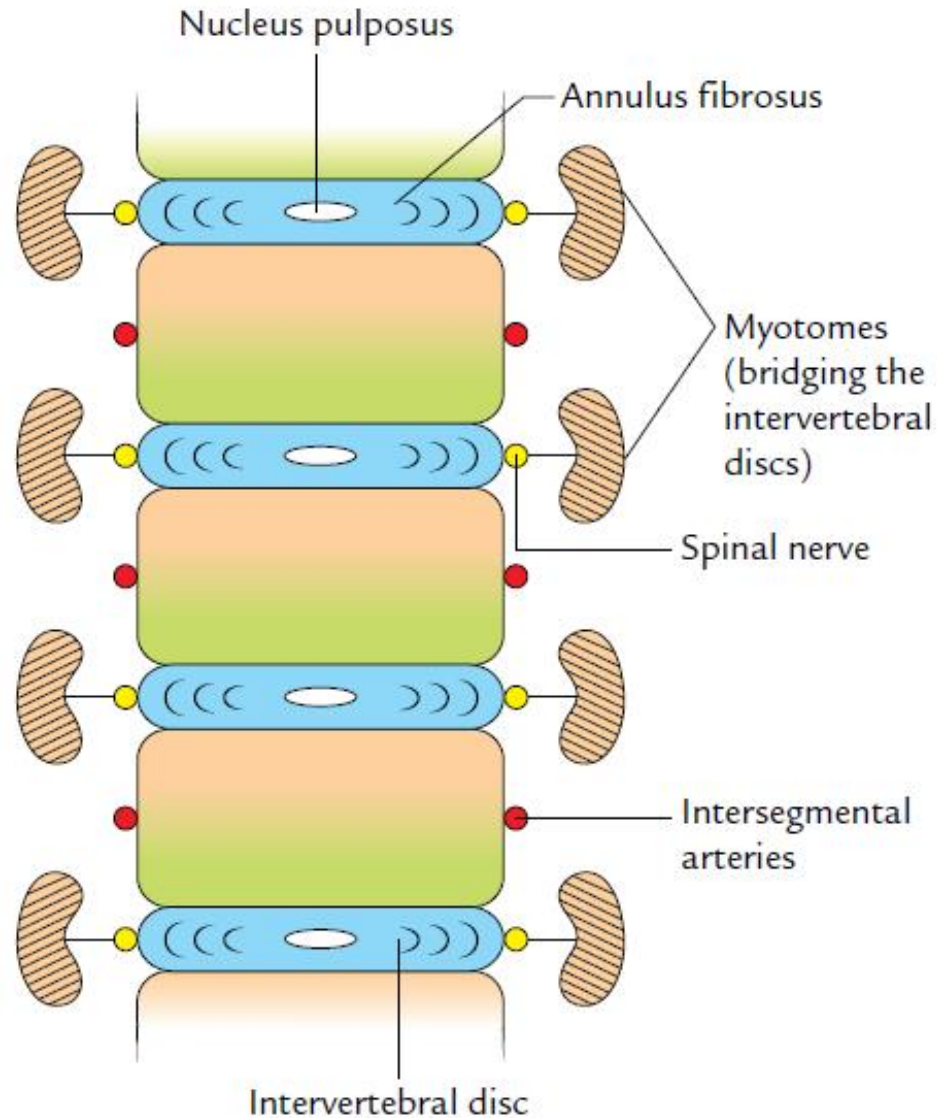
Development of Vertebral Column,,



Development of Vertebral Column,,

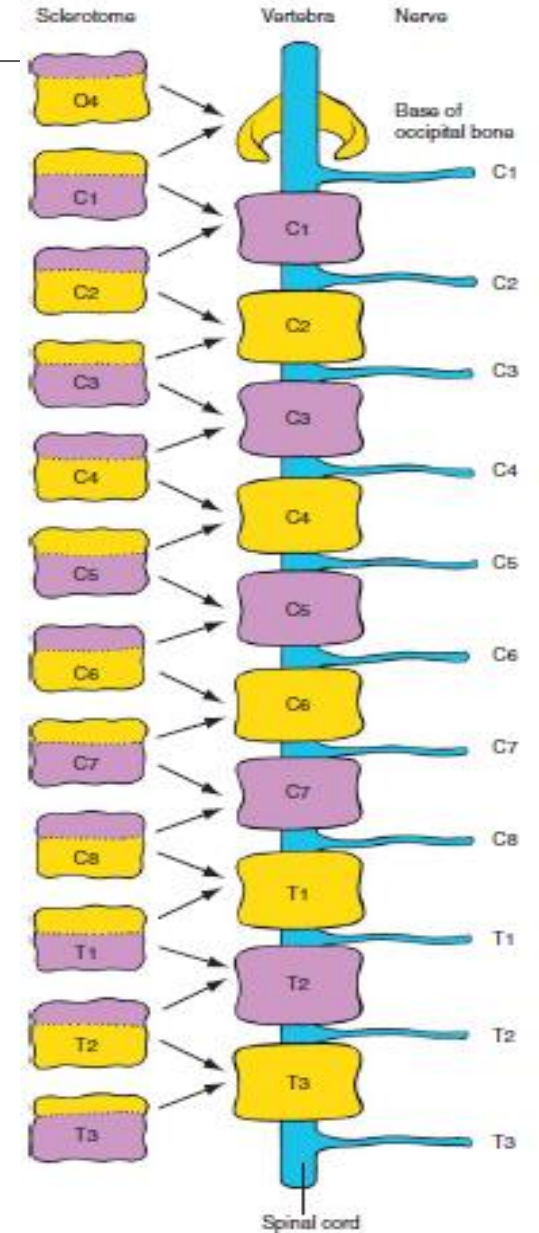
- **The newly formed vertebra is cartilaginous.**
- This cartilaginous model of vertebra is soon converted into bone by ossification.
- **The vertebrae ossify by the three primary centers of ossification:** one for each vertebral (neural) arch and one for the centrum.
- **At birth, each vertebra consists of three bony parts:** two vertebral arches and a centrum connected by a cartilage.
- The vertebral arches fuse posteriorly by 3–5 years of age to form **spinous process**.
- The vertebral arches articulate with the centrum at cartilaginous **neurocentral joints**, which disappear by 3–6 years of age.

Development of Vertebral Column,,,



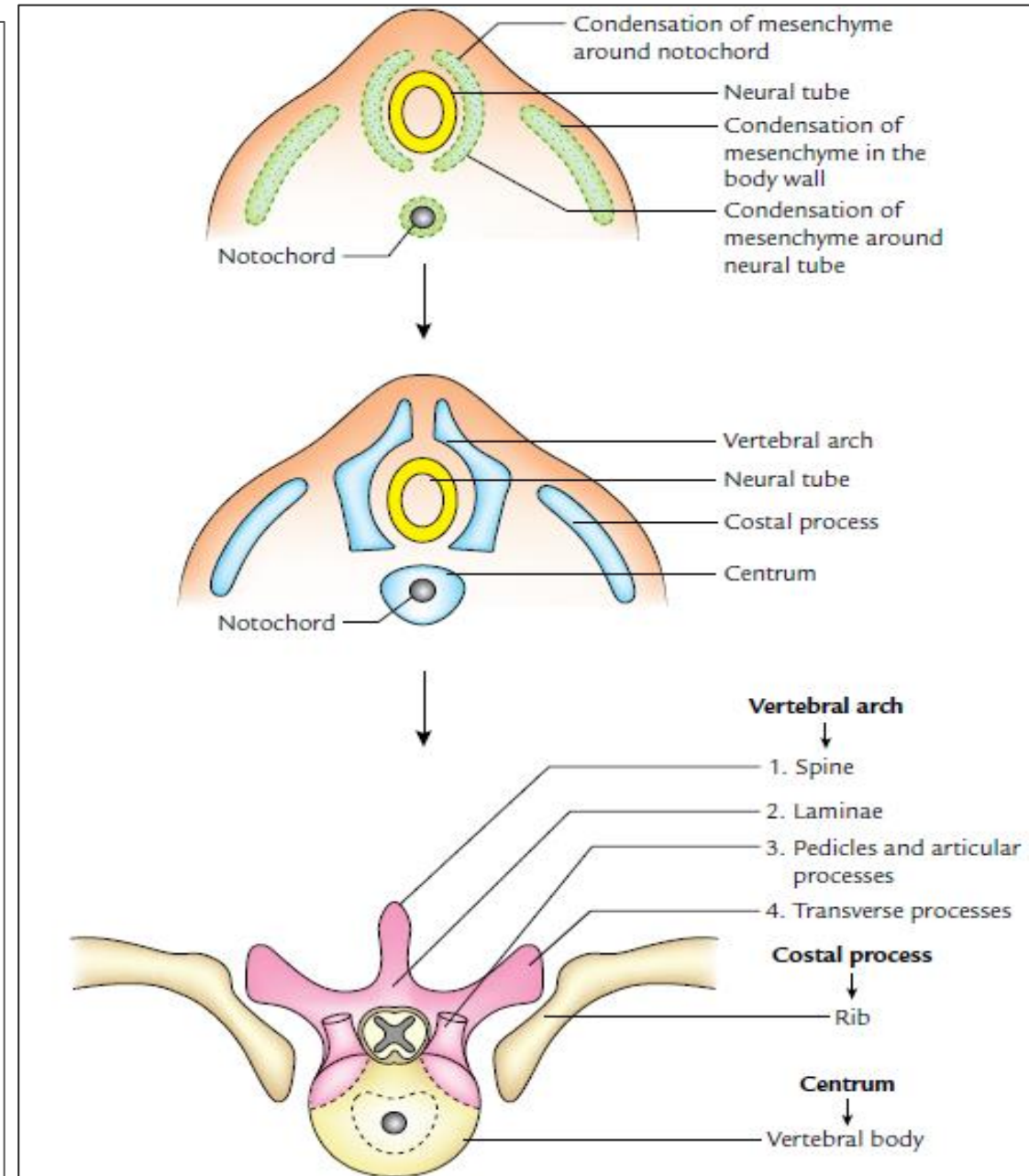
Features of Development of Vertebral Column can be summarized as follows:

- I. Vertebrae are **intersegmental** structures because each vertebra is derived from portions of two adjacent somites (e.g., caudal half of one somite and cranial half of the other somite below it).
 - II. The transverse processes and ribs are **intersegmental**, and hence separate the muscles derived from two adjoining myotomes.
 - III. The spinal nerves are segmental and hence emerge between the two adjacent vertebrae.
- **N.B.** Each vertebra is formed by caudal half of one somite and cranial half of other somite below it.



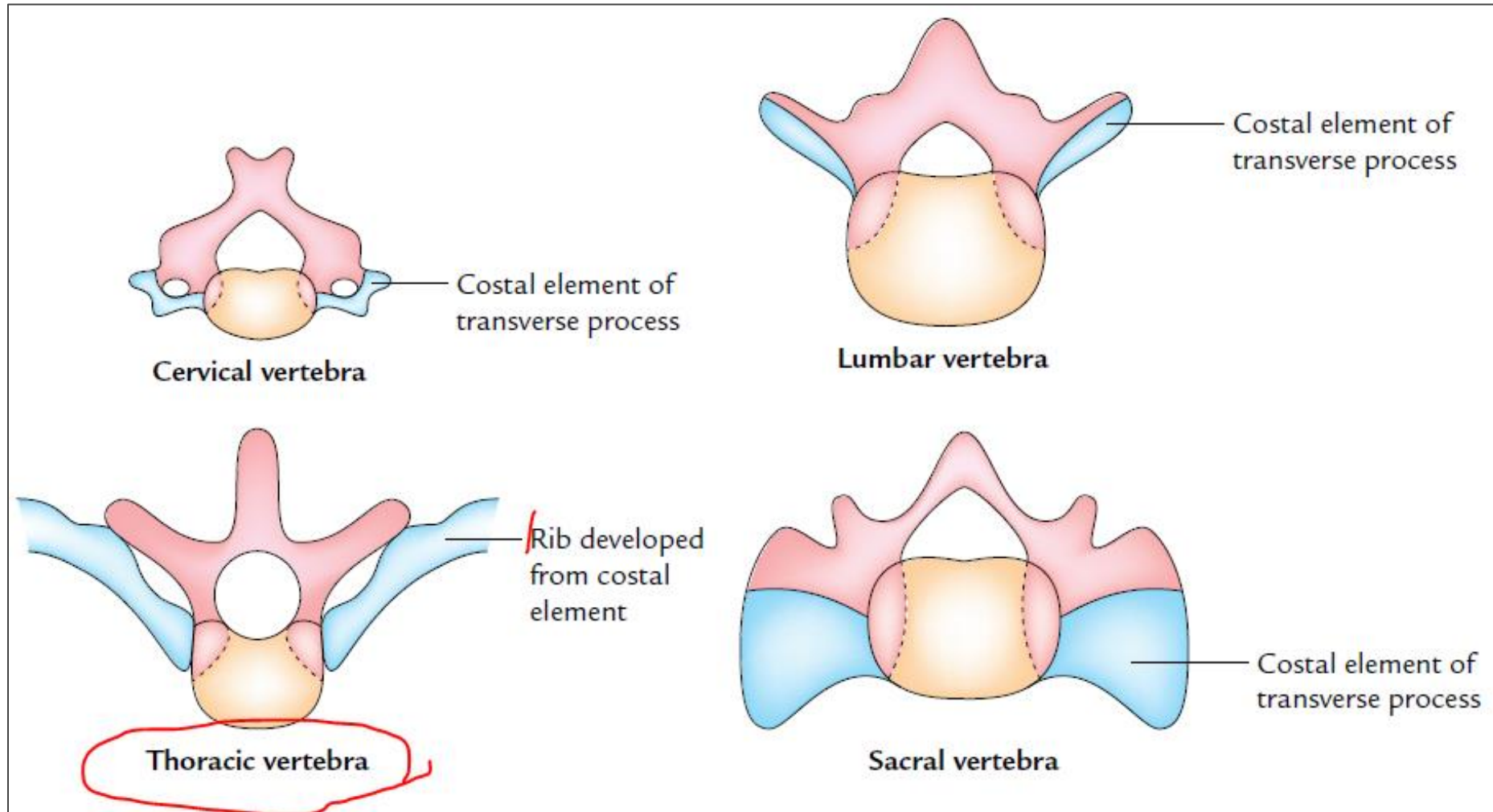
Development of Ribs

- The ribs develop from the mesenchymal **costal processes** of the thoracic vertebrae.
- **They become cartilaginous during the embryonic period and ossify during the fetal period.**
- The original site of union of the costal processes with the vertebra is replaced by costovertebral synovial joints.
- In **cervical, lumbar, and sacral regions**, the costal processes remain rudimentary and are represented by a small part of the transverse process of each vertebra called **costal element**.

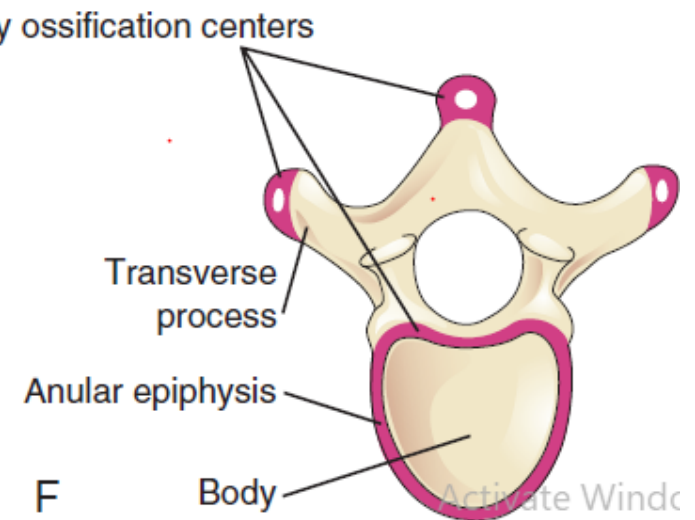
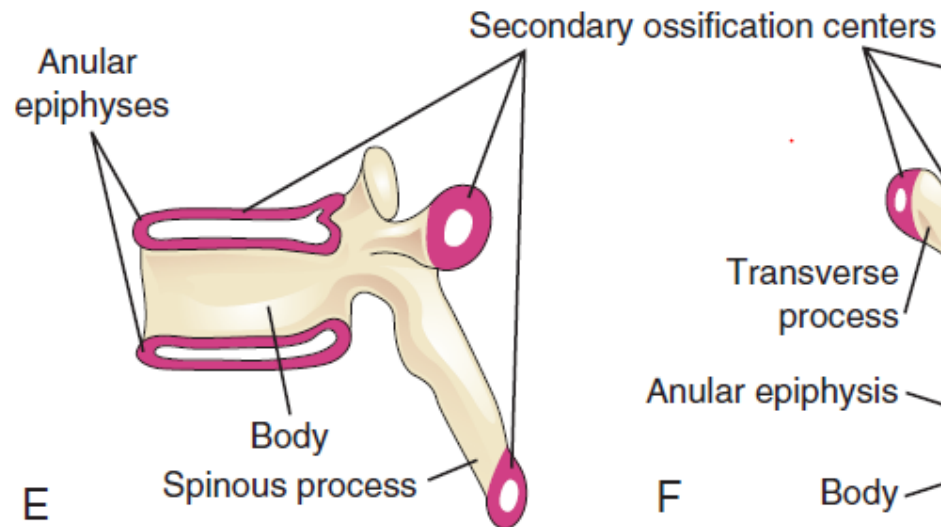
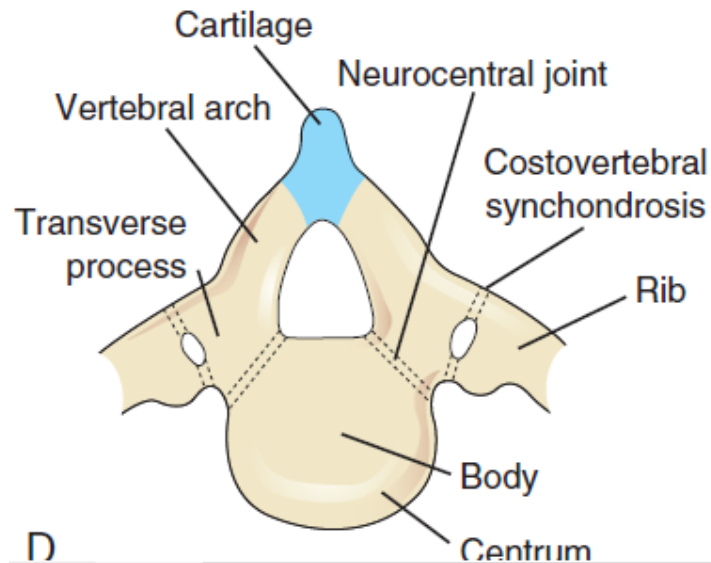
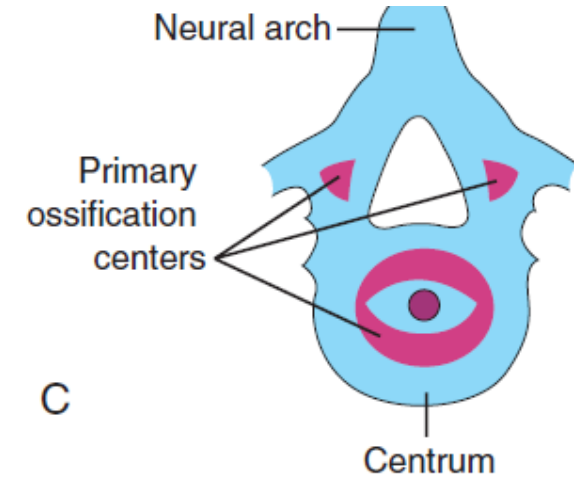
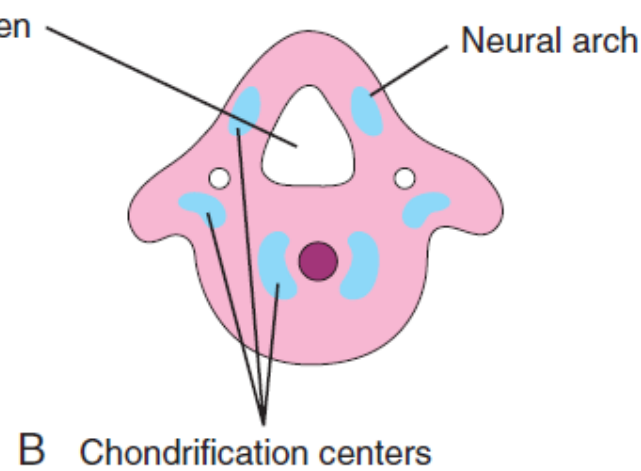
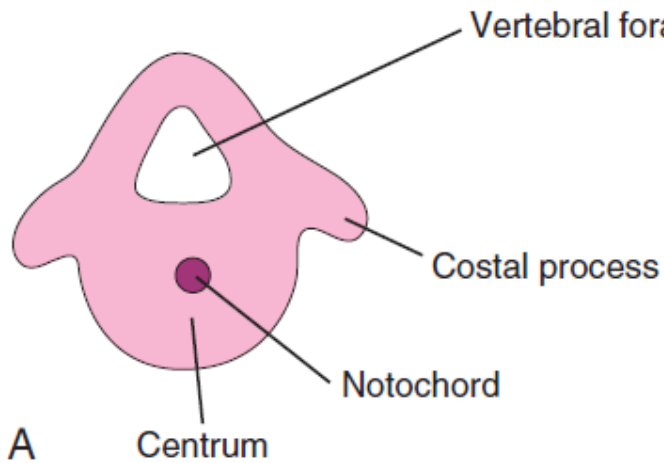


Development of Ribs,,,

- ❑ *Position of costal elements (rudimentary costal processes) in the transverse processes of cervical, lumbar, and sacral vertebrae.*
- ❑ *Note the development of rib from the costal process in the thoracic region.*



Development of Ribs,,,

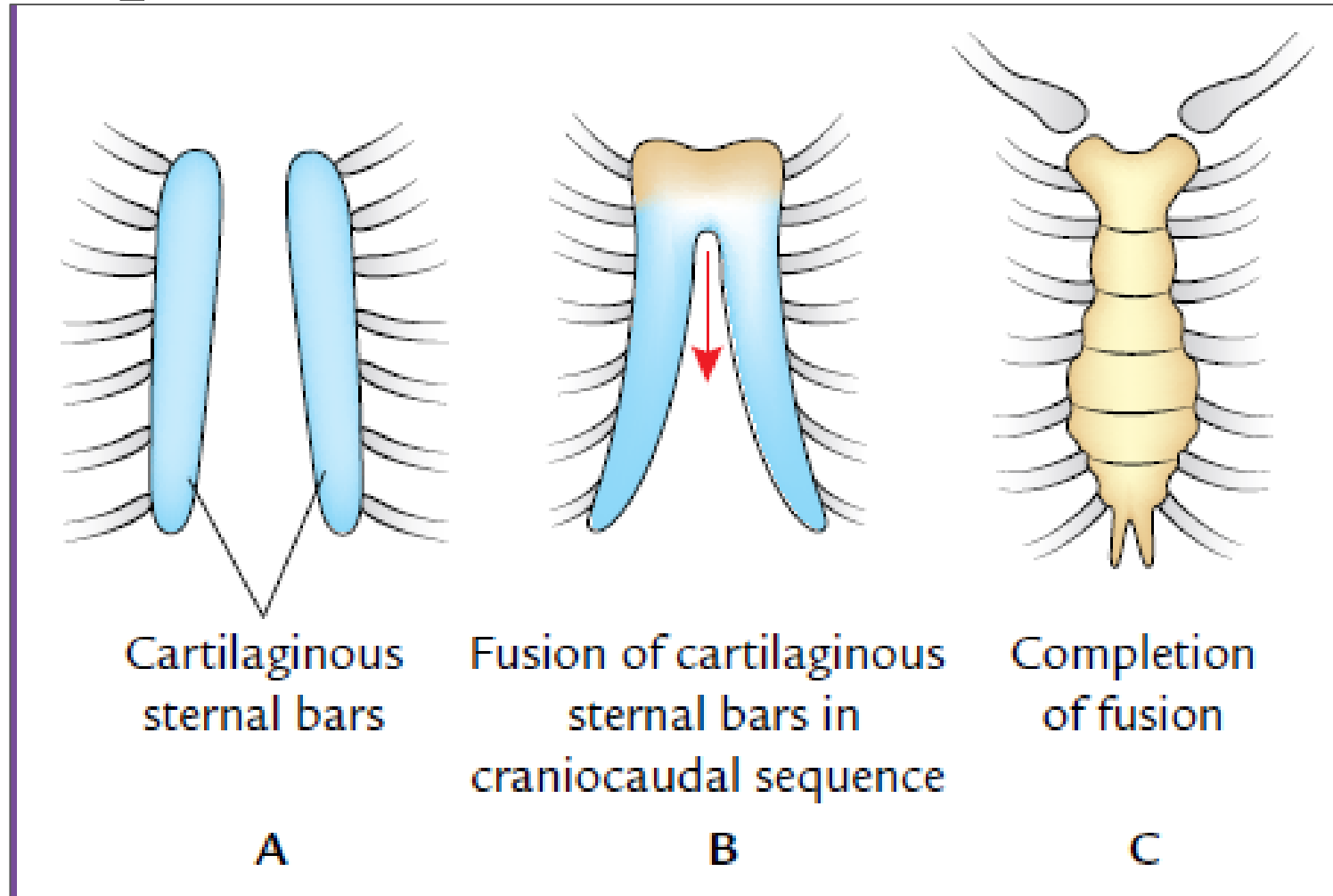


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Development of Sternum

- **At first, two mesenchymal vertical plates**, one on either side of midline, develop in anterior body wall due to condensation of somatic mesoderm.
- These plates are called **mesenchymal sternal bars**.
- Later two sternal bars are converted into cartilage to form **cartilaginous sternal bars**.
- The 2 cartilaginous sternal bars start fusing with each other in a craniocaudal sequence.
- After their fusion, the **cartilaginous model of the sternum** is formed.
- It consists of **manubrium, body** (made up of 4 segments) & **xiphoid process**
- By 10 weeks, they fuse craniocaudally in the median plane to form cartilaginous models of the manubrium, sternebrae (segments of sternal body), and xiphoid process.

Development of Sternum,,



☐ Stages of development of sternum

Development of Sternum,,,

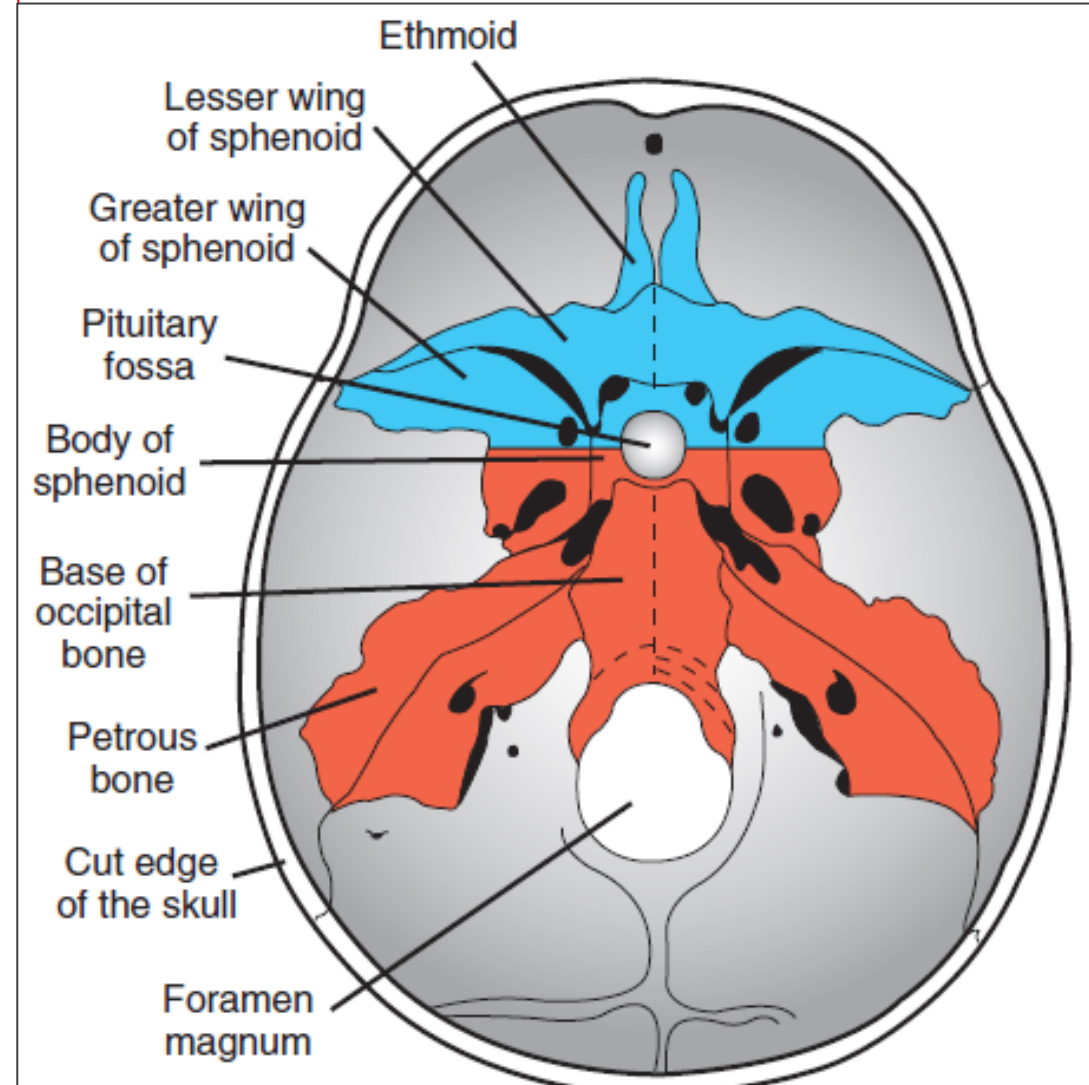
- ❖ **Manubrium and body** ossify by **five double centers** from above downward during fifth, sixth, seventh, eighth, and ninth months.
- The upper pair of centers form **manubrium** and the lower four pairs of centers form four sternebrae that fuse with each other from below upward.
- The fusion is complete by 25 years of age.
- The xiphoid process ossifies late in life.
- The center for xiphoid process appears during the third year or later and fuses with the body of sternum at about 40 years.

Development of Skull

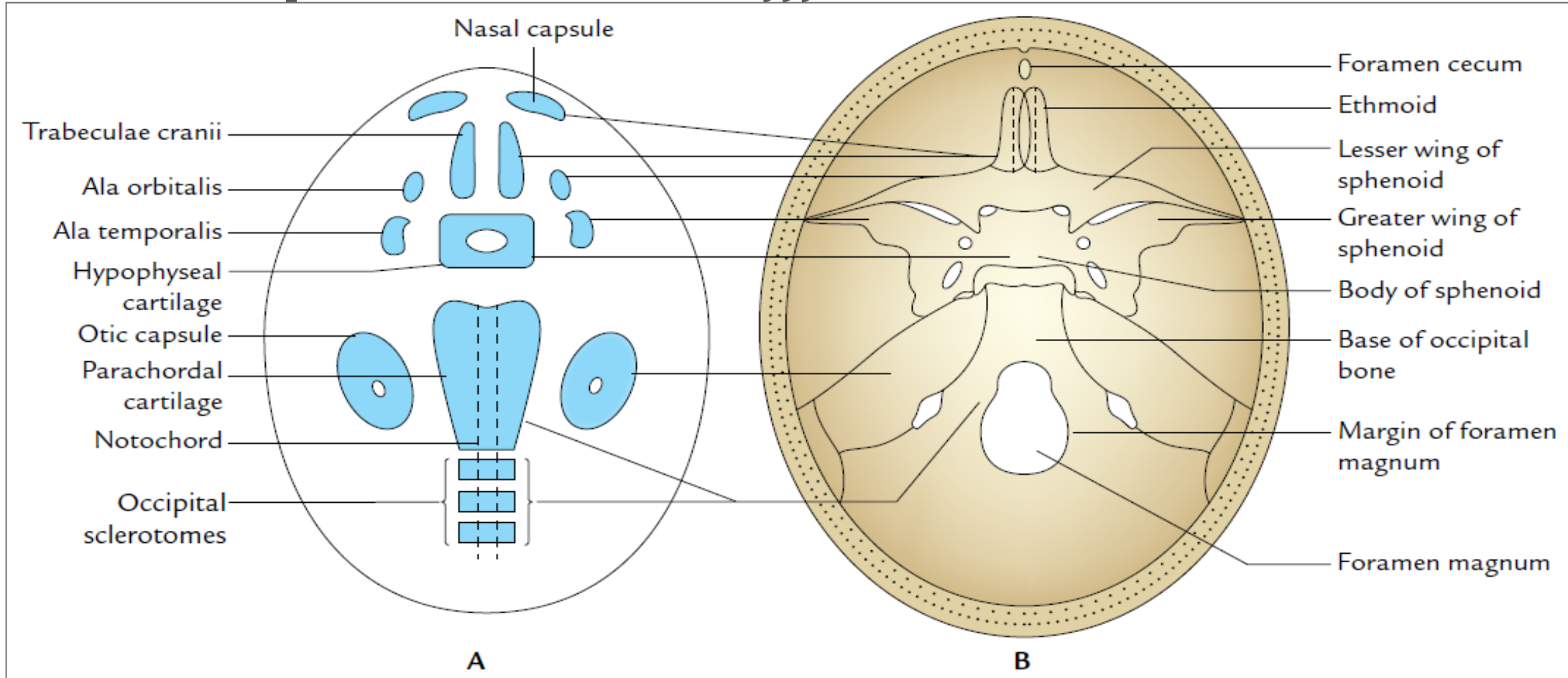
- Anatomically the skull (cranium) is divided in two parts:
 - I. **Neurocranium** that encloses cranial cavity and protects the brain, and
 - II. **Viscerocranium** that forms the skeleton of the face.
- The neurocranium consists of a cartilaginous portion that forms the **base of the skull** and a membranous portion that forms **cranial vault**.
- **Base of skull** : The base of skull, at first, is cartilaginous (cartilaginous neurocranium).
- It is formed by the fusion of several cartilages.
- These cartilages are formed by **chondrification** of mesenchyme below the brain.

Development of Skull,,,

- ❑ Dorsal view of the chondrocranium, or base of the skull, in the adult showing bones formed by **endochondral ossification**.
- ❑ Bones that form rostral to the rostral half of the **sella turcica** arise from **neural crest** and constitute the prechordal (in front of the notochord) chondrocranium (*blue*).
- ❑ Those forming posterior to this landmark arise from **paraxial mesoderm (chordal chondrocranium)** (*red*).



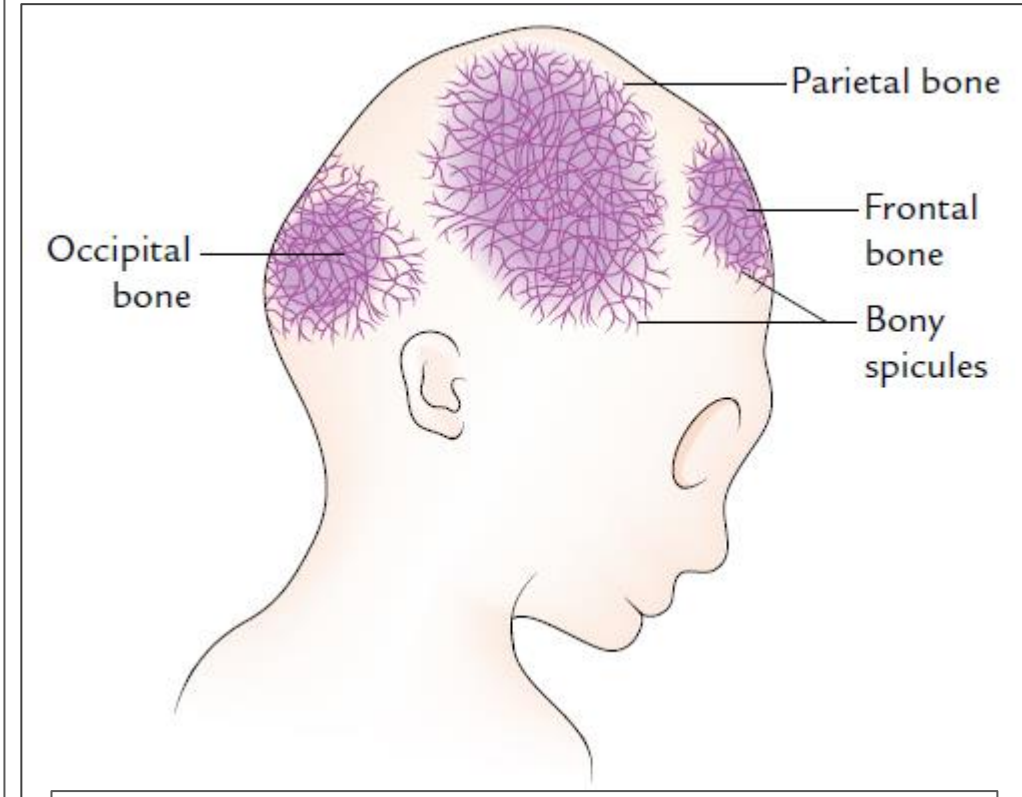
Development of Skull,,,



- Development of the base of the skull (dorsal view).
- ✓ A. Cartilages that form cartilaginous neurocranium.
- ✓ B. Bones of the base of the skull formed by various cartilages of neurocranium by endochondral ossification

Development of Skull,,,

- ❑ **Cranial vault (vault of skull);** The vault of skull develops from **mesenchyme on superior** and lateral aspects of developing brain (**membranous neurocranium**).
- This mesenchyme is derived from **the neural crest cells and paraxial mesoderm**.
- **The vault of the skull is first membranous** in nature that later undergoes membranous ossification to form a number of flat membranous bones, which together form the vault (top) of the skull.

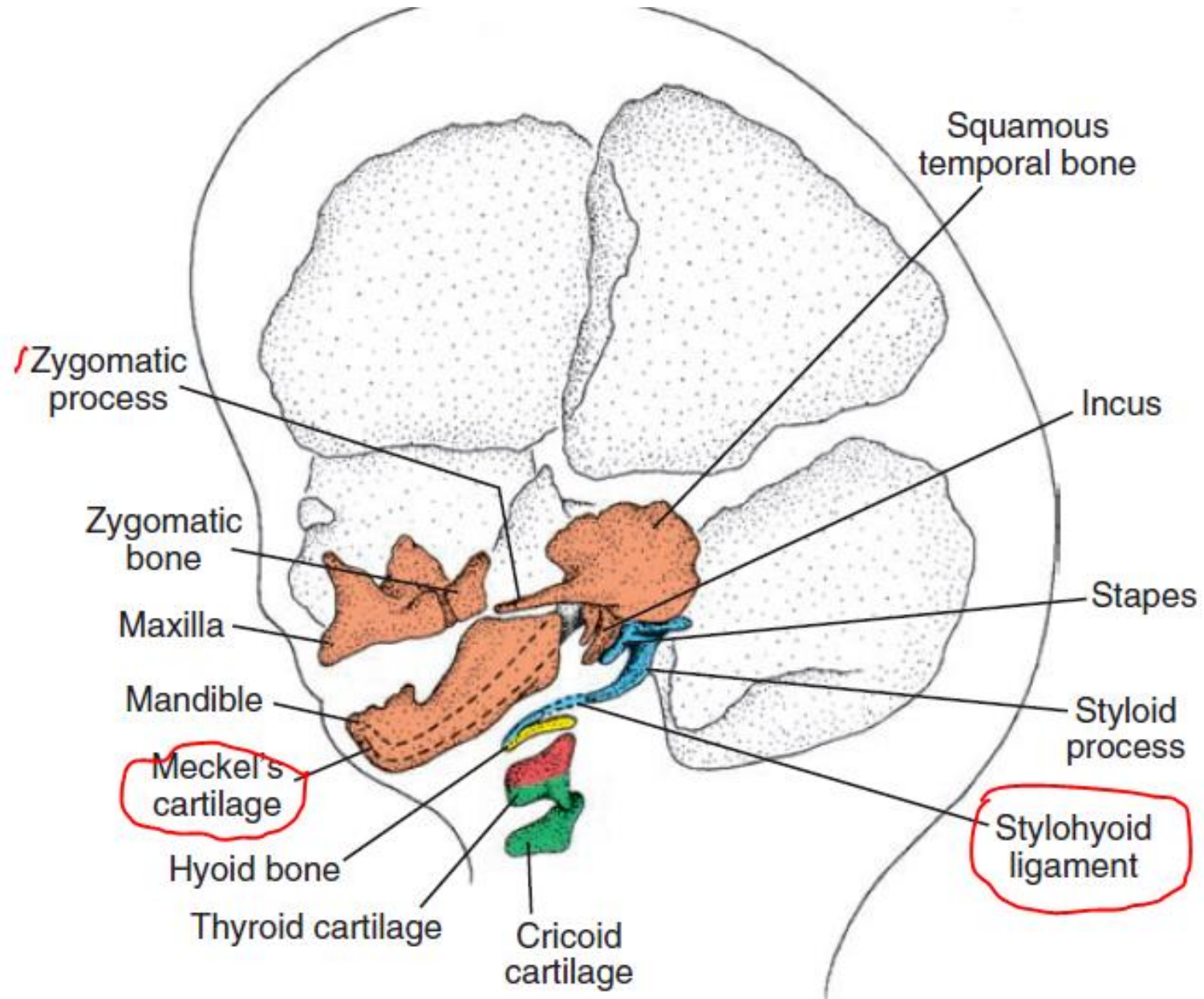


- ❑ Spread of bony spicules from primary ossification centers in the flat bones of skull in 3-month-old fetus.

Viscerocranium

- The major part of viscerocranium develops from the mesenchyme derived from **first (having maxillary and mandibular processes) and 2nd pharyngeal arches.**
- **This mesenchyme undergoes membranous ossification to form the bones of the facial skeleton.**
- The mandibular process of the first pharyngeal arch contains Meckel's cartilage.
- **The mesenchyme around the Meckel's cartilage condenses and ossifies by membranous ossification to form the mandible.**
- The Meckel's cartilage disappears; however its perichondrial sheath remains in the form of **sphenomandibular ligament.**

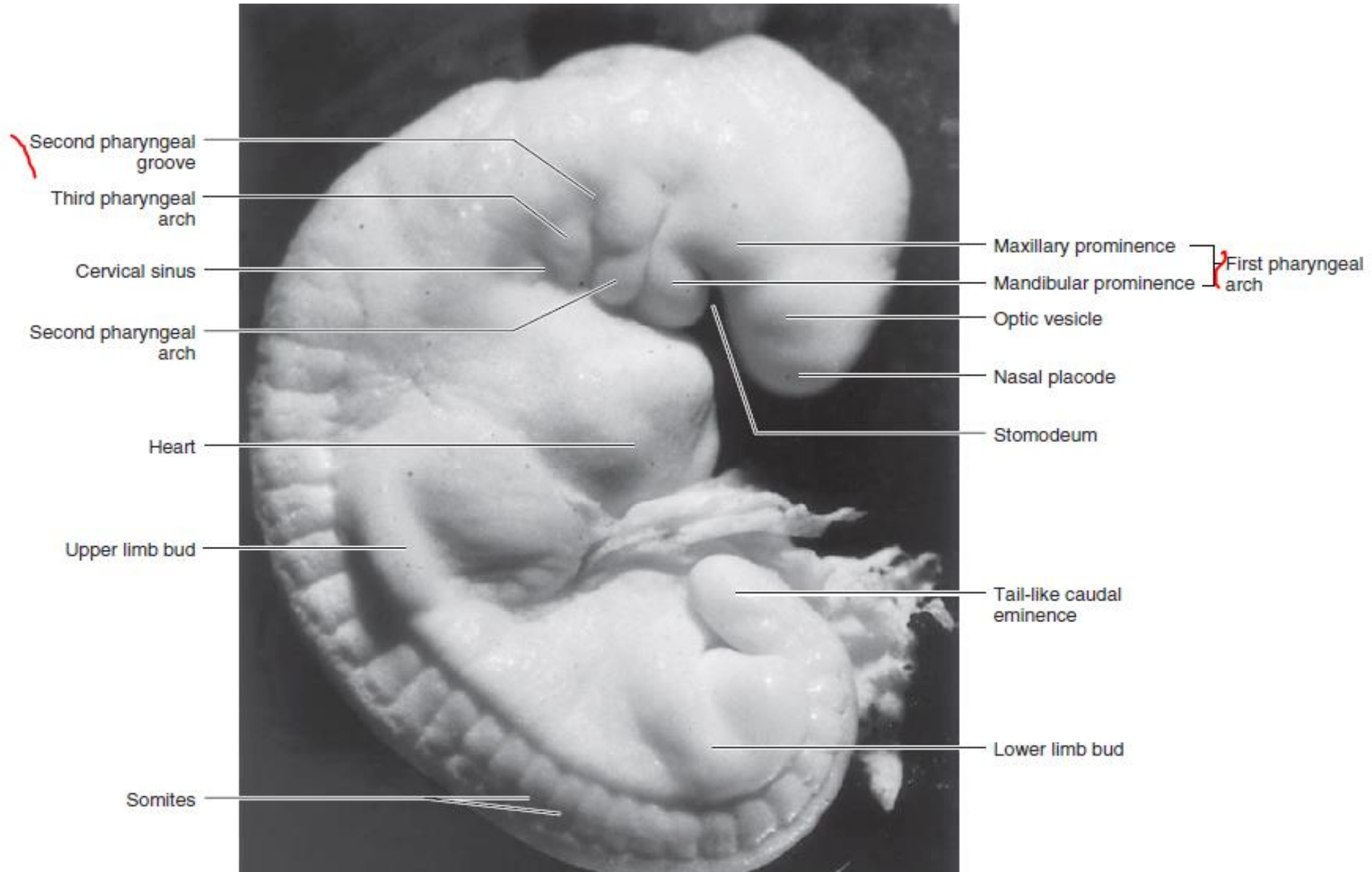
Viscerocranium,,,



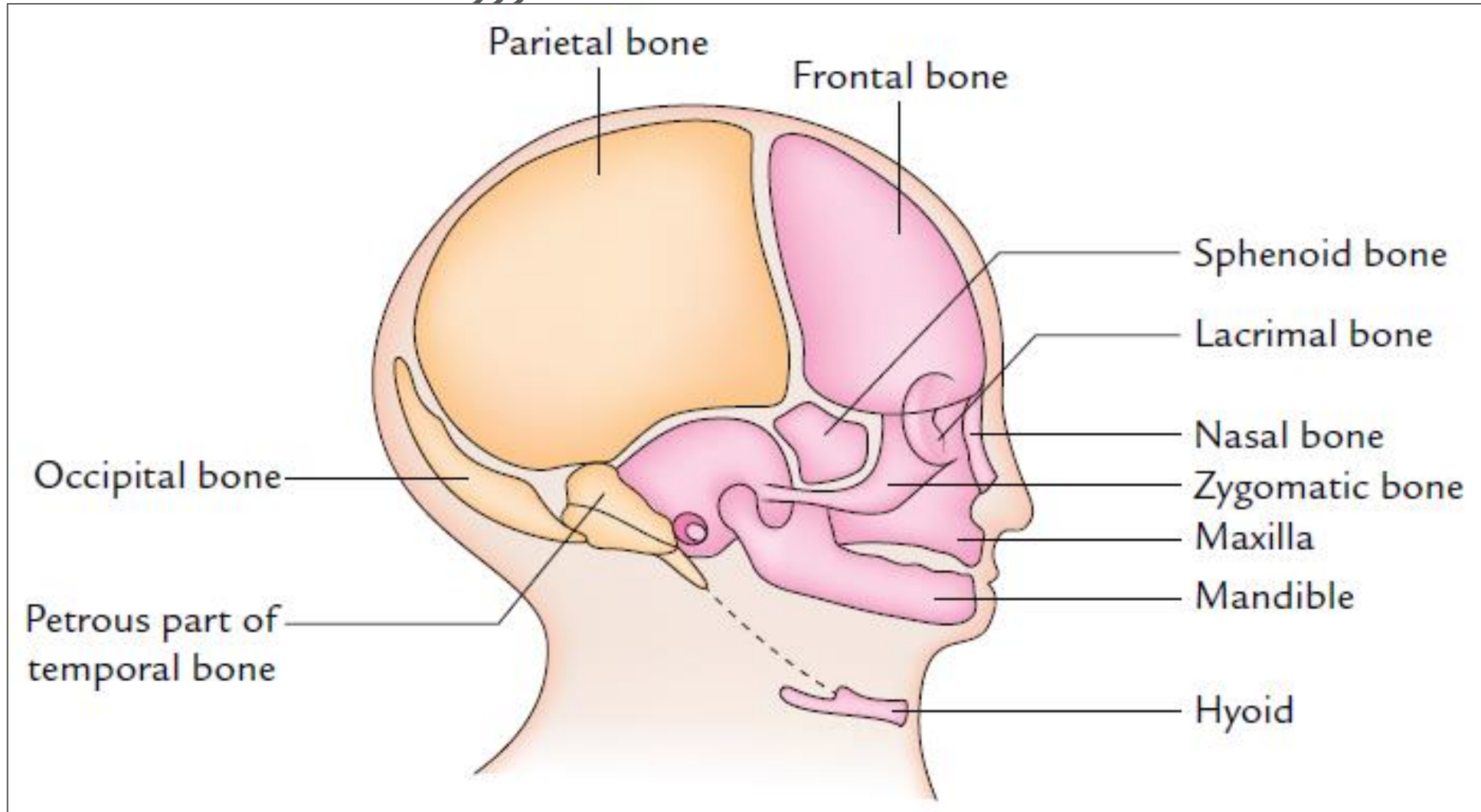
Viscerocranium,,,

- The **first pharyngeal arch** separates into two prominences, **the maxillary and mandibular prominences:**
 - ❑ The maxillary process of first pharyngeal arch gives rise to **maxilla, zygomatic bone, and part of the temporal bone.**
 - ❑ The mandibular prominence forms the mandible and the squamous temporal bone.
 - ❑ **The second pharyngeal arch** (hyoid arch) contributes, along with parts of the third and fourth arches, to the **formation of the hyoid bone.**

Viscerocranium,,,



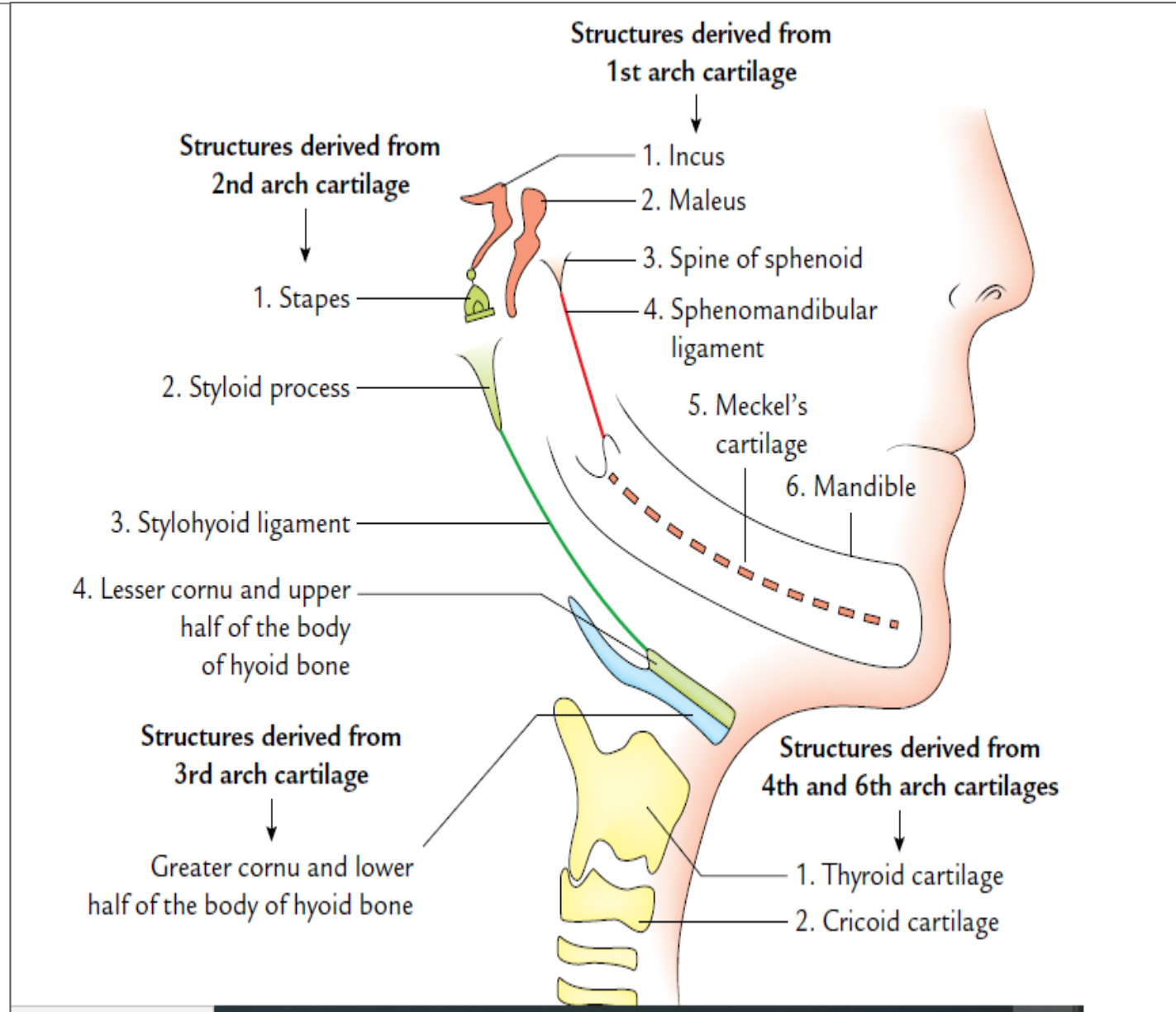
Viscerocranium,,,

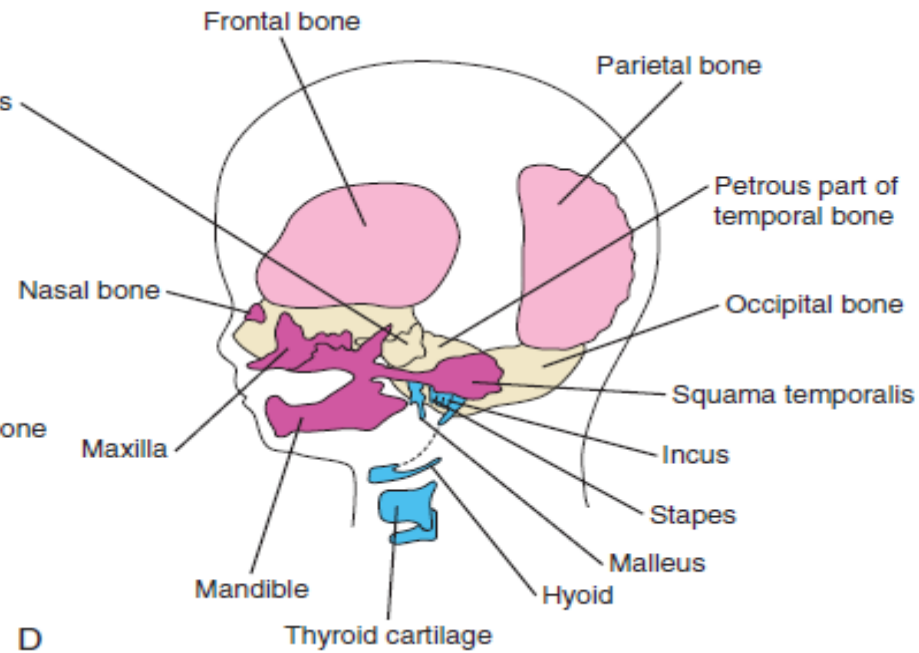
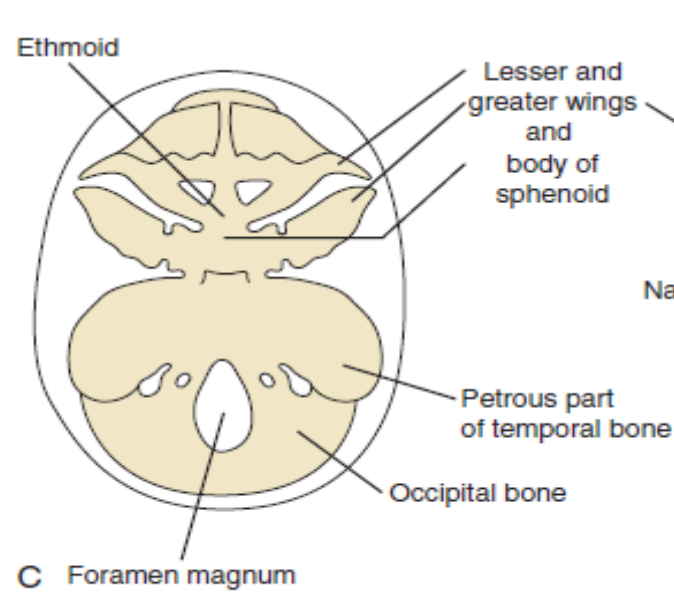
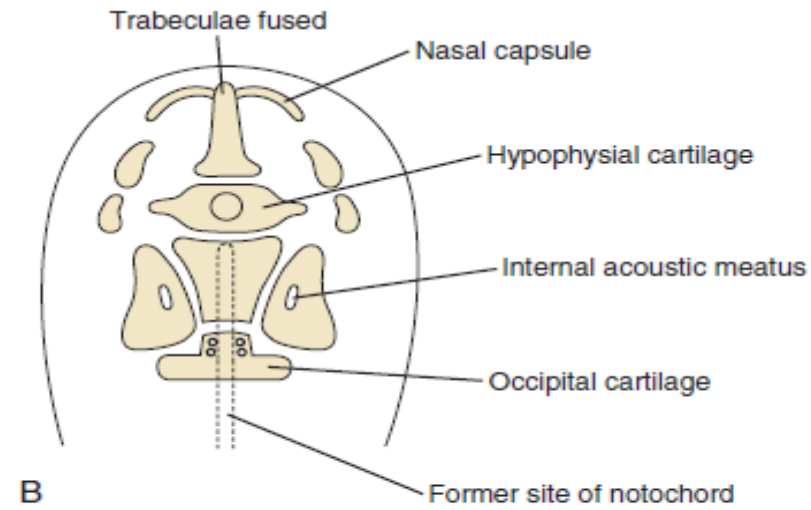
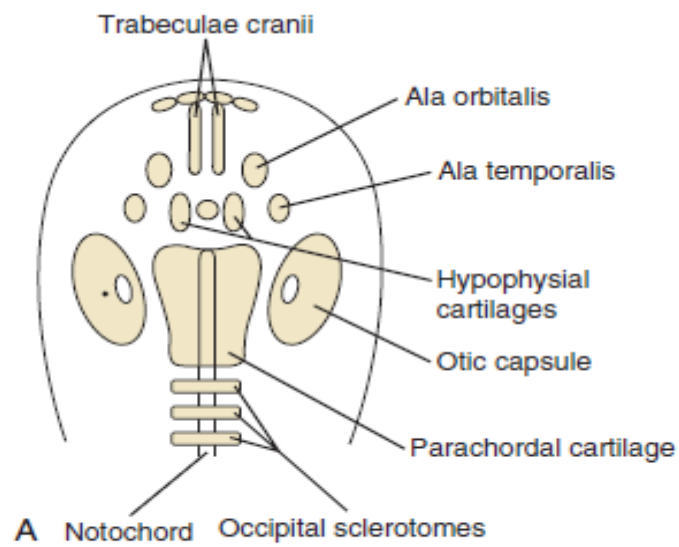


- ❑ Skeletal structures of the head and face.
- ❑ The mesenchyme of these bones is derived **from neural crest (pink)** and **paraxial mesoderm (somites and somatomeres) (orange)**.

Viscerocranium,,,

- The dorsal tip of the mandibular **process along with the second pharyngeal arch** gives rise to three ear ossicles., **malleus, incus, and stapes.**
- **The lacrimal and nasal bones are derived from neural crest cells.**
- **N.B.** Initially, the neurocranium is **larger than viscerocranium (face)** due to the absence of paranasal air sinuses and smaller size of the bones of the face, especially jawbones.





Cartilaginous neurocranium

Membranous neurocranium

Cartilaginous viscerocranium

Membranous viscerocranium

Newborn Skull

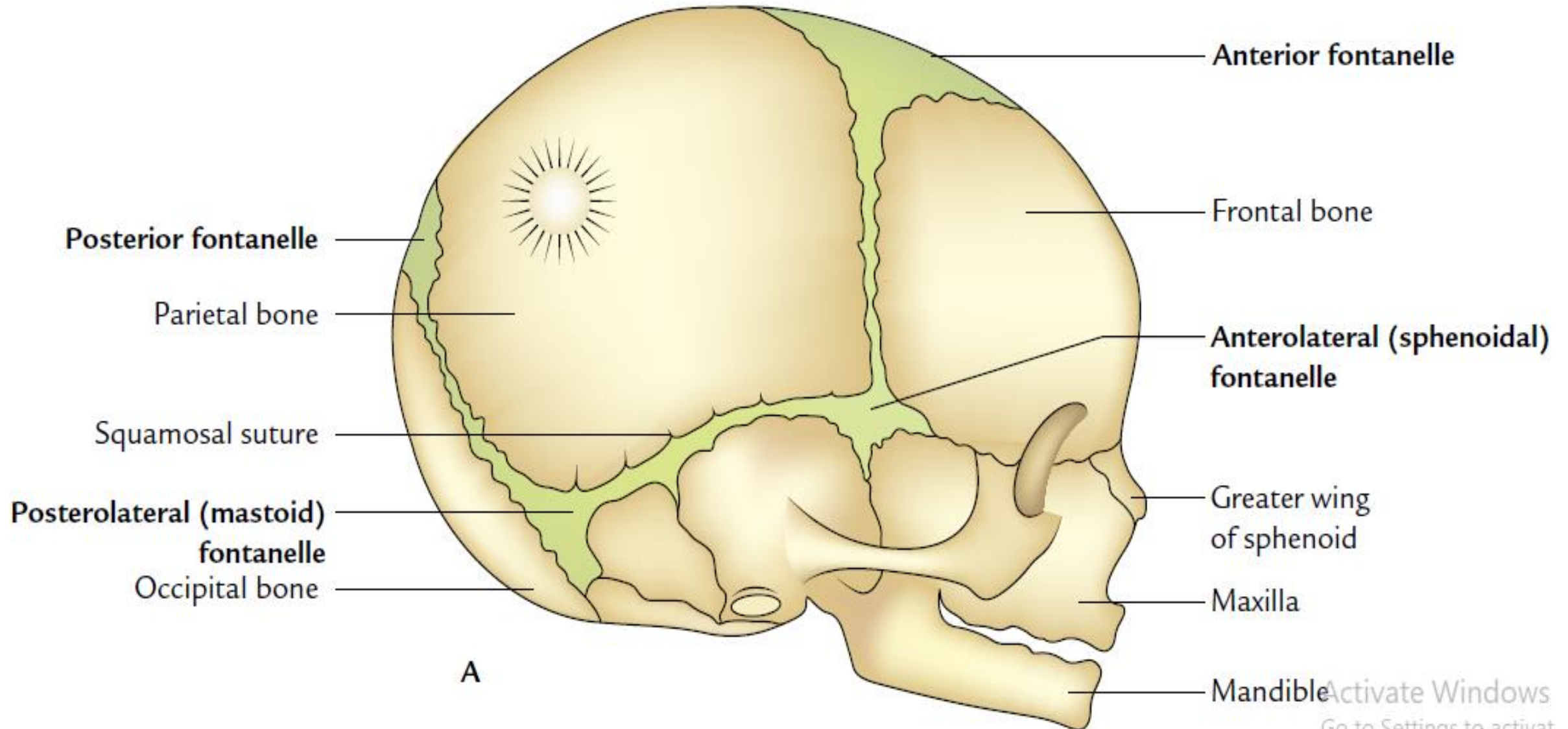
❖ The newborn skull presents two striking features:

- Small viscerocranium (facial skeleton) as compared to neurocranium
- Presence of fontanelles.

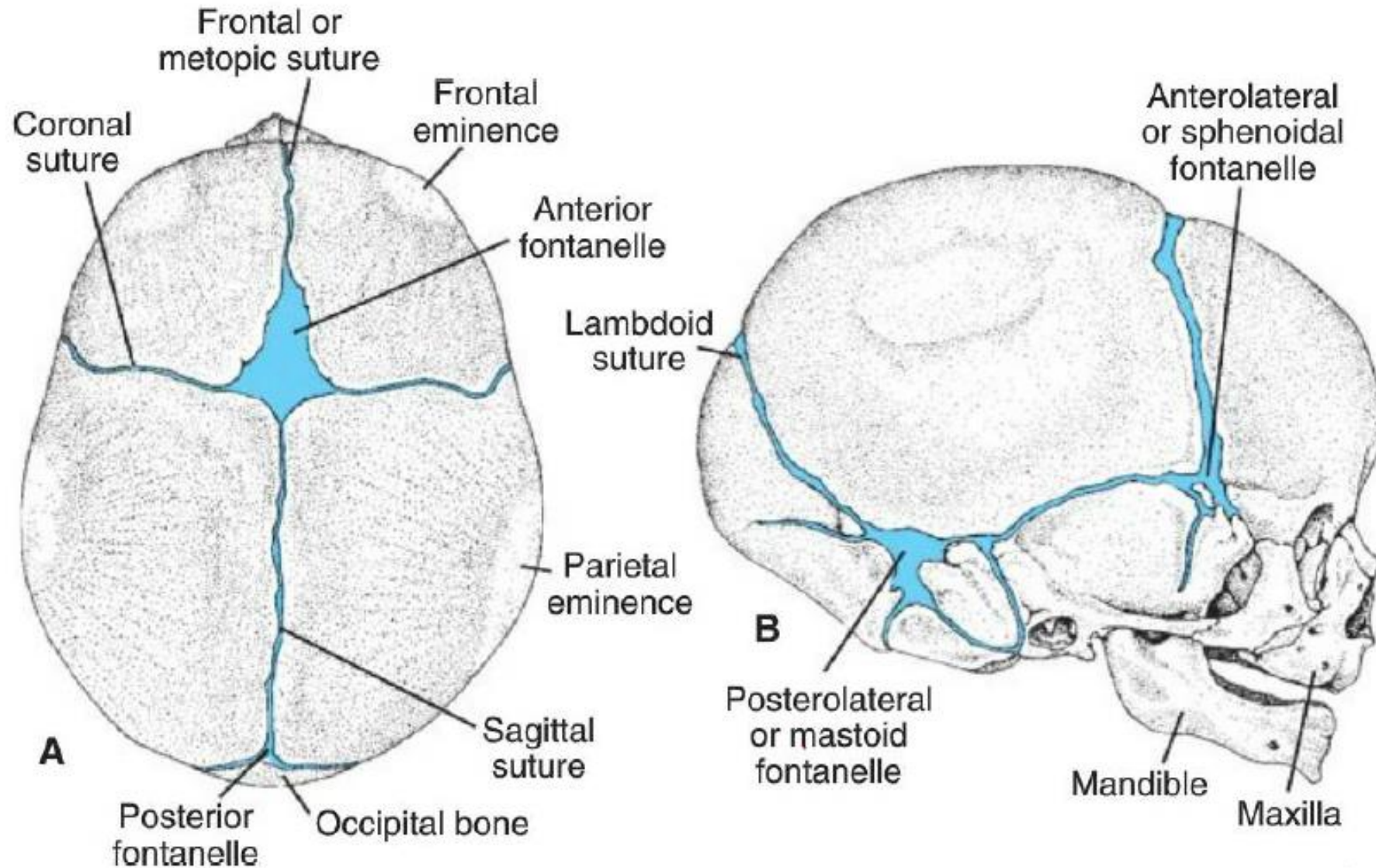
❖ Factors Responsible for Disproportion Between Viscerocranium and Neurocranium is;

- Relatively huge size of **neurocranium** is due to fast and enormous development of brain.
 - The brain reaches 25% of its adult size at birth and 75% by the age of 4 years.
 - The face (**viscerocranium**) is small because of small size of facial bones, particularly the jaws (maxillae and mandibles) and virtual absence of paranasal air sinuses.
- ❑ ***N.B. With the eruption of teeth and development of paranasal air sinuses, the face loses its babyish look.***

Newborn Skull,,,



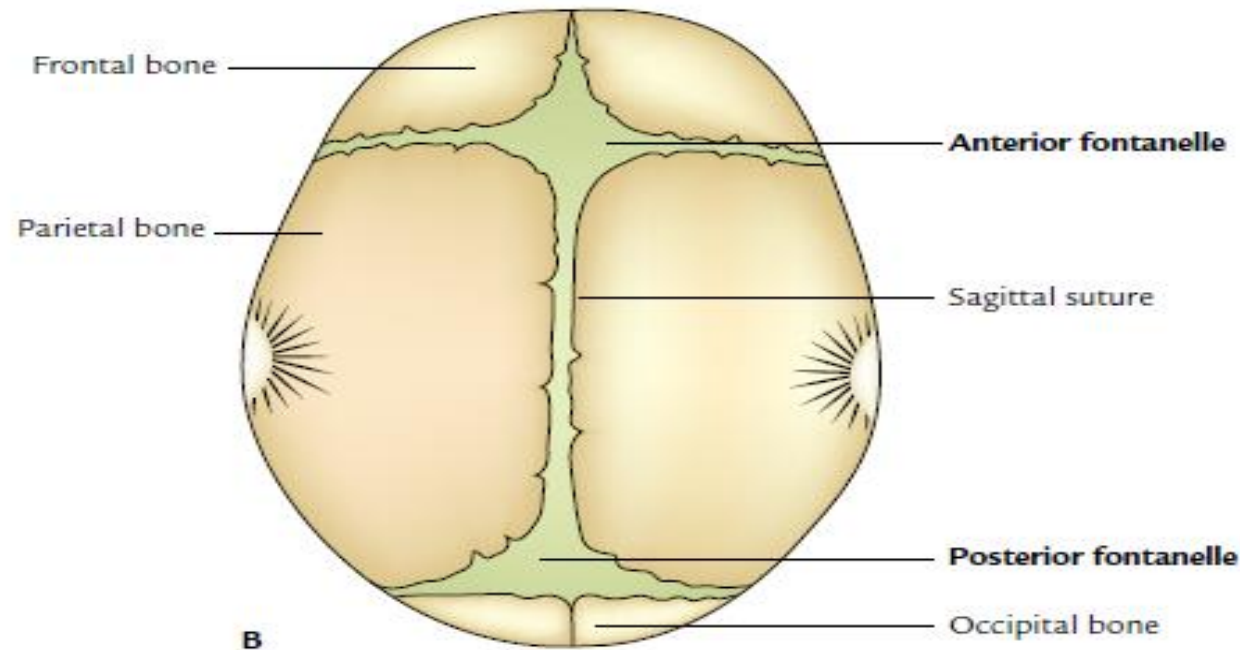
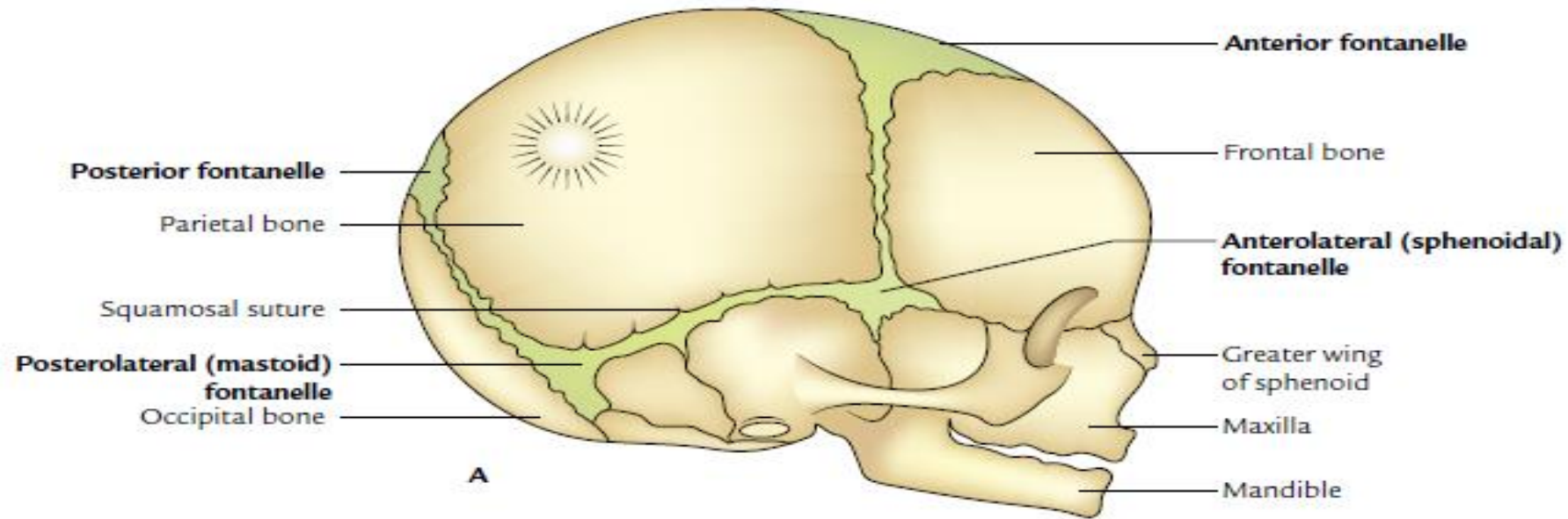
Newborn Skull,,,



Fontanelles

- The fontanelles are soft membranous areas in the vault of newborn skull.
- They are also sometimes termed **soft spots**.
- At birth, the flat bones of the skull are separated from each other by narrow seams of fibrous tissue called **sutures**.
- At sites where more than two bones meet the sutures are wide & are called **fontanelles**.
- ❖ **There are six fontanelles at birth—one at each angle of the parietal bone.**
- Thus, 2 are located in the median plane on top of the skull & 2 on either side of the skull.
- **The two median fontanelles are anterior and posterior, and two paired lateral fontanelles are anterolateral (or sphenoid) and posterolateral (or mastoid).**

Fontanelle,,,



Fontanelle,,,

- The most prominent of these is **anterior fontanelle**.
- All the fontanelles except anterior fontanelle are closed within **three or four months after birth**.
- The anterior fontanelle is usually closed **between second and third year of age**.
- Soon after birth, membranous bones move back to their original positions, and the skull appears large and round.

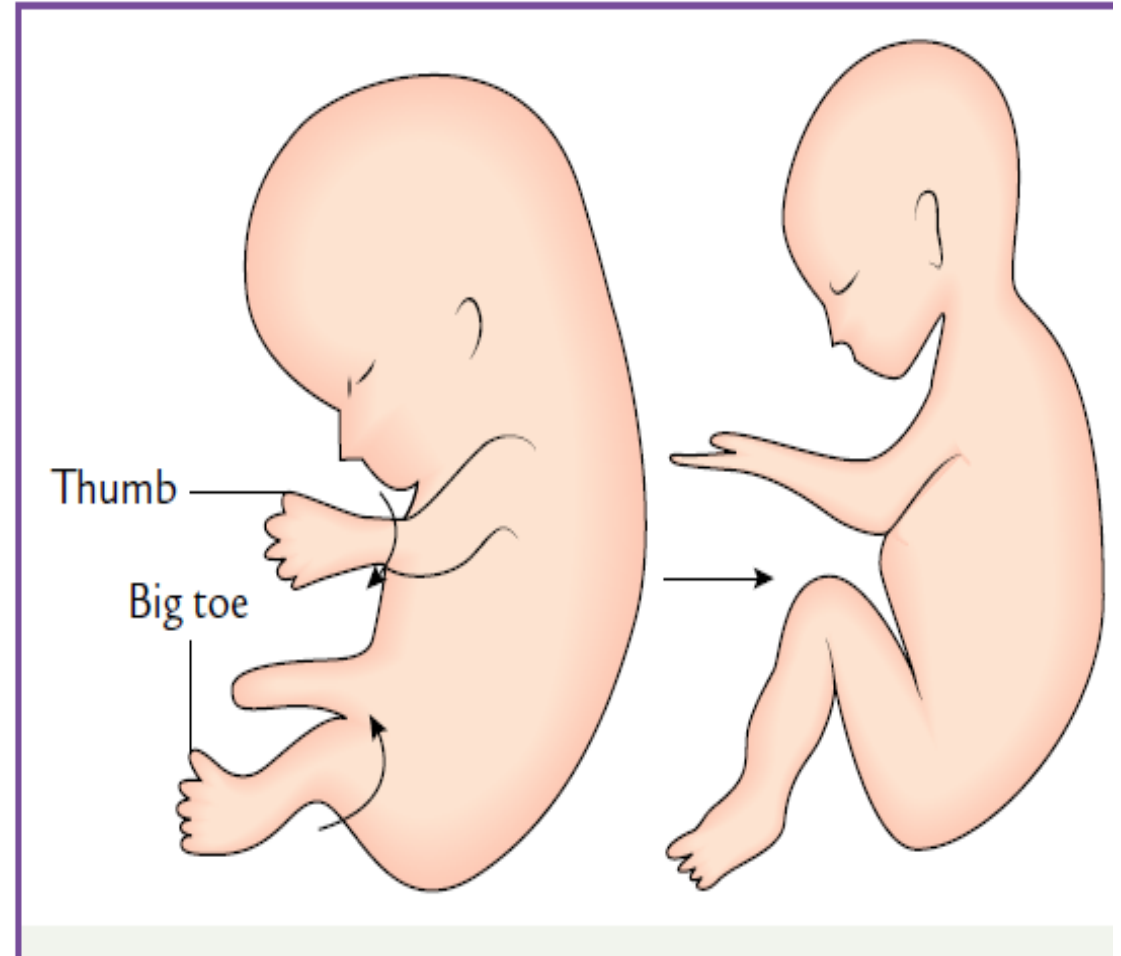
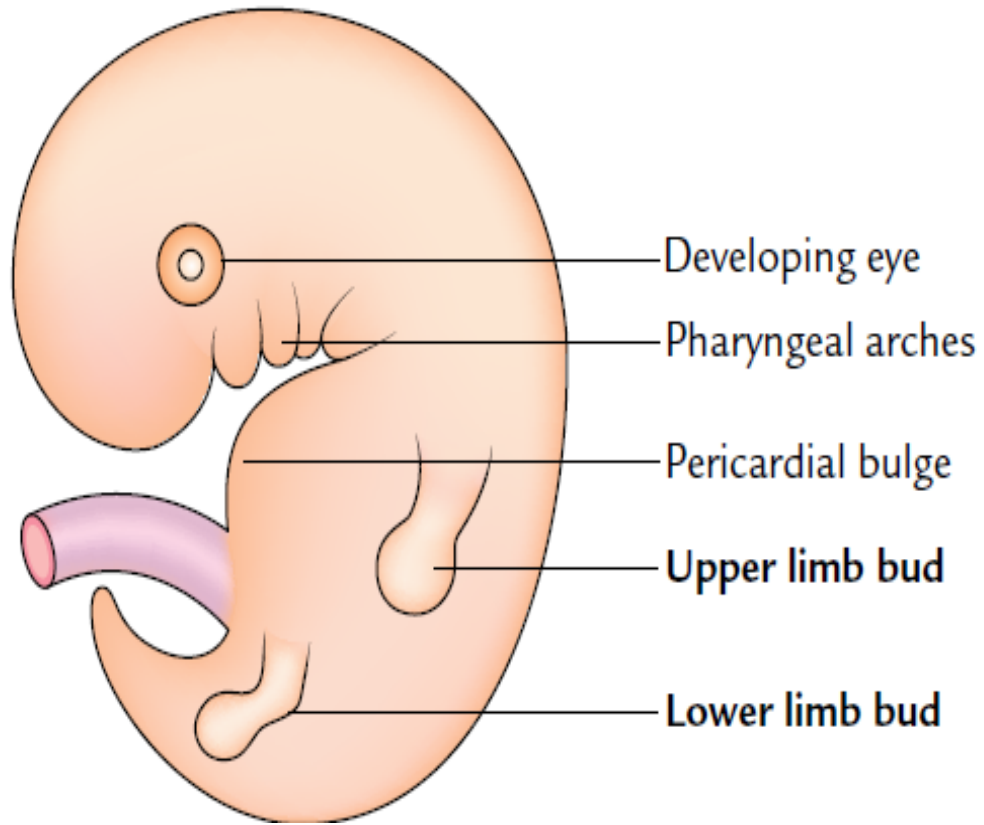
❖ *Functions of fontanelles include;*

- I. Allow bones of the skull to overlap (**moulding**) during parturition to facilitate the baby birth.
- II. Permit postnatal growth of skull bones, especially bones of the vault to increase the cranial capacity.
- III. Allows postnatal development of the brain.

Development of Appendicular Skeleton

- The appendicular skeleton consists of the pectoral and pelvic girdles & the limb bones.
- ❑ **The lateral plate mesoderm migrates** into limb **buds** and condenses along the central axis to form skeletal components of the limbs.
- **Mesenchymal bones** form **during the fifth week** as condensations of mesenchyme appear in the limb buds.
- During the **sixth week**, the **mesenchymal bone models** in the limbs undergo chondrification to form **hyaline cartilage bone models**.
- ❖ **The clavicle initially** develops by membranous ossification and it later forms growth cartilages at both ends.
- The models of the **pectoral girdle** and upper limb bones appear slightly before those of the **pelvic girdle** and lower limb bones.

Development of Appendicular Skeleton,,,



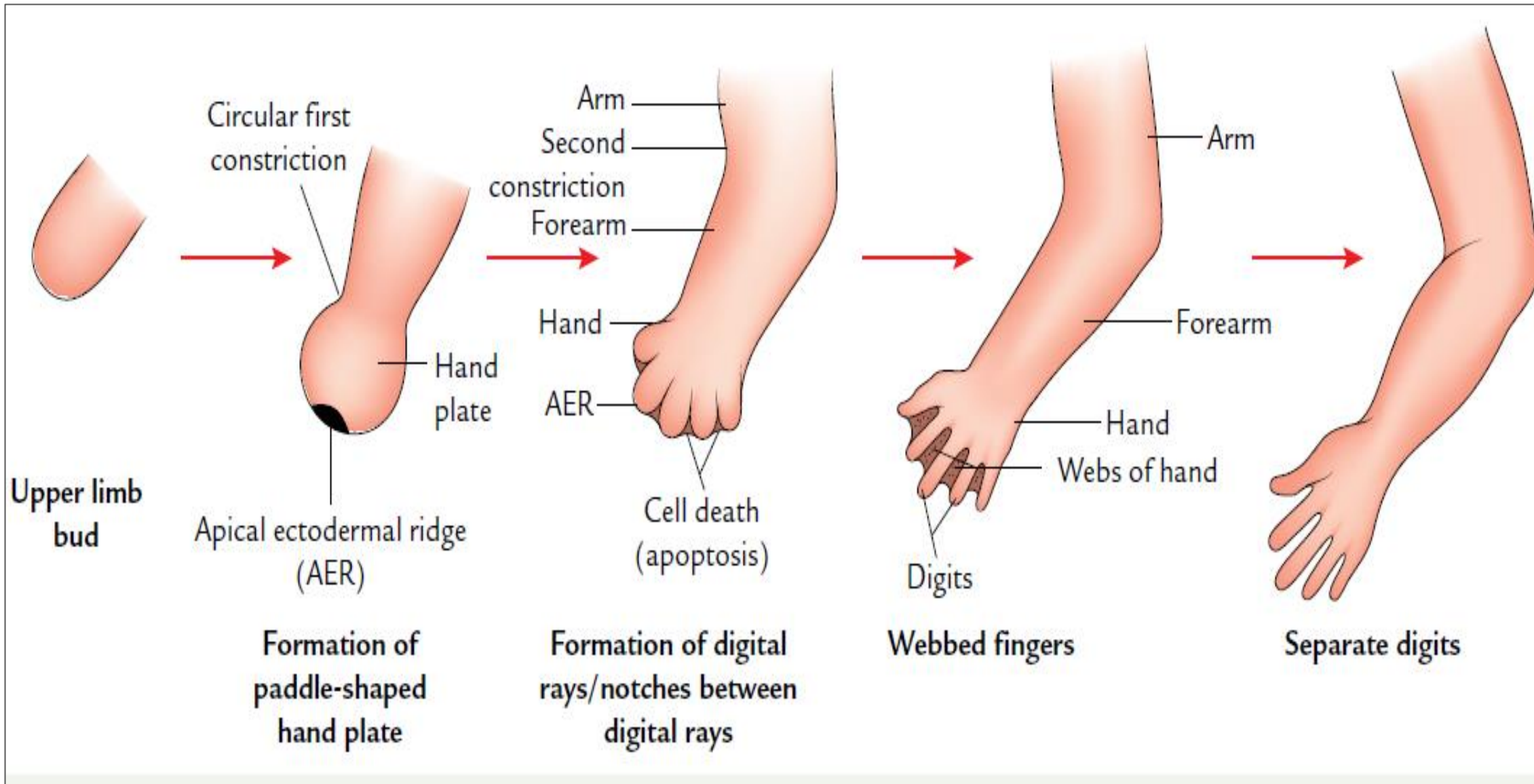
Development of Appendicular Skeleton,,,

- The bone models appear in a proximodistal sequence.
- *Patterning in the developing limbs is regulated by homeobox containing (Hox) genes.*
- **The limb buds appear as outpouchings from** the ventrolateral aspects of the body wall at the end of the **fourth week of** embryonic development.
- The **forelimbs** appear first followed by **hind limbs** 1 or 2 days later.

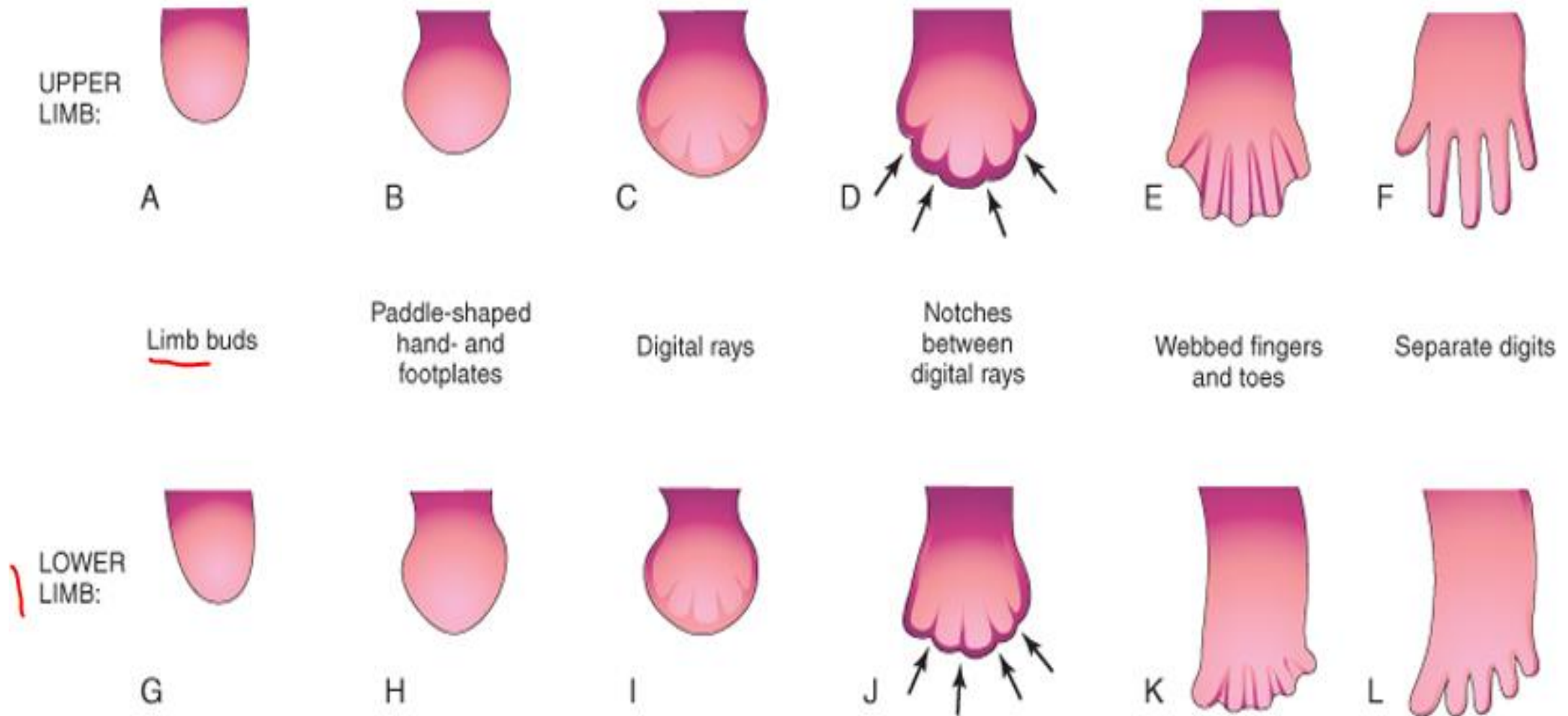
□ **The various stages in the development of upper limb are as follows:**

- The ectoderm at the tip of the limb bud thickens to form **apical ectodermal ridge** (AER).
- **In the sixth week of** embryonic development, terminal part of the limb bud becomes flattened to form **hand and foot plates**, and are separated from the rest of the limb bud by a circular constriction.

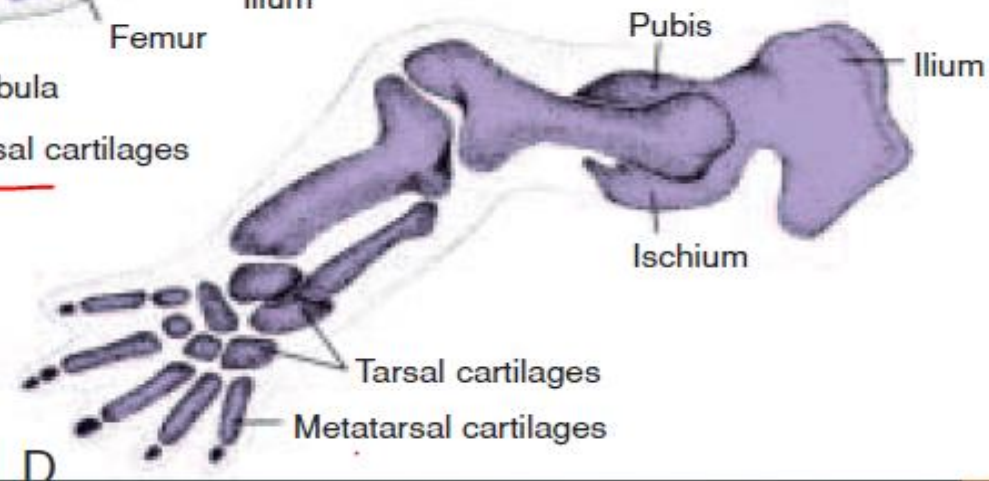
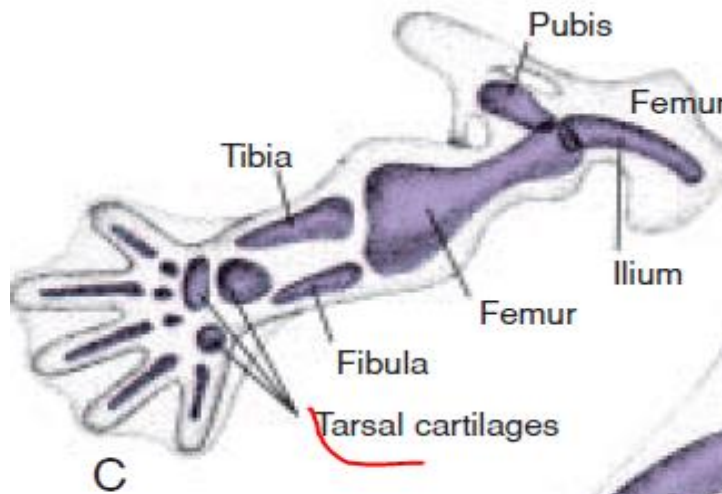
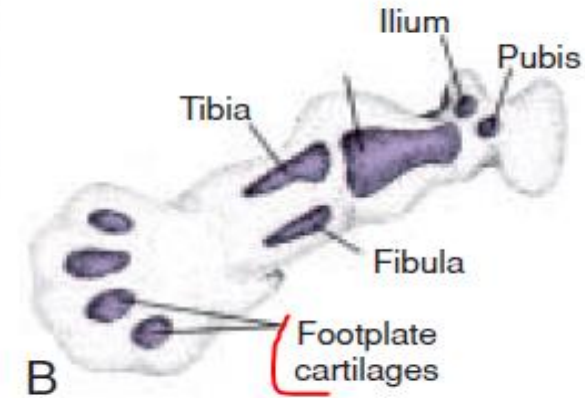
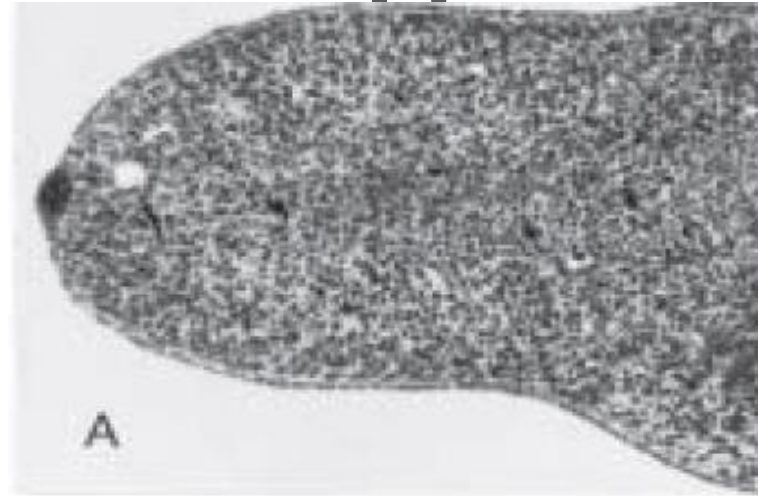
Development of Appendicular Skeleton,,,



Development of Appendicular Skeleton,,,



Development of Appendicular Skeleton,,



Rotation of Limb Buds

- Initially the limb buds are paddle shaped and each bud has **preaxial** and **postaxial border**, with former being directed towards the cranium (head).
- The digit formed along the preaxial border is **thumb** in the upper limb and **great toe** in the lower limb.

❖ The limbs now rotate:

- The **upper limb rotates 90° laterally** and as a result, its preaxial border and thumb come to lie on the lateral side.
- On the other hand, the **lower limb rotates 90° medially** and as a result, its preaxial border and great toe come to lie on the medial side.
- **N.B.** The development of upper & lower limbs is similar except following facts:
 - I. The development of lower limb starts 1 or 2 days later to that of the upper limb.
 - II. During the seventh week of gestation, upper limb rotates 90° laterally whereas the lower limb rotates 90° medially.

Rotation of Limb Buds....

Rotation of the Limbs Changes in position of limbs before birth



At 5 weeks. Upper and lower limbs have formed as finlike appendages pointing laterally and caudally



At 6 weeks. Limbs bend anteriorly, so elbows and knees point laterally, palms and soles face trunk



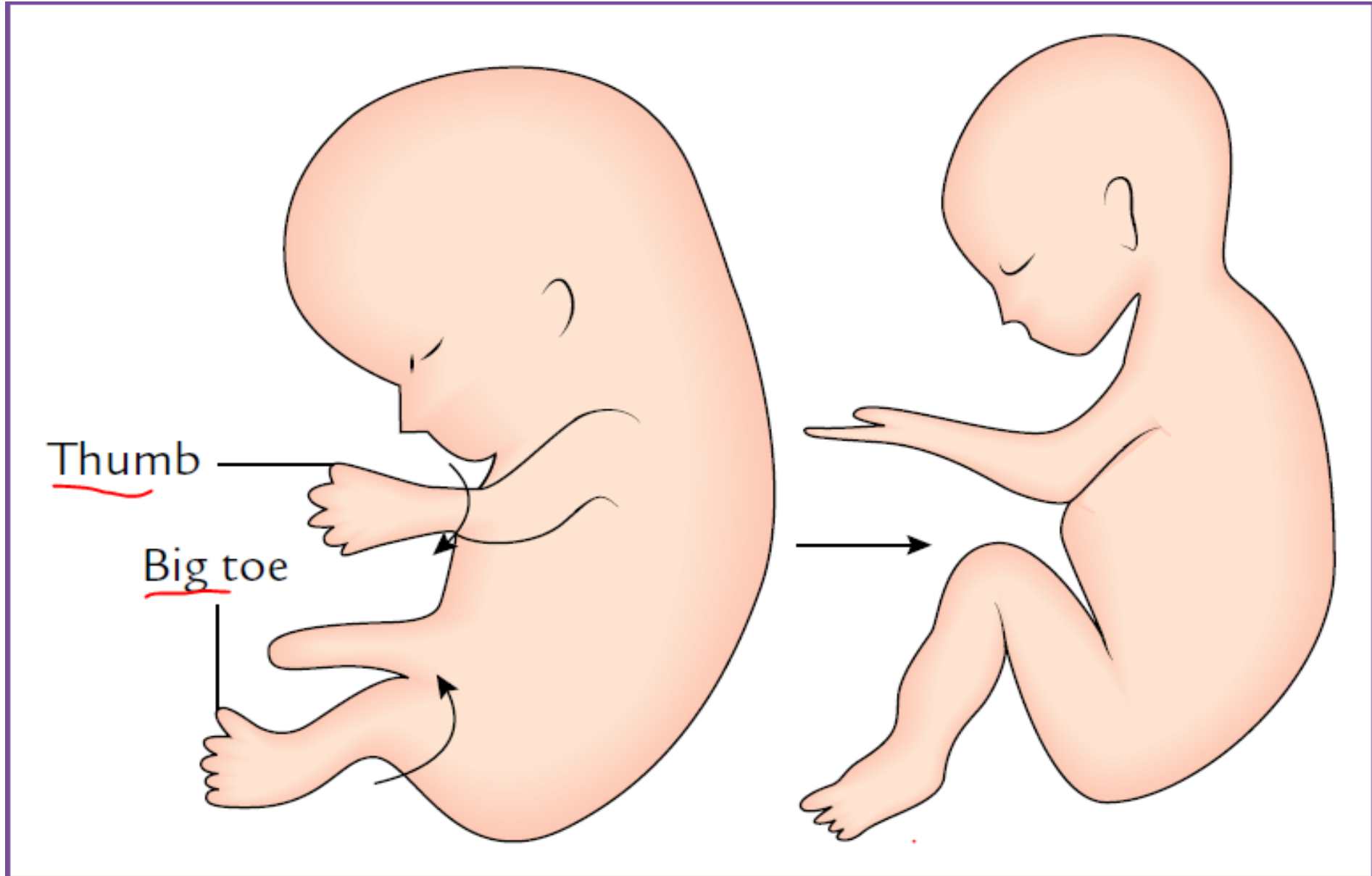
At 7 weeks. Upper and lower limbs have undergone 90-degree torsion about their long axes, but in opposite directions, so elbows point caudally and knees cranially



At 8 weeks. Torsion of lower limbs results in twisted or "barber pole" arrangement of their cutaneous innervation

F. Netter M.D.
1871

Rotation of Limb Buds....

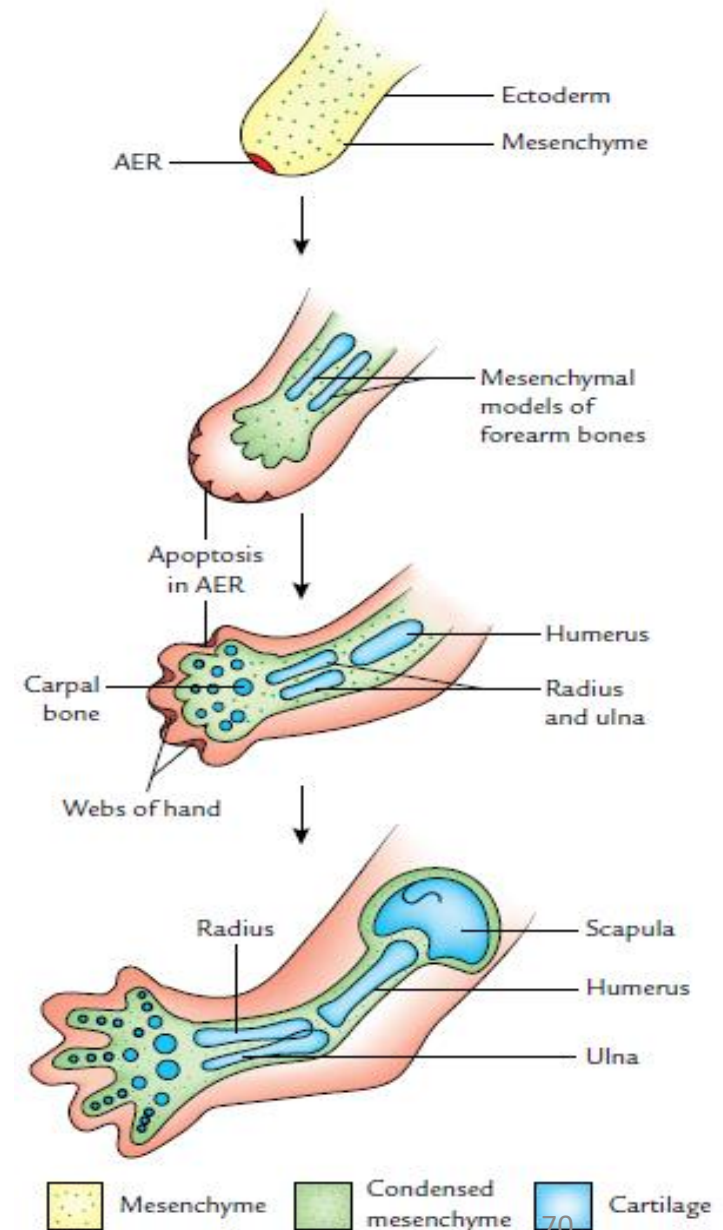


Development of Upper Limb Bones

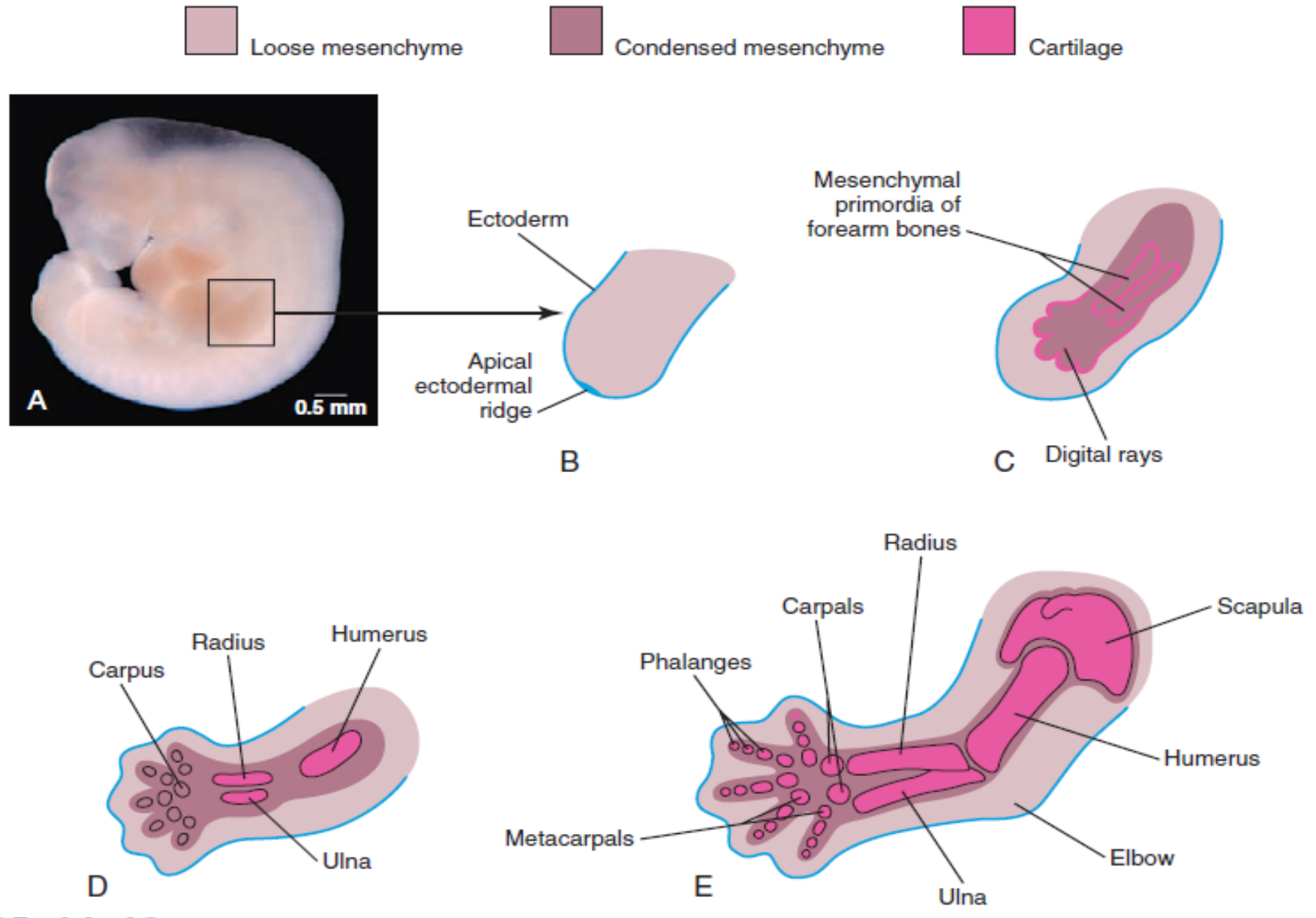
- All the bones of upper limb, develop from somatic layer of **lateral plate mesoderm**.

□ The sequence of events of formation of upper limb bones;

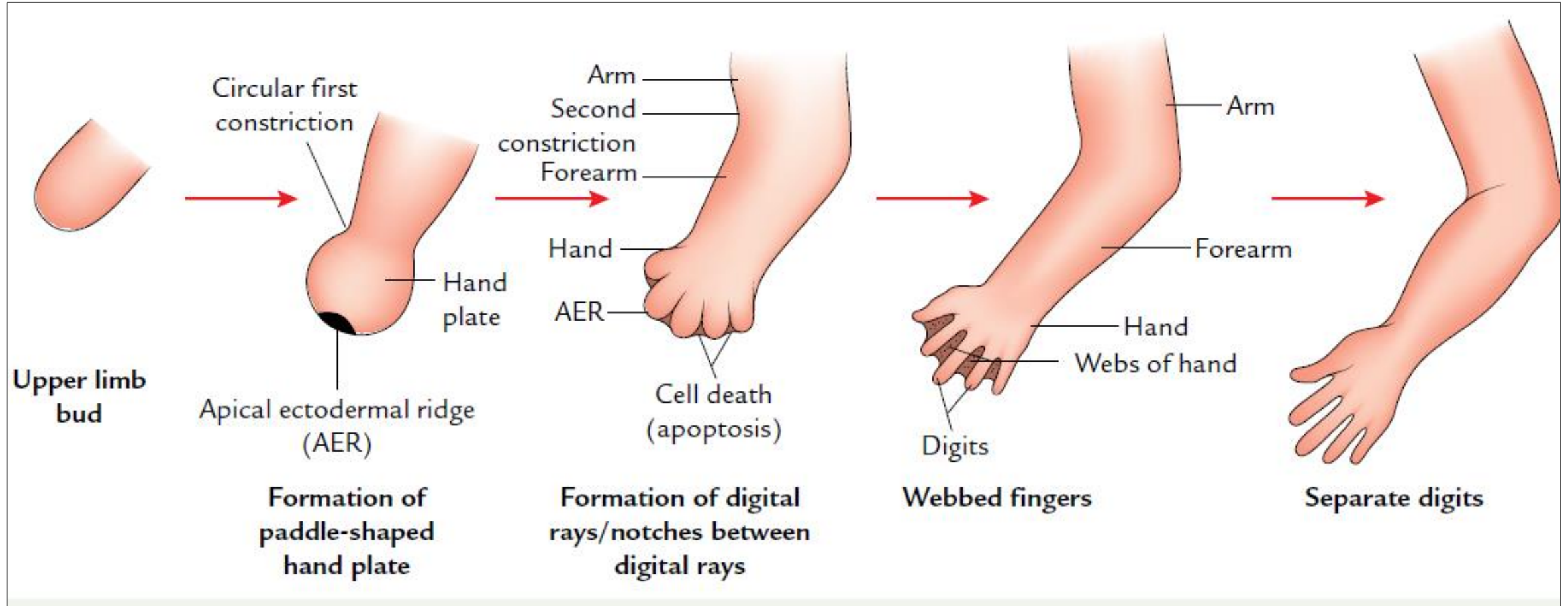
- I. During the fifth week, the mesenchyme derived from lateral plate mesoderm migrates along the central axis of limb bud and condenses to form **mesenchymal models of bones**.
- II. During the 6th week, the **mesenchymal models of bones** undergo chondrification to form **hyaline cartilage models**.
- III. The hyaline cartilage models undergo **ossification** to form **bones**.



Development of Upper Limb Bones,,,



Development of Upper Limb Bones,,



Stages in the development of upper limb

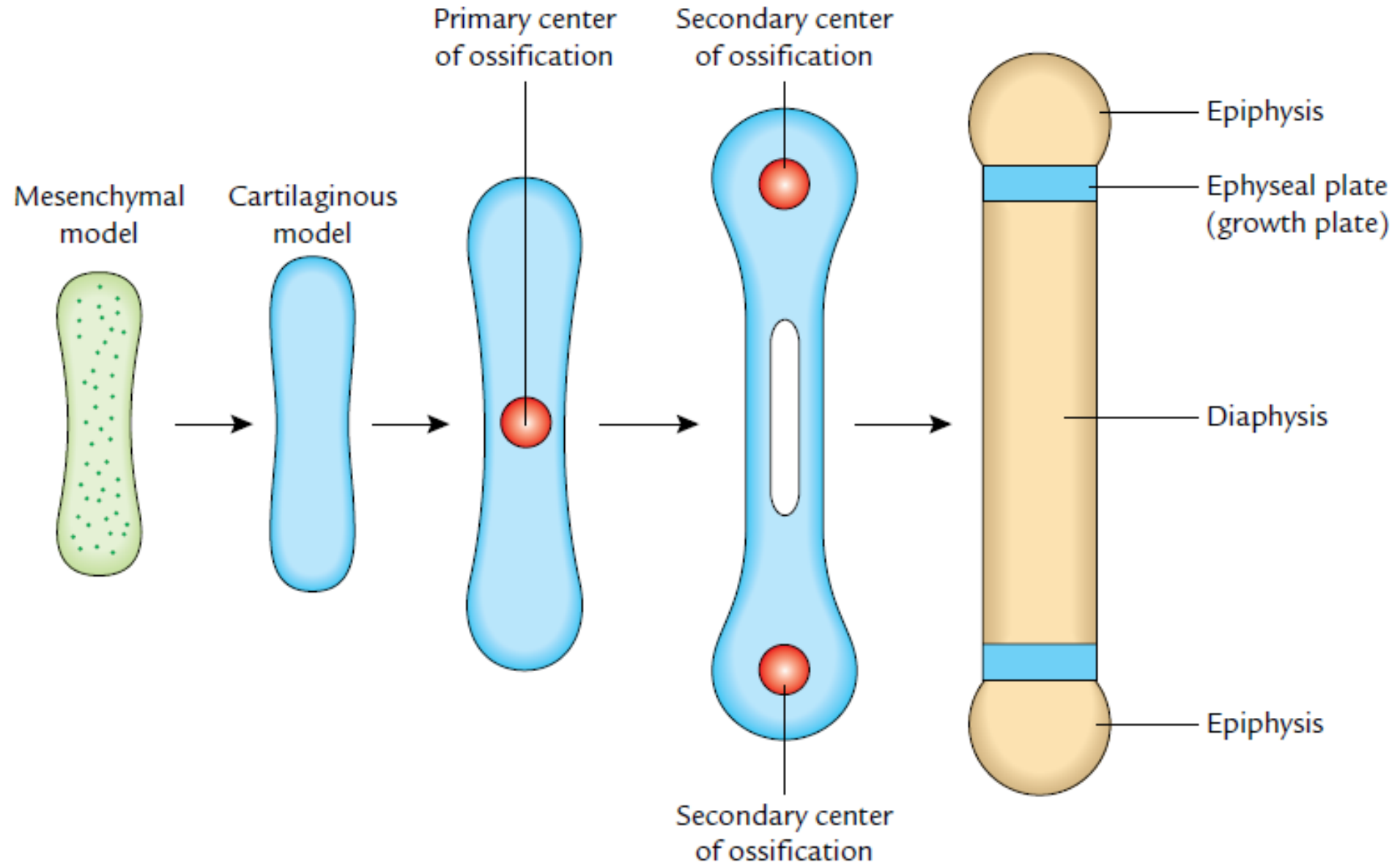
Development of Upper Limb Bones,,,

- ❖ *All the bones of the upper limb are formed by **endochondral ossification**; however clavicle is formed by both membranous (mainly) and endochondral ossification.*
- ❖ *The clavicle is the first bone to ossify in the entire body.*

❑ **The ossification of a typical long bone occurs as under:**

- I. During the eighth week, primary center of ossification appear in the middle of the long bone.
- II. During childhood, secondary centers appear at the ends of long bone.
- III. The primary center forms the diaphysis of long bone whereas the secondary centers form the epiphyses of long bone.
- IV. The bone formed by the primary center in the diaphysis does not fuse with bone formed by the secondary center in the epiphysis.

Development of Upper Limb Bones,,,



Stages of ossification of a typical long bone.

Development of Lower Limb Bones

- Like upper limb all the bones of lower limb, hip bone, femur, tibia, fibula, tarsals, metatarsals, and phalanges develop from **somatopleuric layer of lateral plate mesoderm**.
- The sequence of events involved in development of bones of lower limb is same as in upper limb.

❖ Ossification;

- The bones of the lower limb ossify in the same fashion as those of the upper limb.
- **N.B.** The primary centers of ossification of limb bones appear at different times in different bones, **but most of them appear between 7th and 12th** week, i.e., virtually all primary centers of ossification are present at birth and most of the secondary centers appear after birth.

Development of the muscular system

- The muscular system develops from **intraembryonic mesoderm**, except for the muscles of the **iris**, which develop from **neuroectoderm**(ectoderm of optic cup), and the **muscles of the esophagus**, which are believed to develop by transdifferentiation from smooth muscle.
- **Muscles are classified into three types:** skeletal (striated or **voluntary**), smooth (**involuntary**), and **cardiac** (structurally striated but functionally smooth muscle).
 - I. The **skeletal muscles** mostly develop from **somites (paraxial mesoderm)**.
 - II. The **smooth muscles** mostly develop from **splanchnic mesoderm**.
 - III. The **cardiac muscle** develops from **splanchnic mesoderm surrounding** developing heart tube.

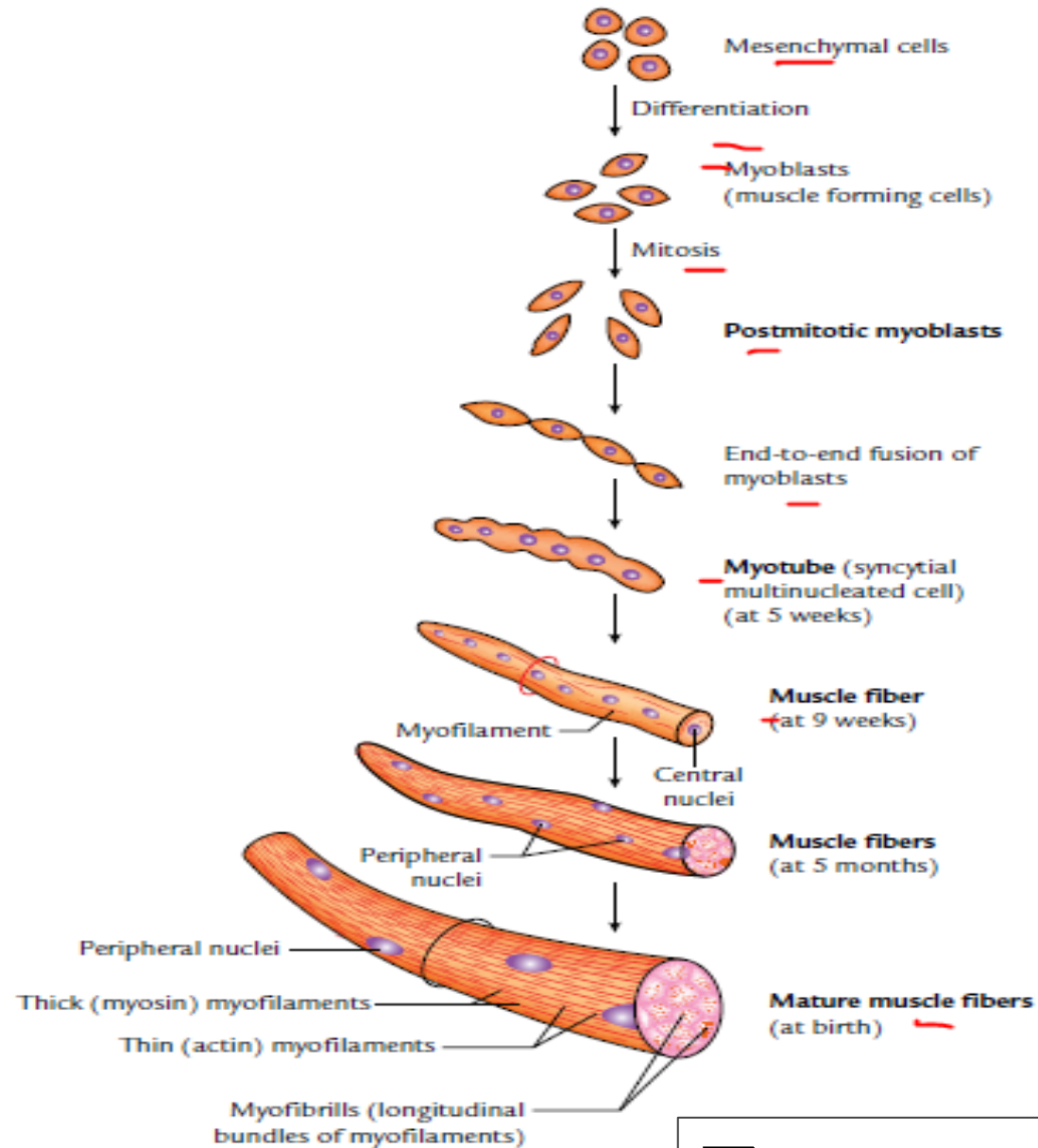
Development of the muscular system,,,

- **Myoblasts** (embryonic muscle cells) are derived from **mesenchyme** (embryonic connective tissue).
- ❖ **Myogenesis (muscle formation) of skeletal muscle:**
 - The skeletal muscles develop from mesenchym derived from myotomal **mesoderm**.
 - ❖ **The first indication of myogenesis (muscle formation) is the elongation of the nuclei and cell bodies of mesenchymal cells** as they differentiate into myoblasts.
- Soon these **primordial muscle cells** fuse to form elongated, multinucleated, cylindrical structures—**myotubes**.

Development of the muscular system,,,

- ✓ The mesenchymal cells differentiate into primordial muscle cells called **myoblasts**.
- ✓ **The myoblasts fuse with each other end-to-end to** form an elongated multinucleated cylindrical syncytial cell called **myotube**.
- ✓ The myotubes synthesize **actin, myosin, and other muscle proteins**.
- ✓ These proteins aggregate to form **myofilaments** and **myofibrils**.
- ✓ Now, the myotubes are called **muscle fibers**.

Development of the muscular system ,,

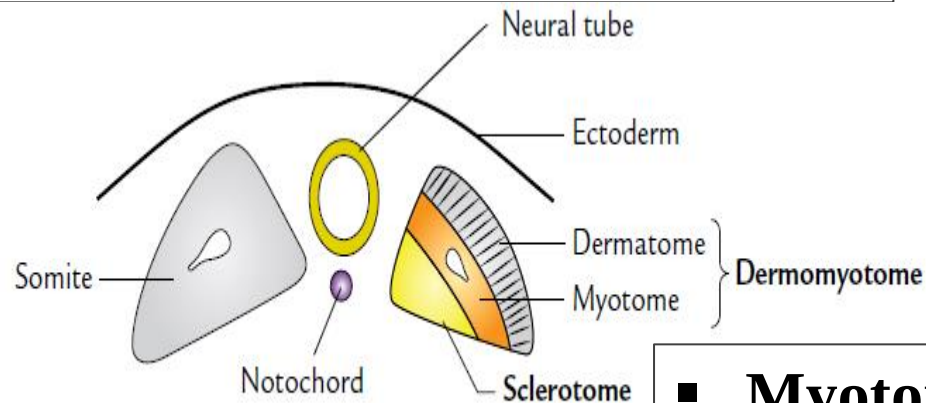


Myogenesis of a skeletal muscle.

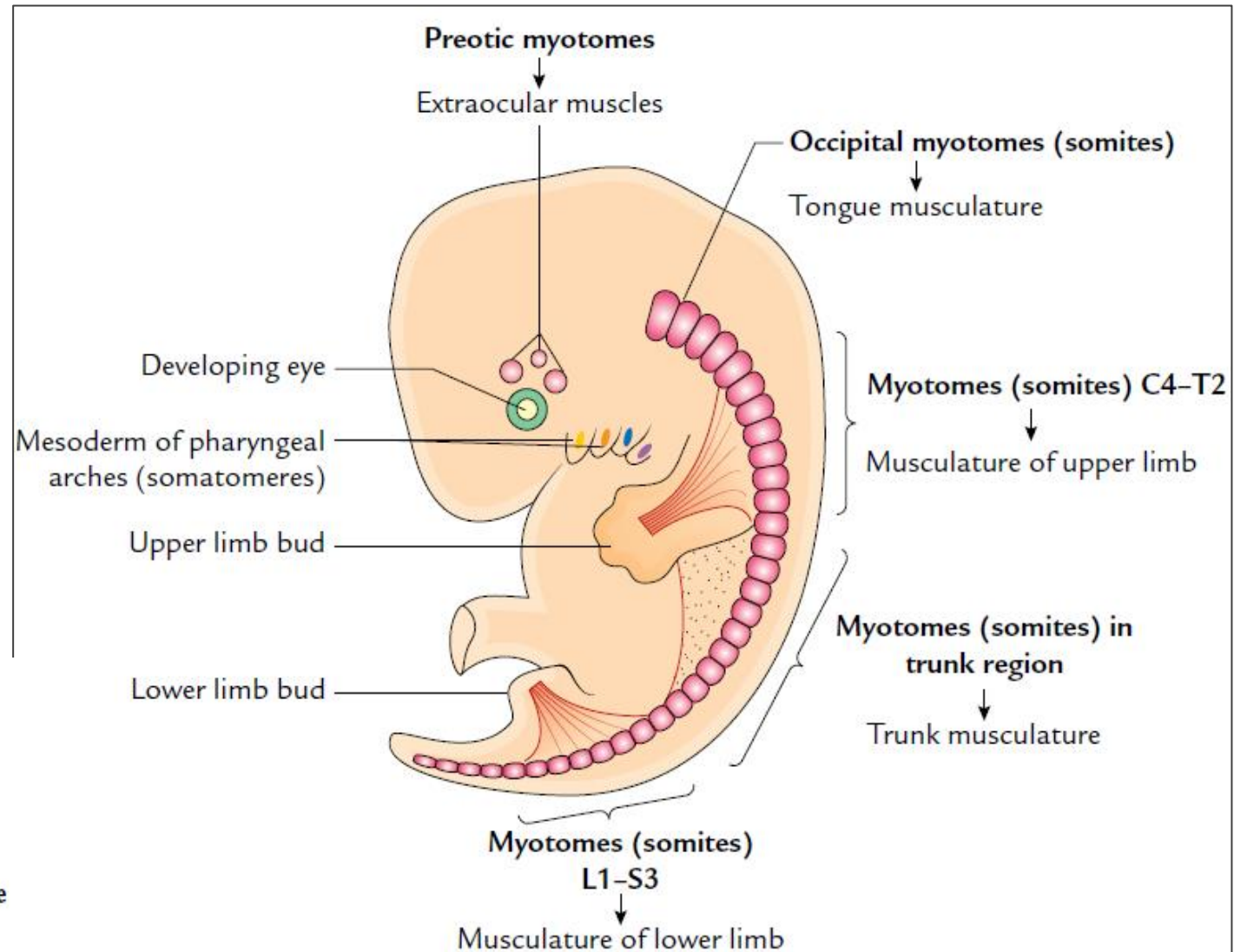
❑ *Myogenesis of a skeletal muscle.*

Development of Skeletal Muscles

- The skeletal muscles of axial skeleton, body wall, limbs, and head develop from **somites** and **somatomeres** that extend from occipital to sacral region.



Somite (on the left side) and its subdivisions (on the right side).

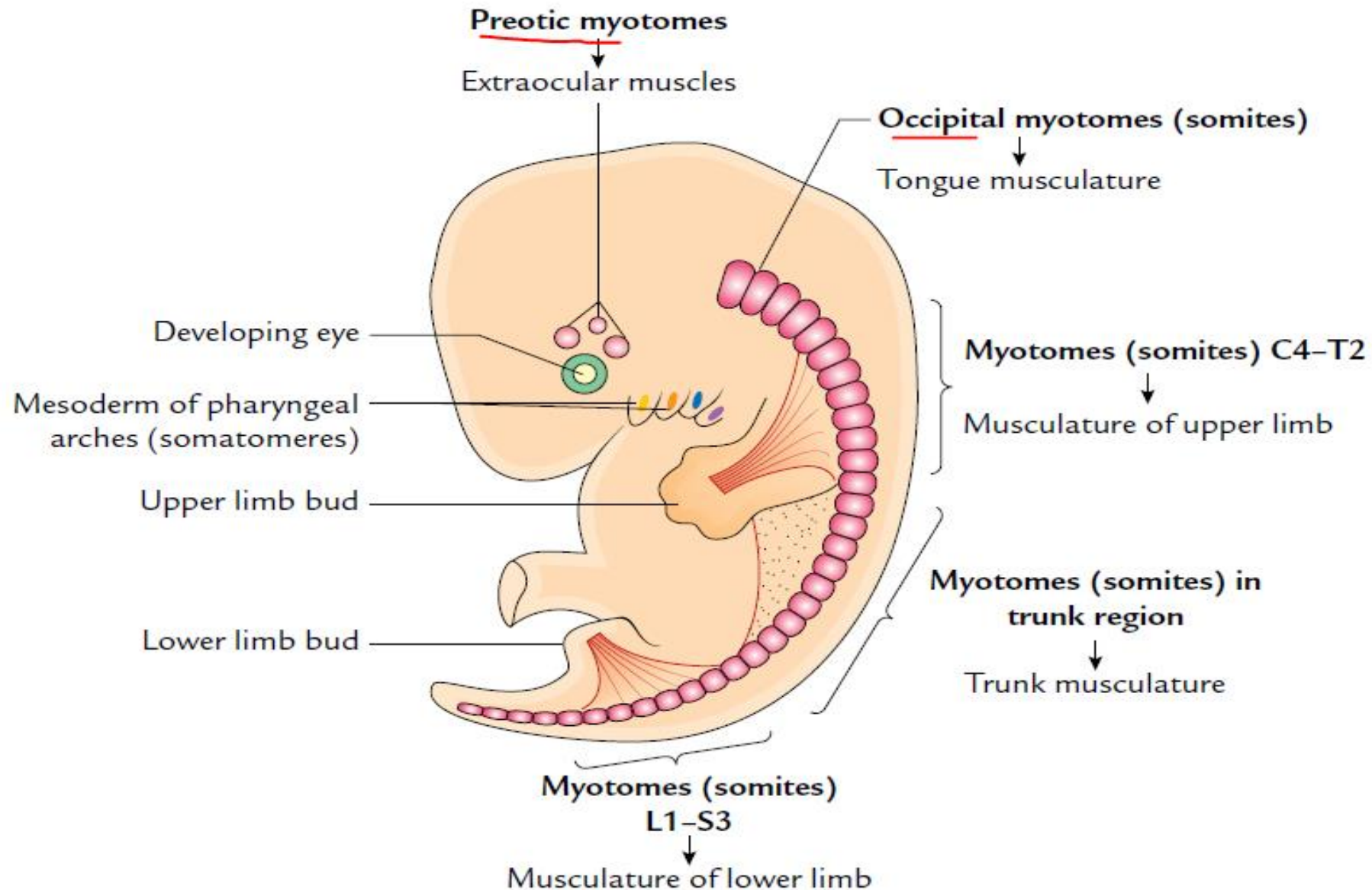


- Myotomes derived from somites and somatomes in different regions of the body which form skeletal muscles.**

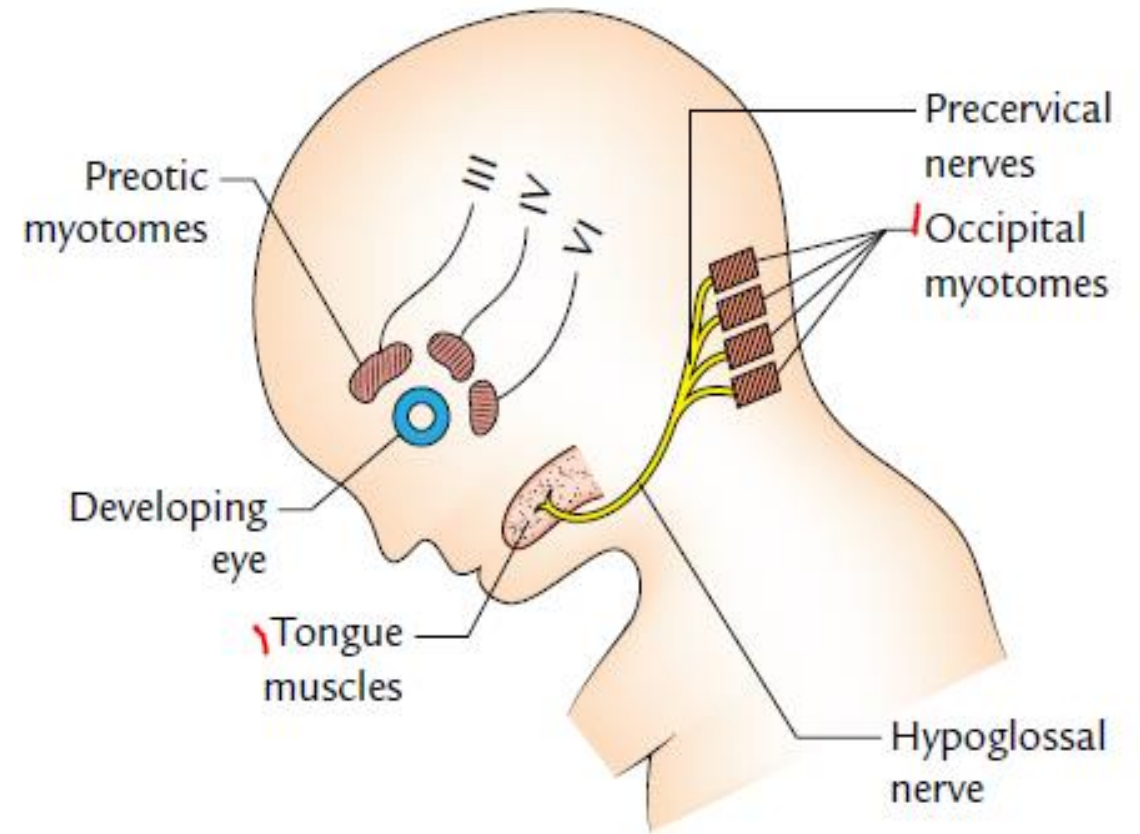
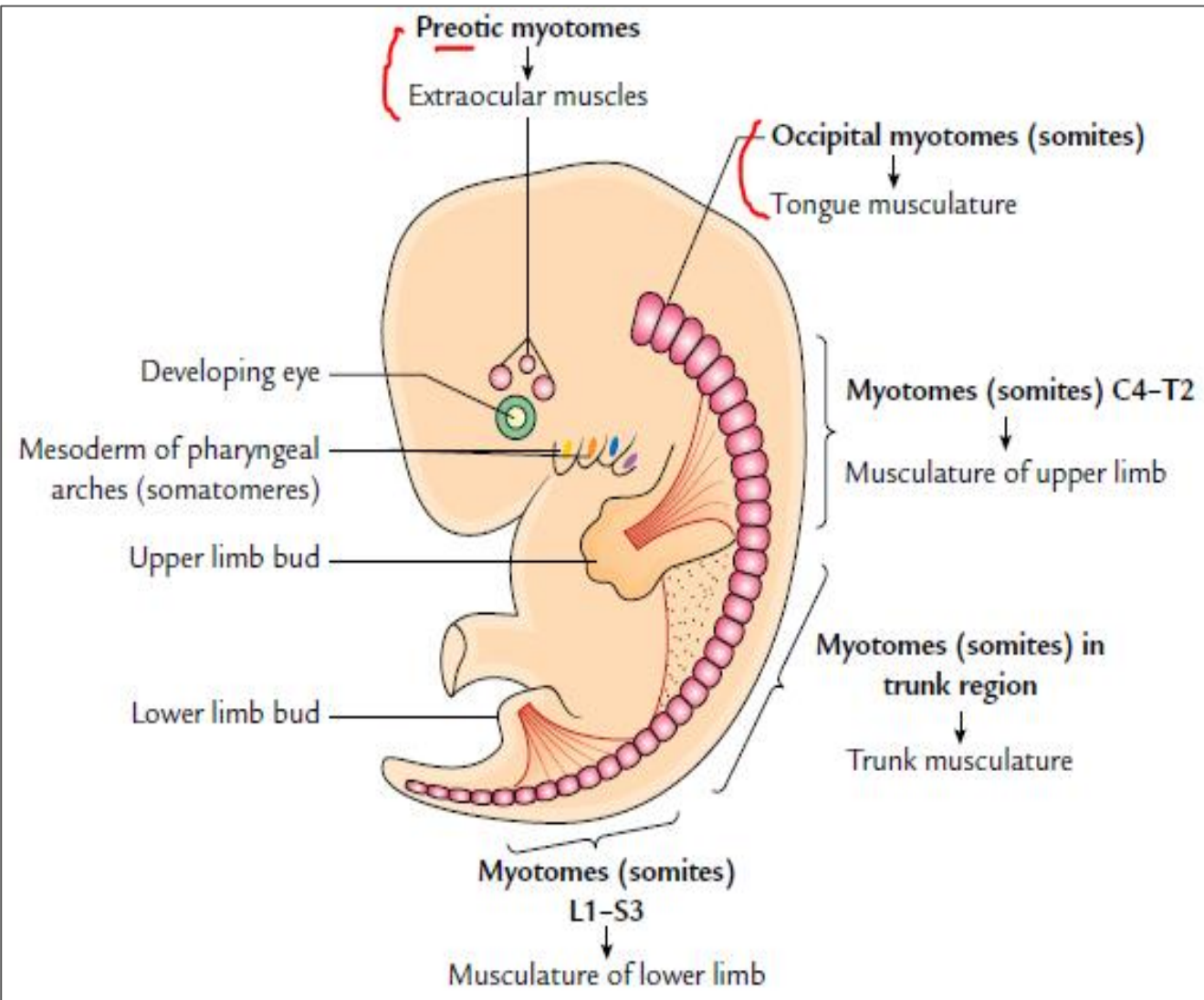
Development of Skeletal Muscles,,,

- **Each somite differentiates** into two distinct zones: **sclerotome** and **dermomyotome**.
 - The sclerotome forms bones of axial skeleton.
 - **The dermomyotome** consists of two components: a deeper part called **myotome** and a superficial part called **dermatome**.
 - **The myotome** forms muscular tissue whereas the dermatome forms dermis of skin.
- ❑ **Muscles of eyeball (extraocular muscles)** develop from **preotic myotomes**,
- ❑ **Muscles of tongue** from **occipital myotomes**, muscles of limbs from myotomes present in upper and lower limb bud regions, and muscles of the body wall from myotomes in the trunk region.

Development of Skeletal Muscles,,,



Development of Skeletal Muscles,,,

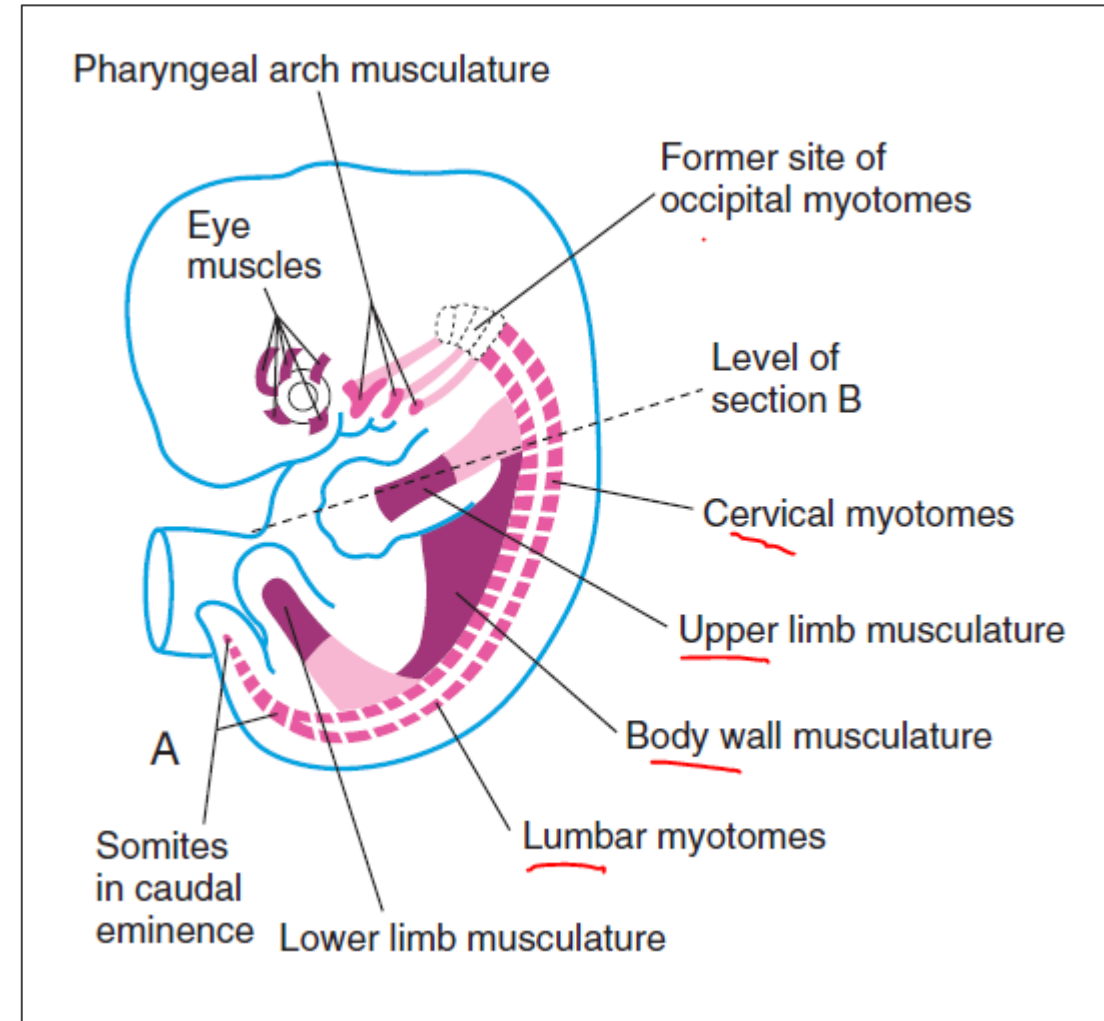
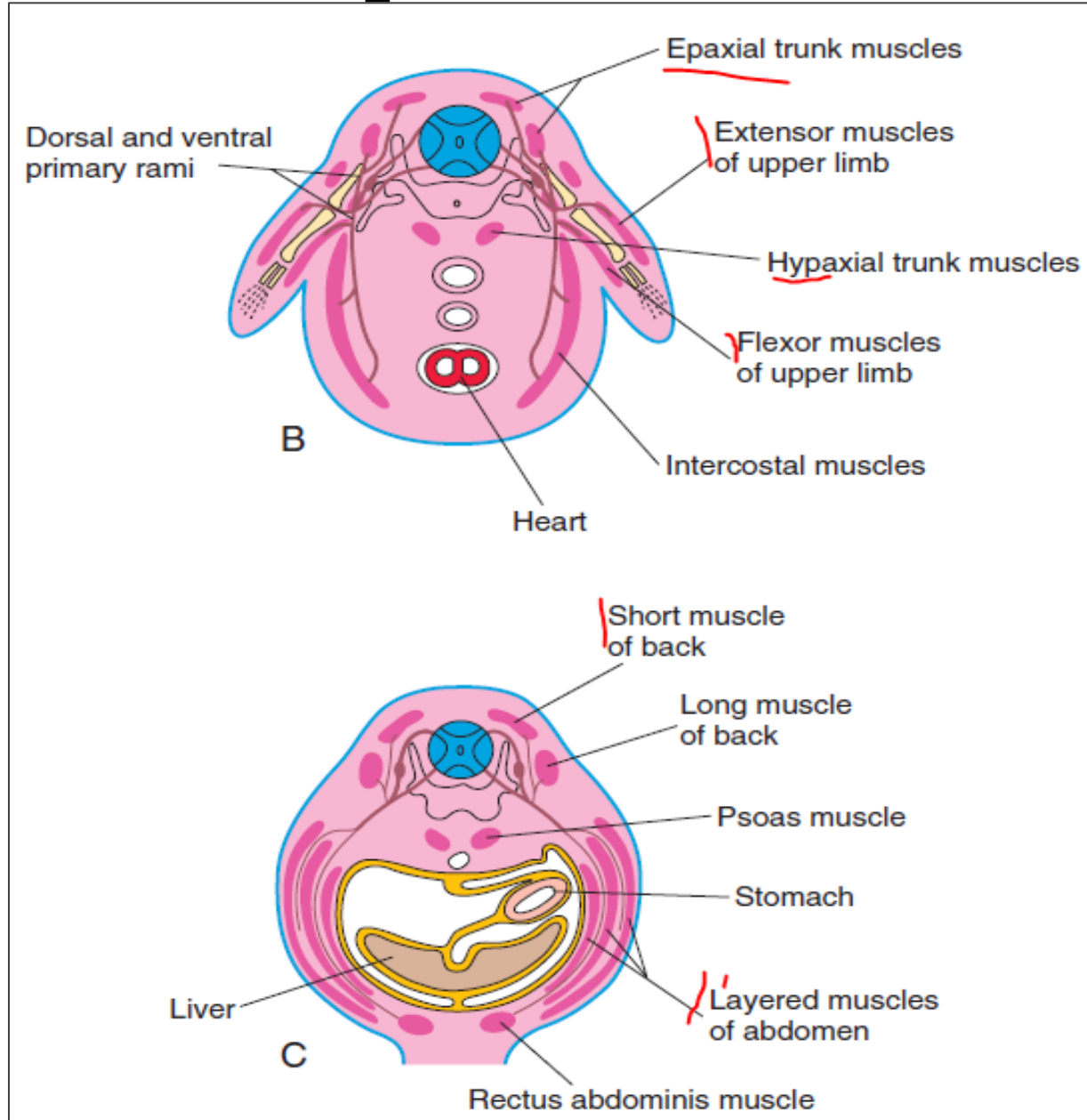


- ***Development of the muscles of tongue and extrinsic muscles of the eye.***

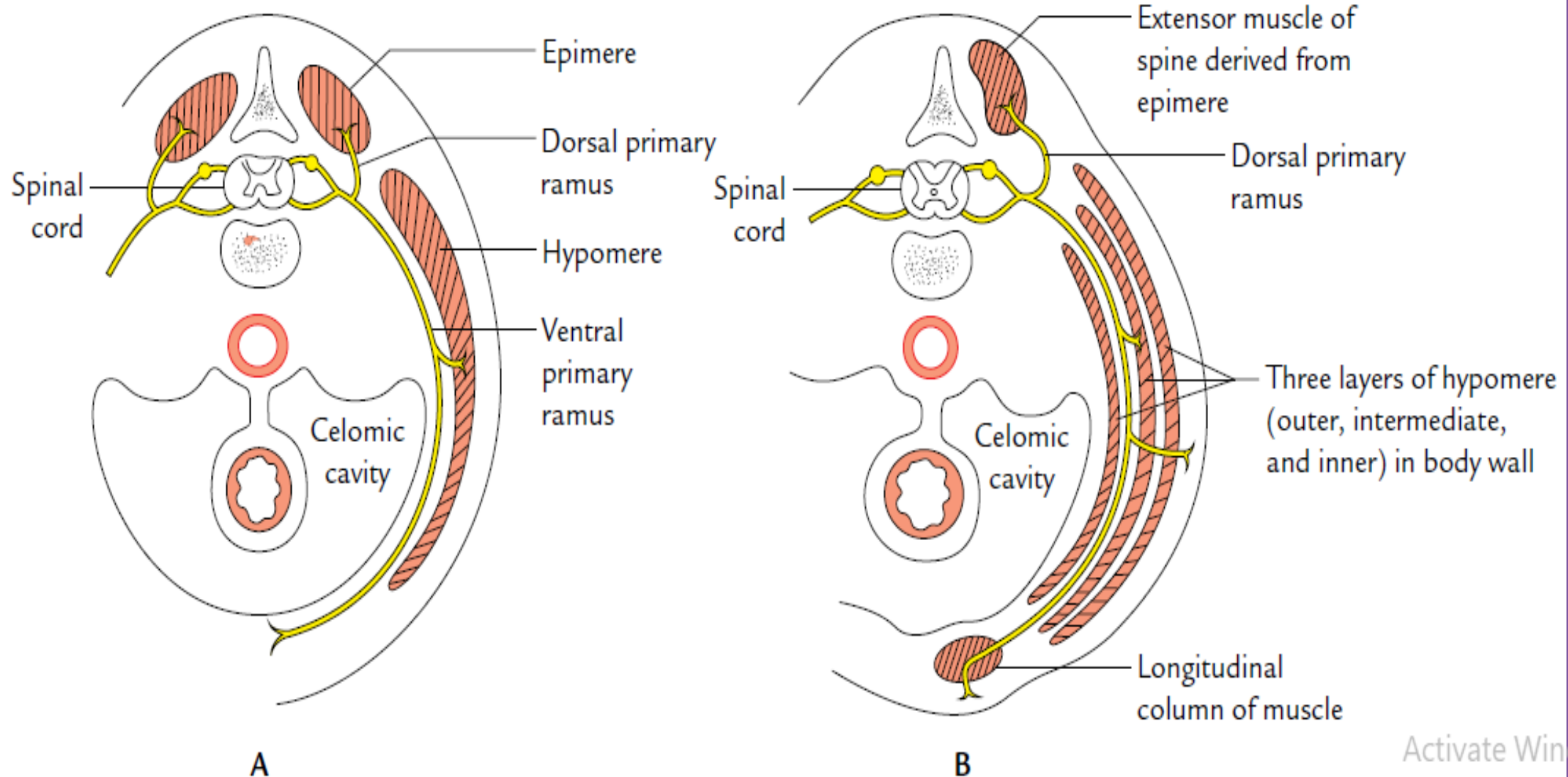
Development of Muscles of the Body Wall

- These are derived from **myotomes** (somites) of the trunk region.
- ❖ **Each myotome divides into two parts:** a smaller **dorsal** part called **epaxial part (epimere)** and a larger ventral part called **hypaxial part (hypomere)**.
- The muscles derived from **epimere** are supplied by dorsal ramus of spinal nerve
- The muscles derived from **hypomere** are supplied by ventral ramus of spinal nerve.
- The **epimeres of myotomes** form extensor muscles of the vertebral column (e.g., erector spinae).

Development of Muscles of the Body Wall,,



Development of Muscles of the Body Wall,,



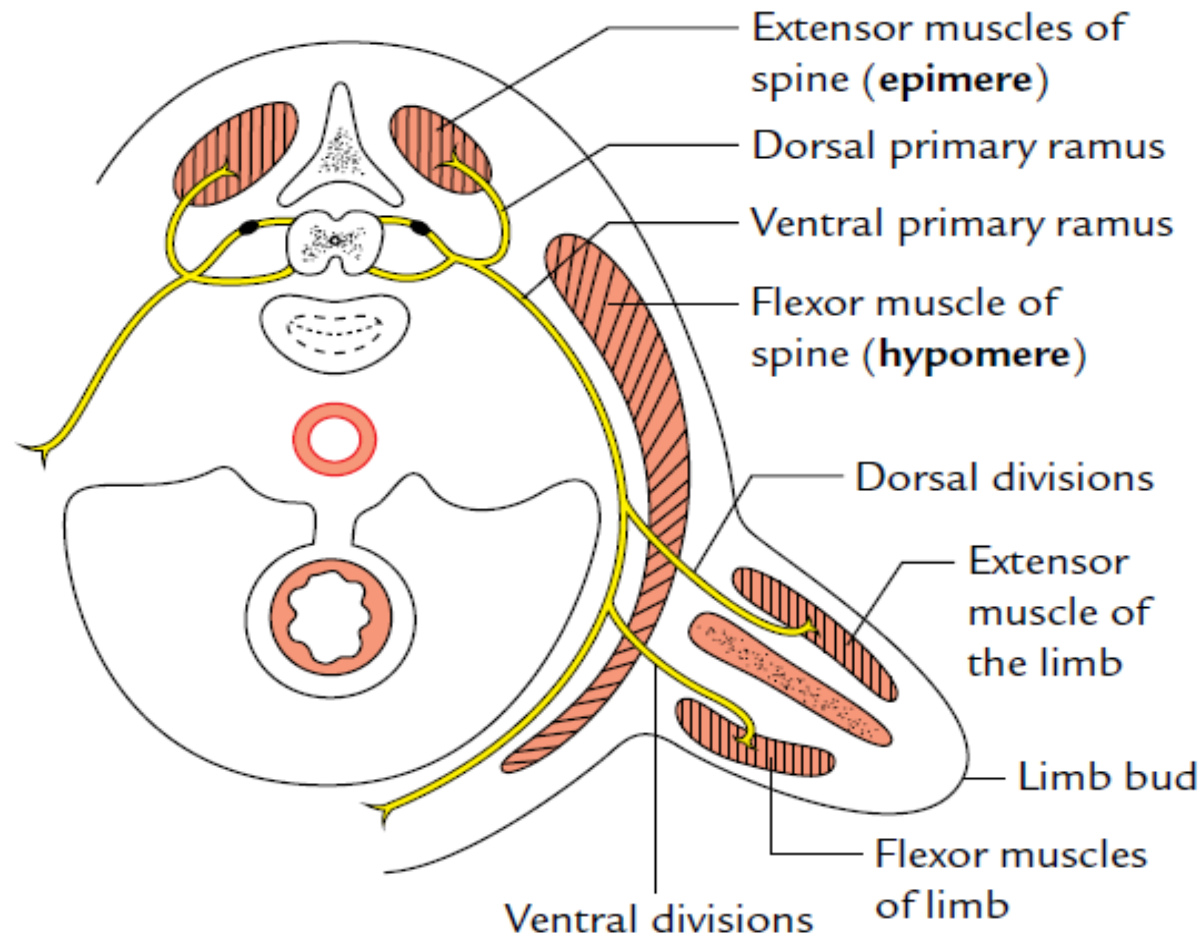
Development of Muscles of the Body Wall,,

- **The hypomeres of myotomes extend ventrolaterally** along the somatopleuric layer of celomic cavity and form following muscles of the body wall.
 - I. They form three layers of muscles in the thorax; **external intercostal, internal intercostal, and intercostal intimus (transversus thoracis).**
 - II. In the abdomen also they form three layers of muscles; **external oblique, internal oblique, and transversus abdominis.**
 - III. In the neck, they form longus colli, longus capitis, and scalene muscles.
- On each side of midline on the ventral aspect of body, three primitive muscles of the body wall fuse to form the longitudinal column of muscle.

Development of Muscles of Limbs

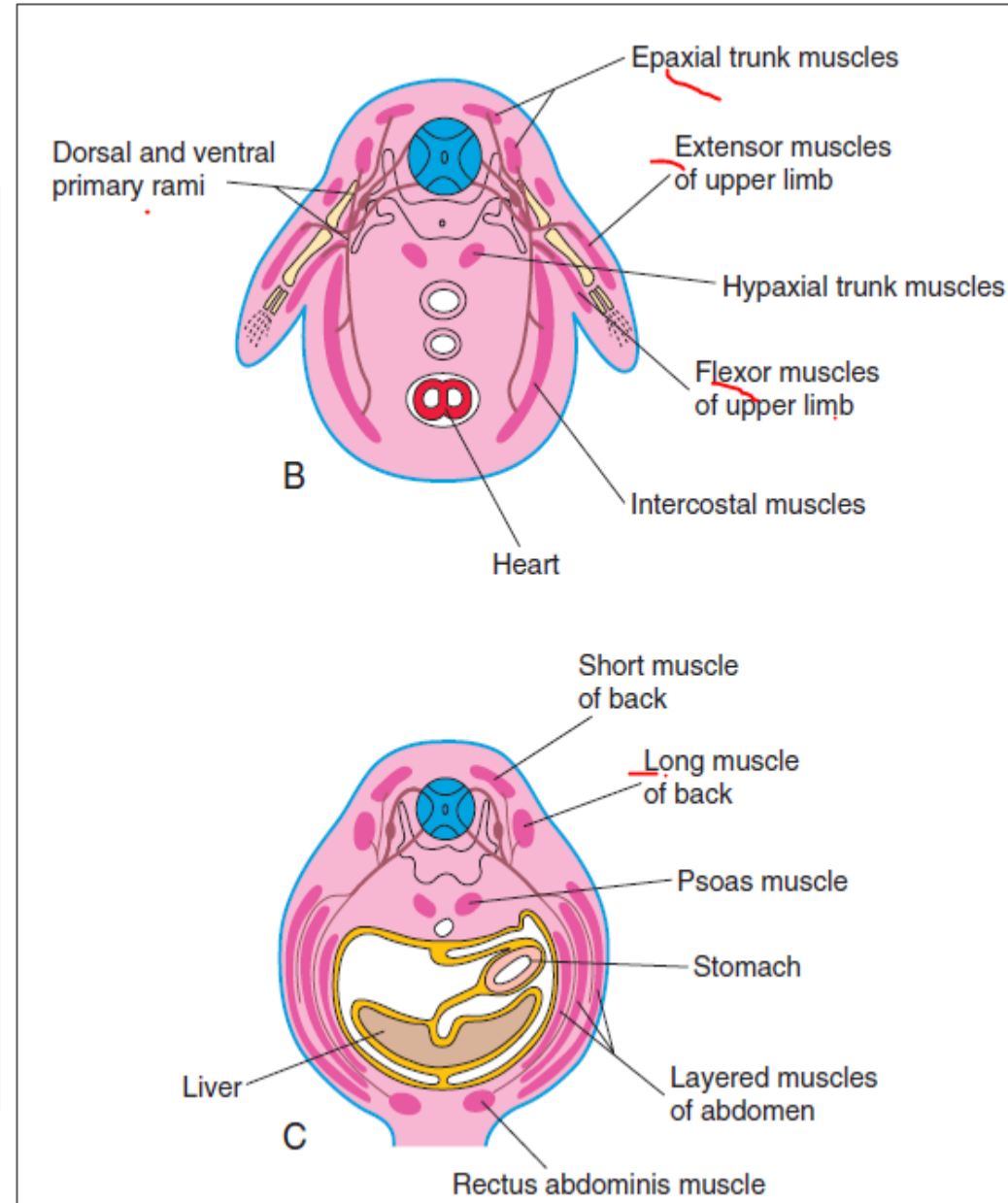
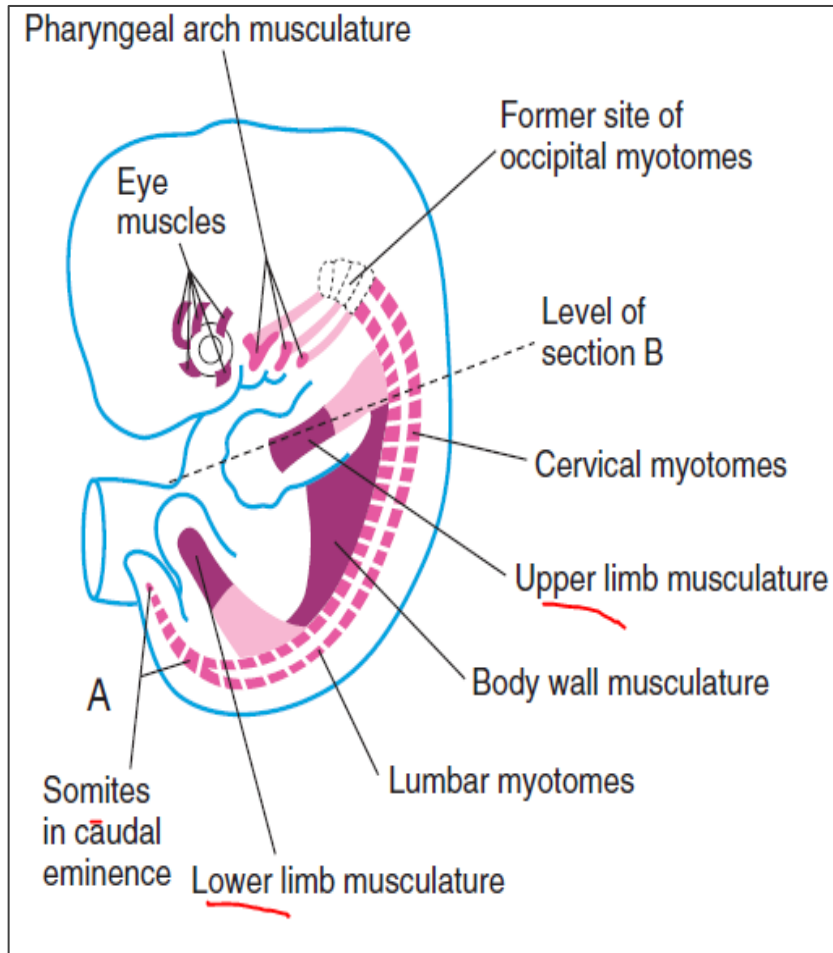
- They develop from myotomes present in the upper and lower limb bud regions.
 - The mesoderm derived from these myotomes migrates into the limb bud during the fifth week and forms **anterior** and **posterior condensations**.
- ❑ **The anterior condensation** of mesoderm gives rise to **flexor and pronator muscles of the upper limb** and **extensor and adductor muscles of the lower limb**.
- The **posterior condensation** of mesoderm gives rise to **extensor and supinator muscles of the upper limb** and **flexor and adductor muscles of the lower limb**.

Development of Muscles of Limbs



Development of musculature into the limb bud. Note the extensor (dorsal) and flexor (ventral) components of limb musculature.

Development of Muscles of Limbs



Development of the Integumentary system

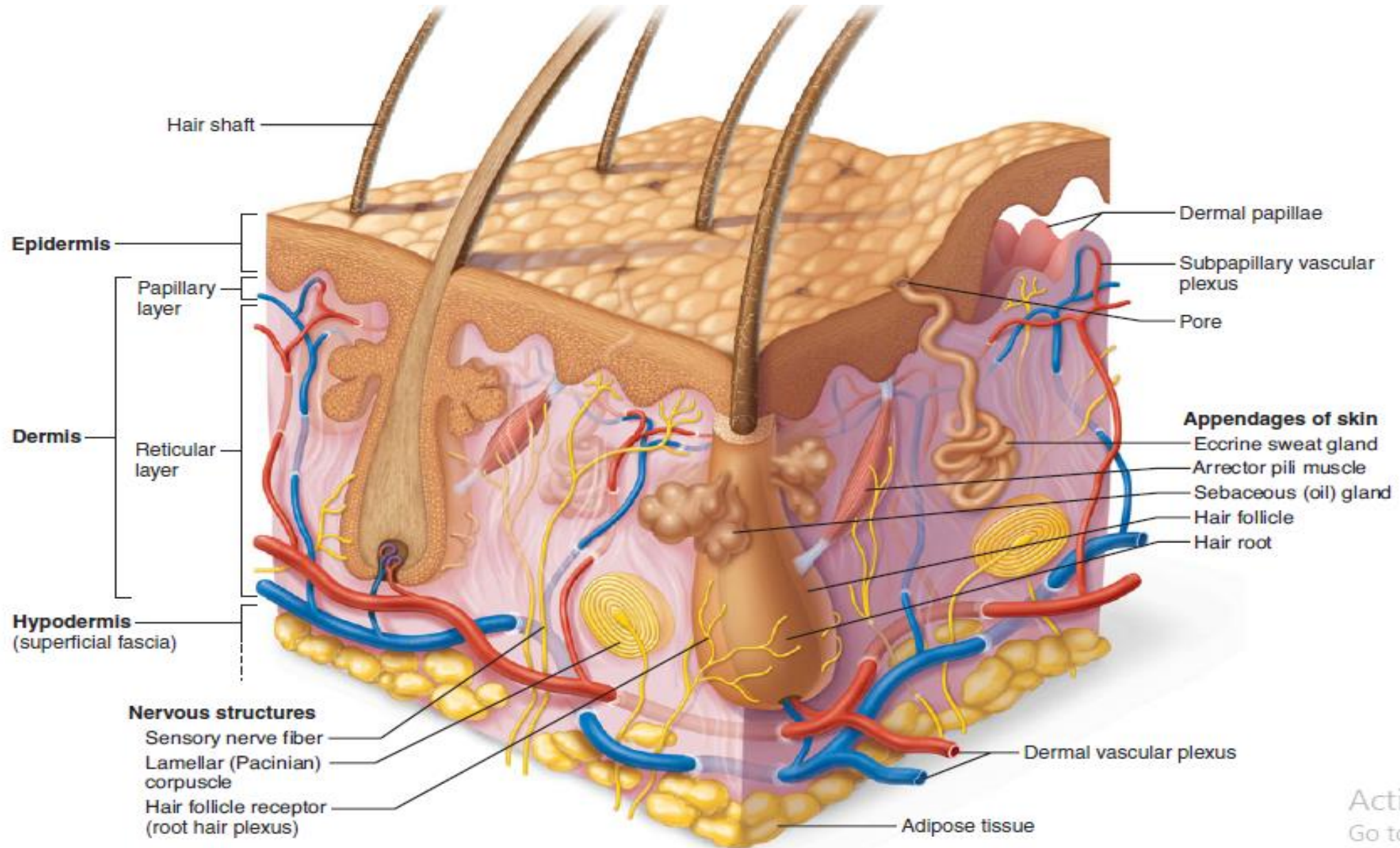
Introduction to anatomy of the integumentary system

- Anatomically the skin and its accessory structures make up the **integumentary system**, which provides the body with overall protection.
- **The skin is made of multiple layers** of cells and tissues, which are held to underlying structures by connective tissue.
- The deeper layer of skin is well vascularized (has numerous blood vessels).
- It also has numerous sensory, and autonomic and sympathetic nerve fibers ensuring communication to and from.

Introduction to anatomy of the integumentary system,,

- Like other systems the integumentary is composed of organs such as the skin and accessory structures of the skin including:
 - ✓ Nails,
 - ✓ Hair
 - ✓ Sensory receptors.
 - ✓ Cutaneous glands
 - Sebaceous glands (oil glands) and
 - Sweat glands(sudoriferous glands),
 - Ceruminous glands
 - Mammary glands

Introduction to anatomy of the integumentary system,,



Structure of The Skin

❖ Structurally, the skin consists of two main parts.

I. The superficial, thinner portion, which is composed of **epithelial tissue**, **is the epidermis**.

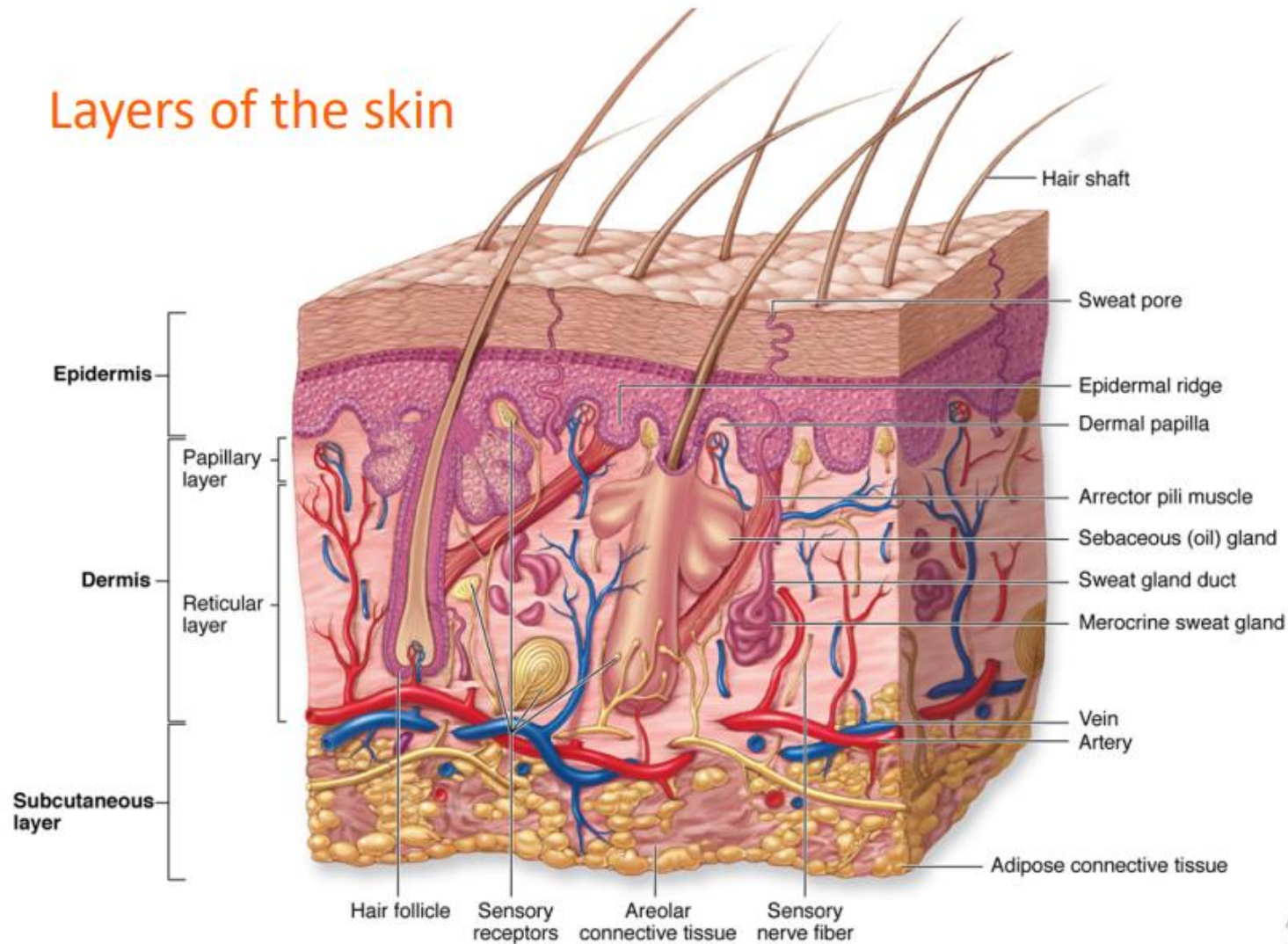
▪ **It is avascular**. For this reason, if you scratch the epidermis there is **no bleeding**.

II. The deeper, thicker, **connective tissue portion is the dermis**.

❖ It is **vascular** and a cut in the dermis produces bleeding.

❖ **Skin appendages**: Merocrine (eccrine) sweat glands, apocrine sweat glands, sebaceous glands, hair follicles, nails are located in this layer.

Structure of The Skin,,

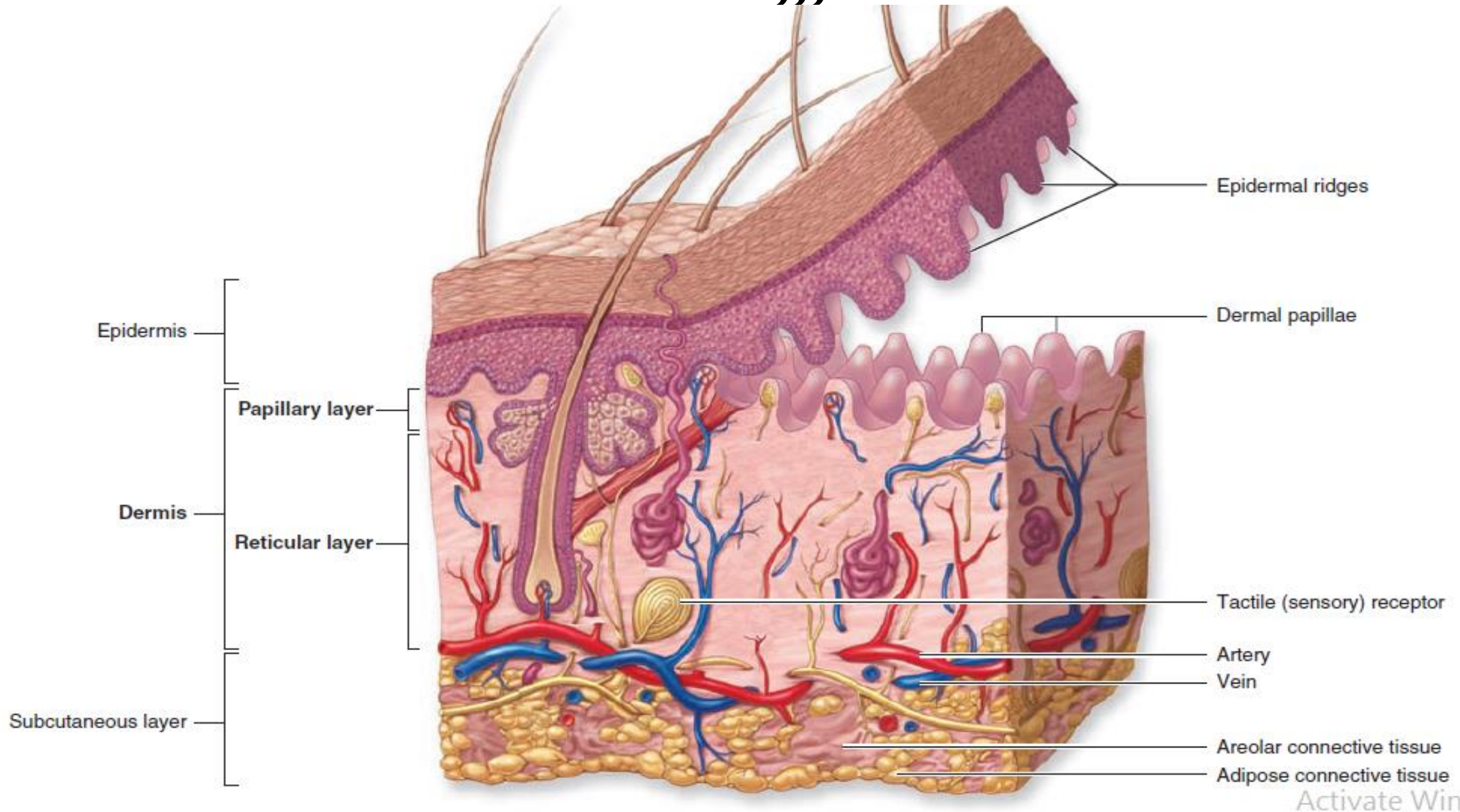


Skin structure:, accessory structures, and the underlying hypodermis.

Structure of The Skin,,,

- **The epidermis** , an epithelial layer of **ectodermal** origin, and **the dermis** is, a layer of **mesodermal** connective tissue
- At the irregular junction between the **dermis and epidermis**, projections **called dermal papillae interdigitate** with invaginating epidermal ridges to strengthen adhesion of the two layers.
- Beneath the dermis lies the subcutaneous tissue or hypodermis, a loose connective tissue layer usually containing pads of adipocytes.
- The subcutaneous tissue binds the skin loosely to the underlying tissues.

Structure of The Skin,,,

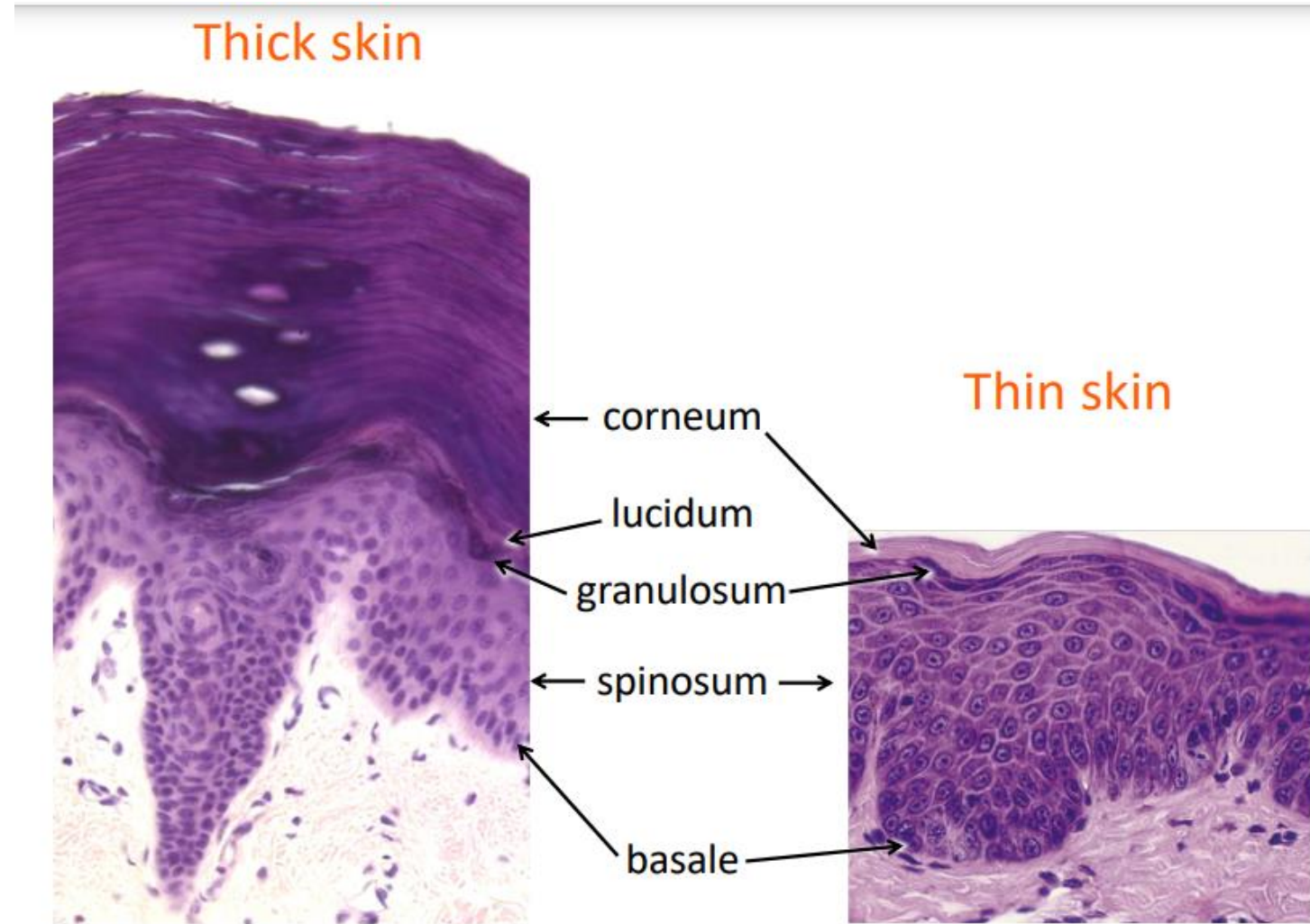


Structure of The Skin,,,

- The skin ranges in thickness from 0.5 mm on the eyelids to 6.0 mm on the **heels**.
However, over most of the body it is 1–2 mm thick.
- The difference is due mainly to **variation in the thickness of the dermis**, although skin is **classified as thick or thin based on the relative thickness of the epidermis alone.**
- ❖ **Thick skin:** The skin is thickest in areas subjected to wear and tear (abrasion), such as the soles of the feet, where it may be 6 mm in thickness.
- ❖ Its epidermis alone is about 0.5 mm thick, due to a very thick surface layer of dead cells called **the stratum corneum.**

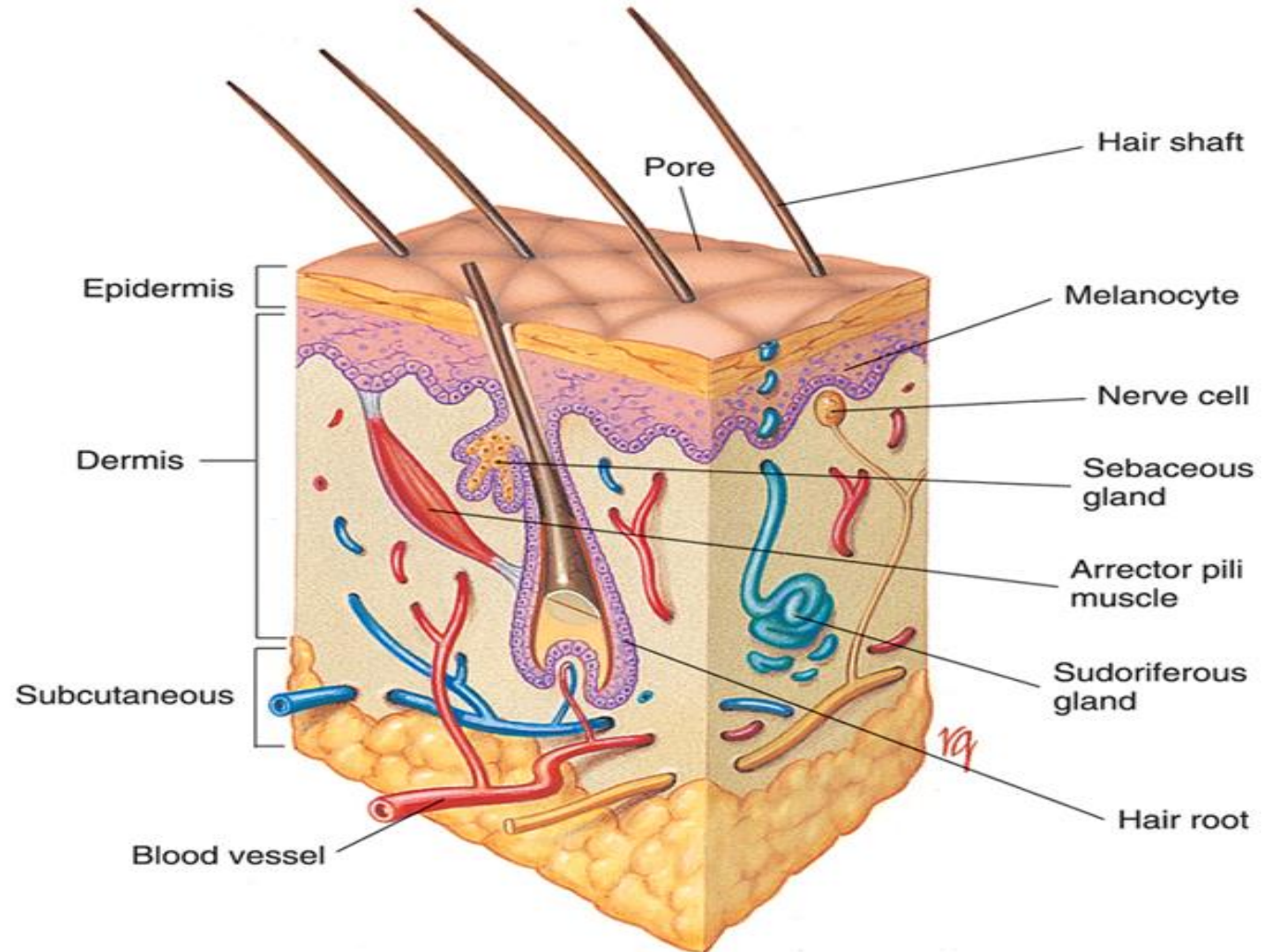
Structure of The Skin,,

- ❖ Thick skin has sweat glands but no hair follicles or sebaceous (oil) glands.
- ❖ The rest of the body is covered with **thin skin**, which has an epidermis about 0.1 mm thick, with a thin stratum corneum.
- ❖ It possesses hair follicles, sebaceous glands, and sweat glands.

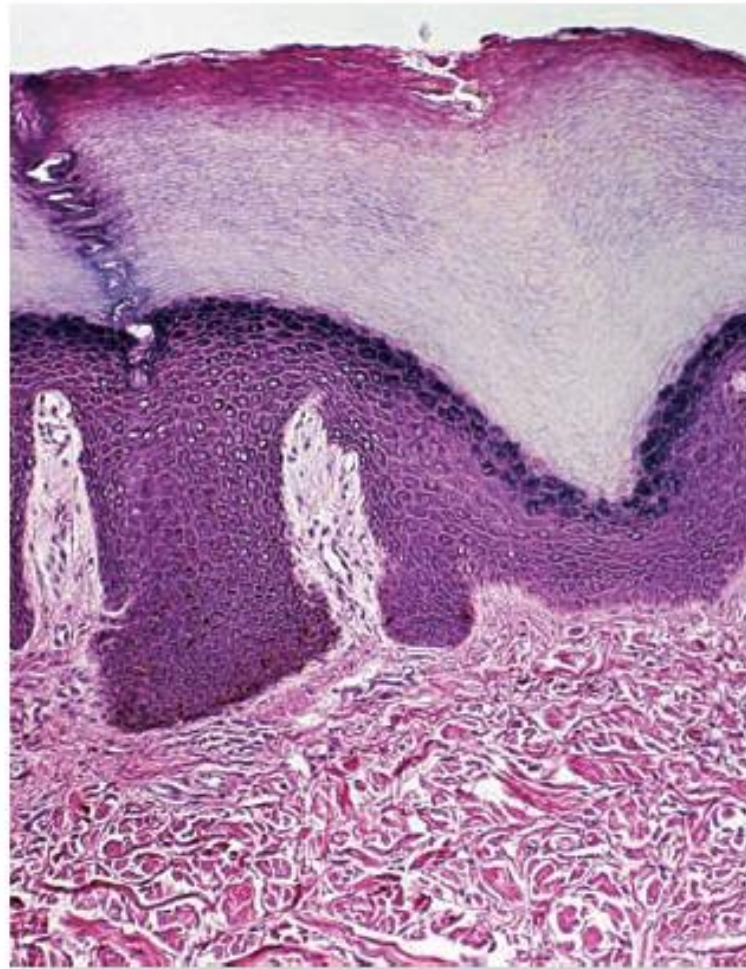


Epidermis

- ❖ The epidermis is a **keratinized stratified squamous epithelium**.
- ❖ its surface consists of **dead** cells packed with the tough protein **keratin**.
- ❖ It is avascular, meaning it lacks blood vessels.
- ❖ Since the epidermis is prone to injury, the lack of blood vessels prevents unnecessary bleeding.

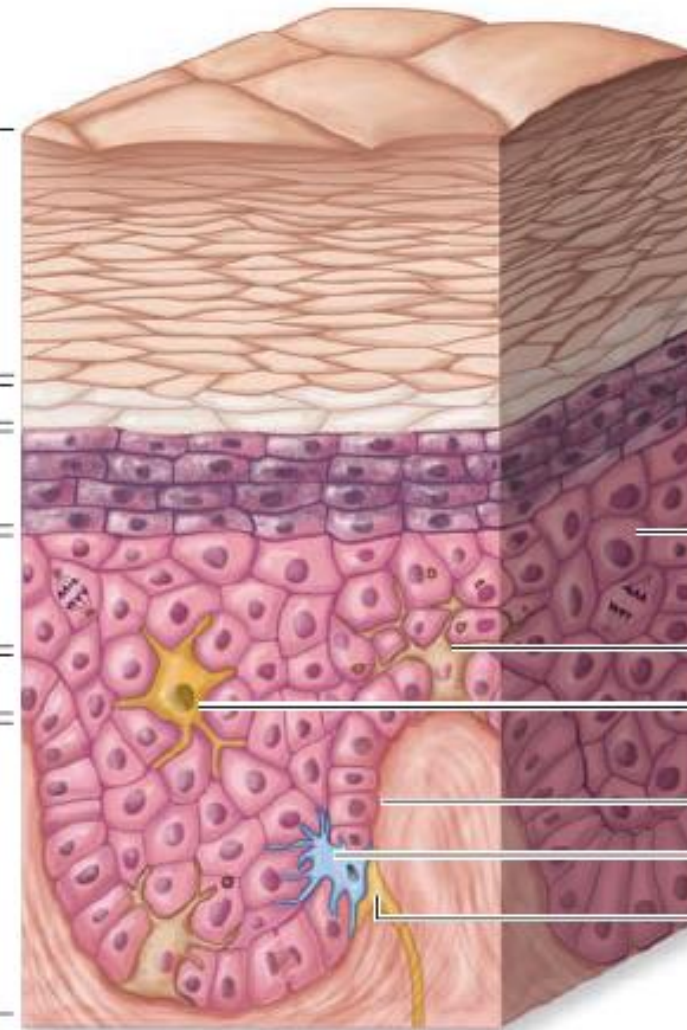


Epidermis,,,



a

Stratum corneum
Stratum lucidum
Stratum granulosum
Stratum spinosum
Stratum basale
Dermis

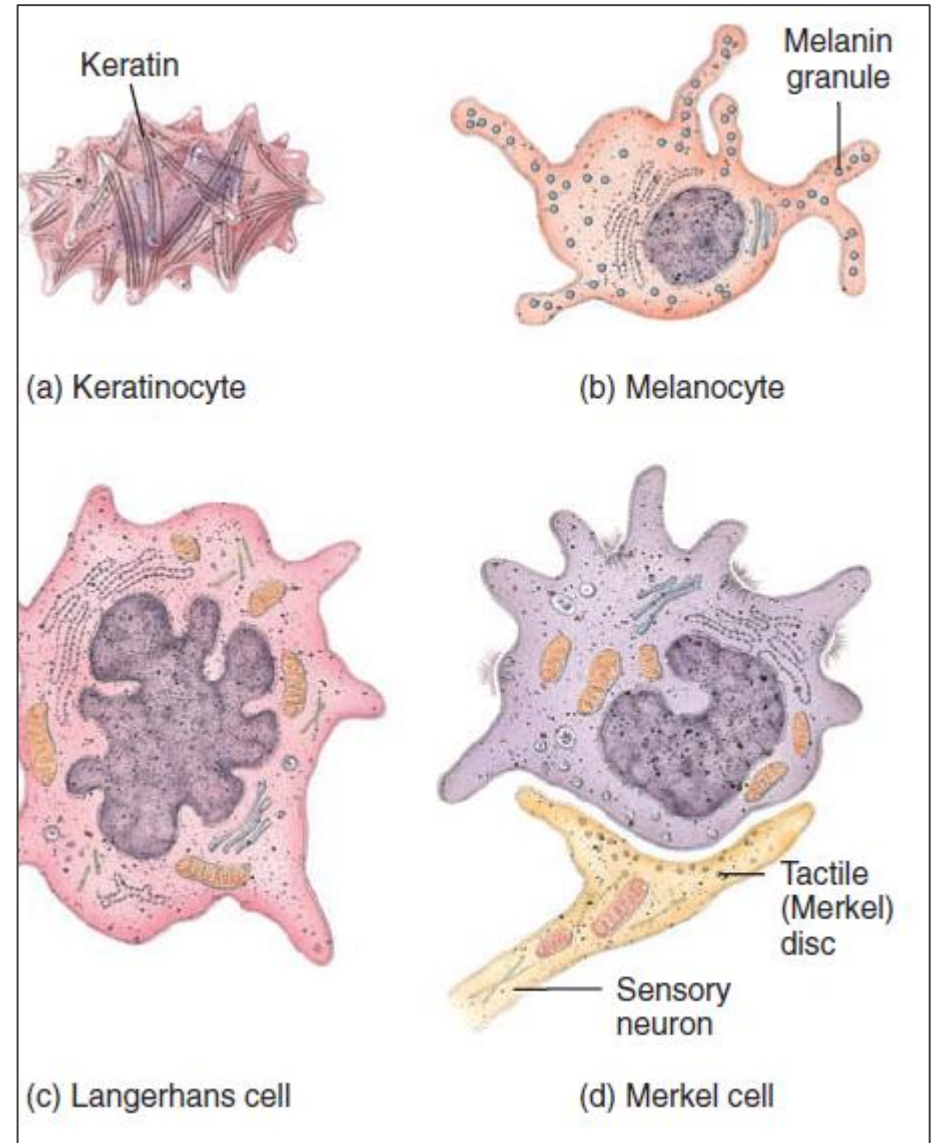


b

Dead keratinocytes
Living keratinocyte
Melanocyte
Epidermal dendritic cell
Basement membrane
Tactile cell
Sensory nerve ending

Epidermis,,,

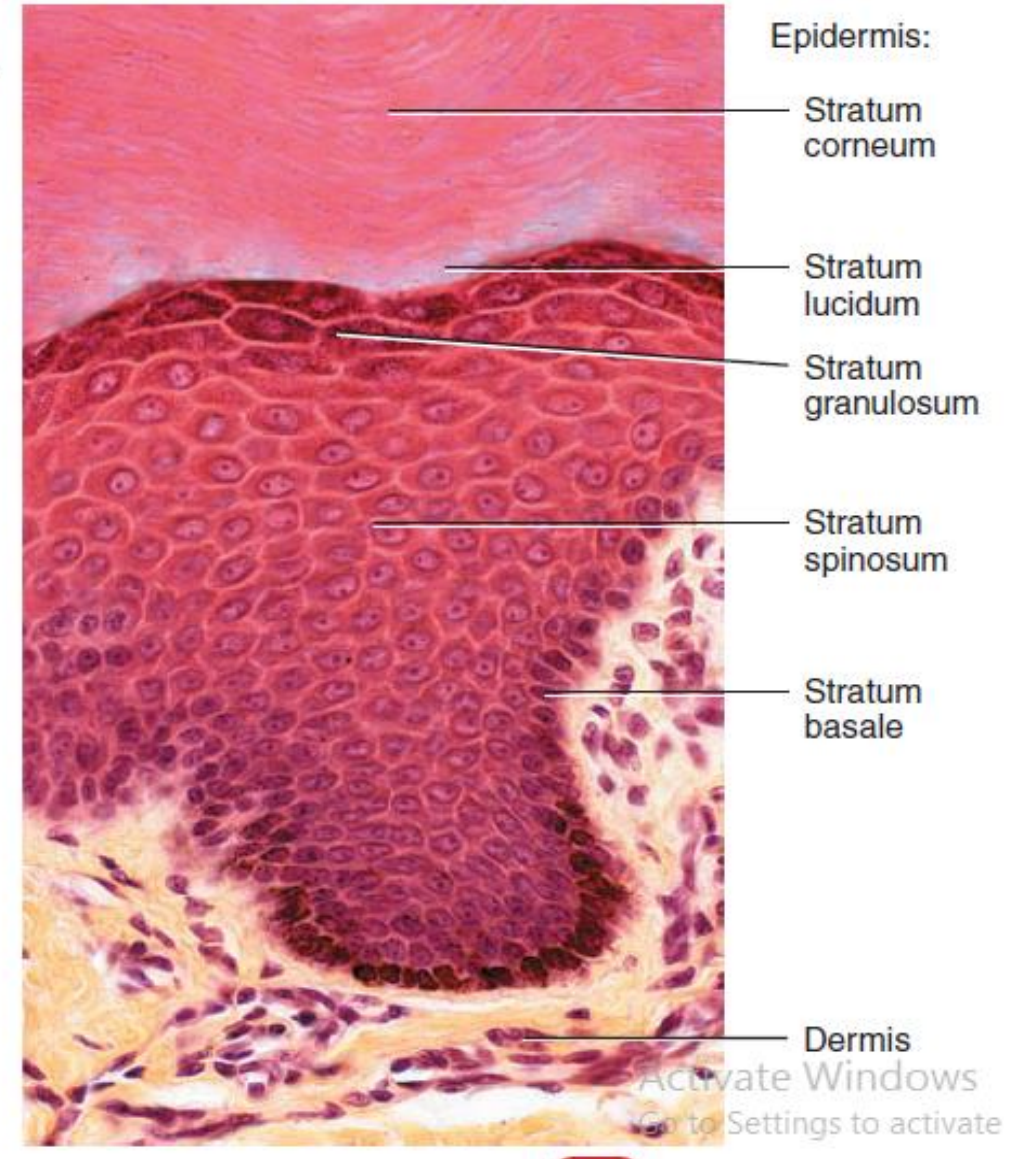
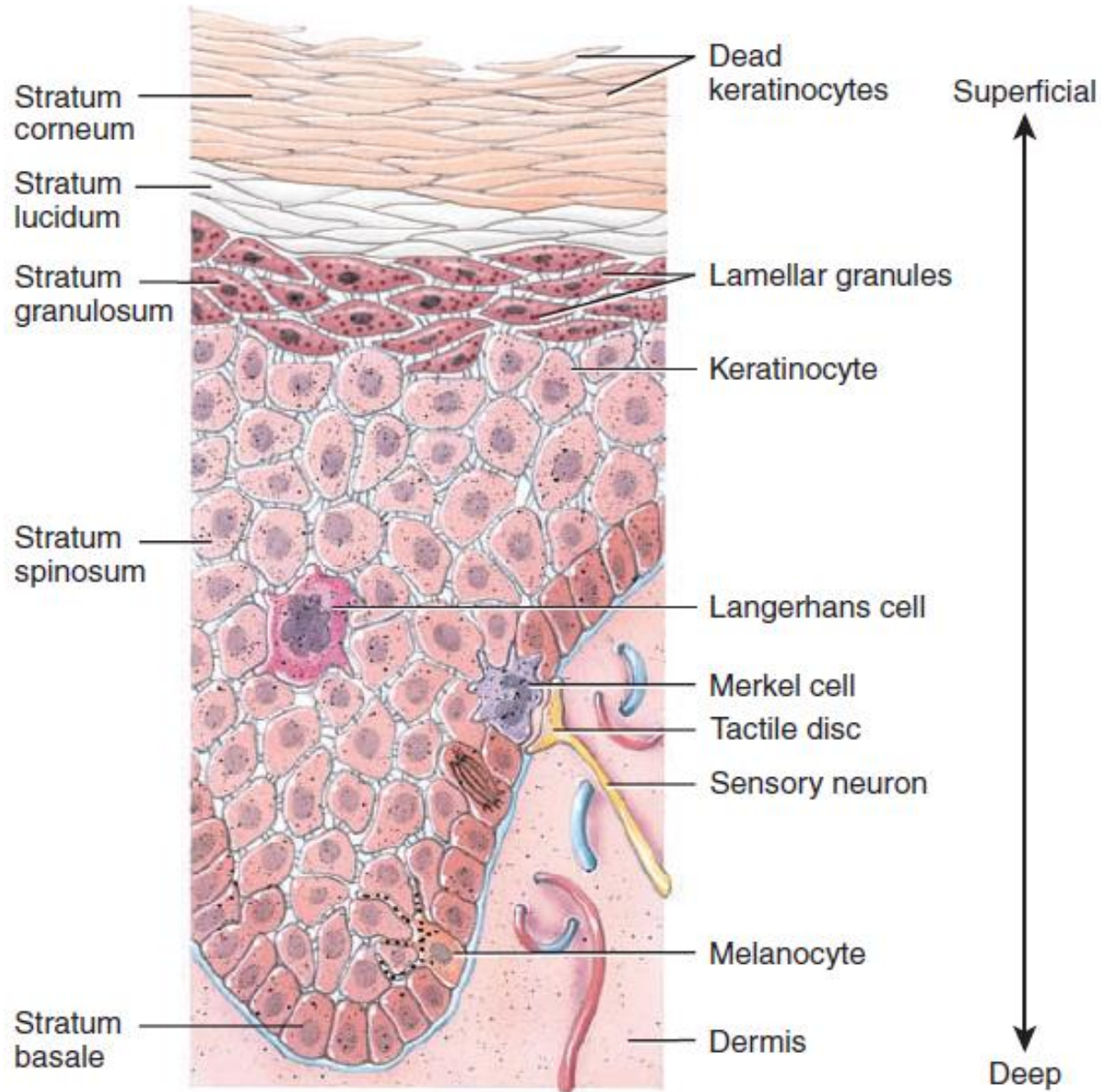
- ❖ The epidermis is composed of keratinized stratified squamous epithelium.
- ❖ It contains five principal types of cells:
 - I. Stem cells
 - II. keratinocytes,
 - III. melanocytes,
 - IV. Langerhans cells, and
 - V. Merkel cell



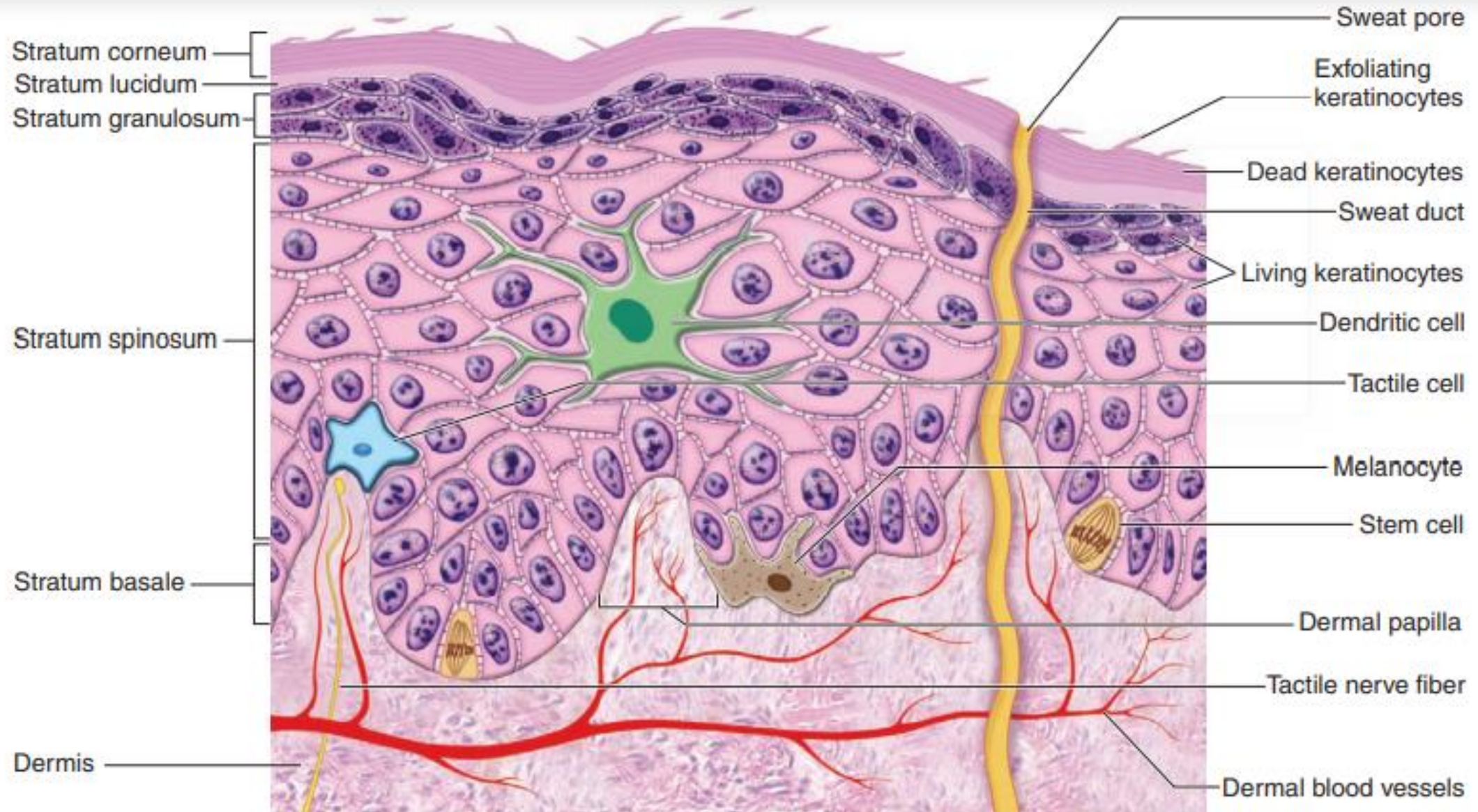
Layers of epidermis

- In most regions of the body the epidermis has **four** strata or layers: **stratum basale, stratum spinosum, stratum granulosum, and a thin stratum corneum. This is called thin skin.**
- ❖ **Where exposure to friction** is greatest, such as in the fingertips, palms, and soles, the epidermis has **five layers: stratum basale, stratum spinosum, stratum granulosum, stratum lucidum, and a thick stratum corneum. This is called thick skin.**

Layers of epidermis,,



Layers of epidermis,,,



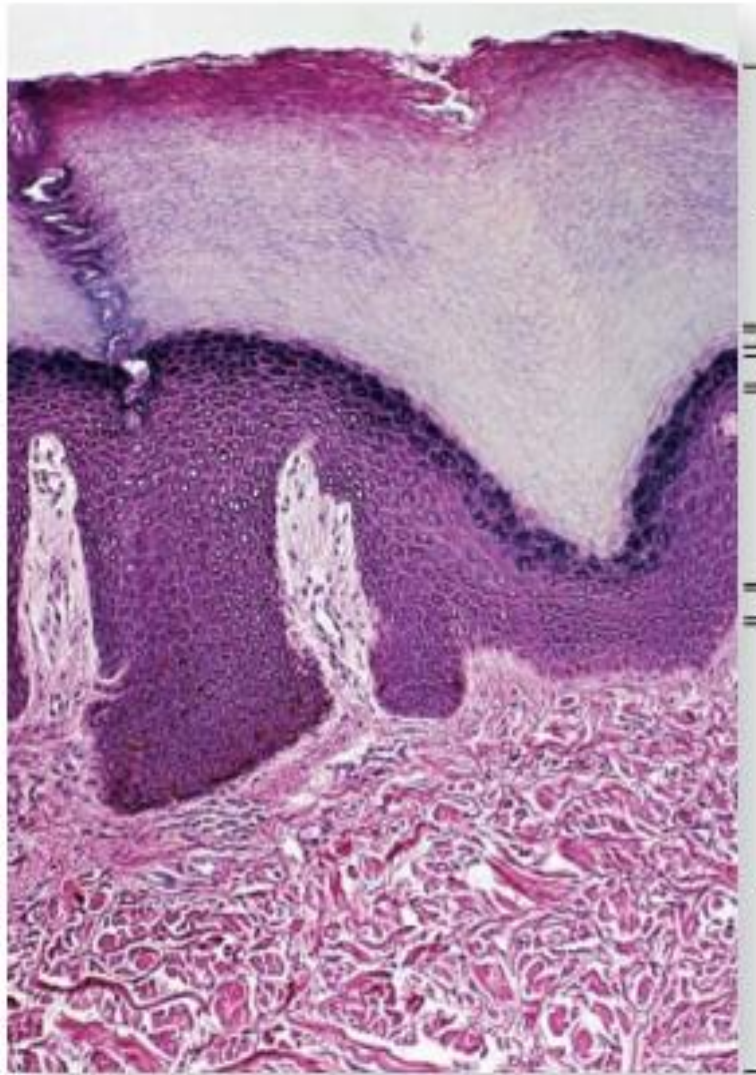
Strata and Cell Types of the Epidermis.

Layers of epidermis,,

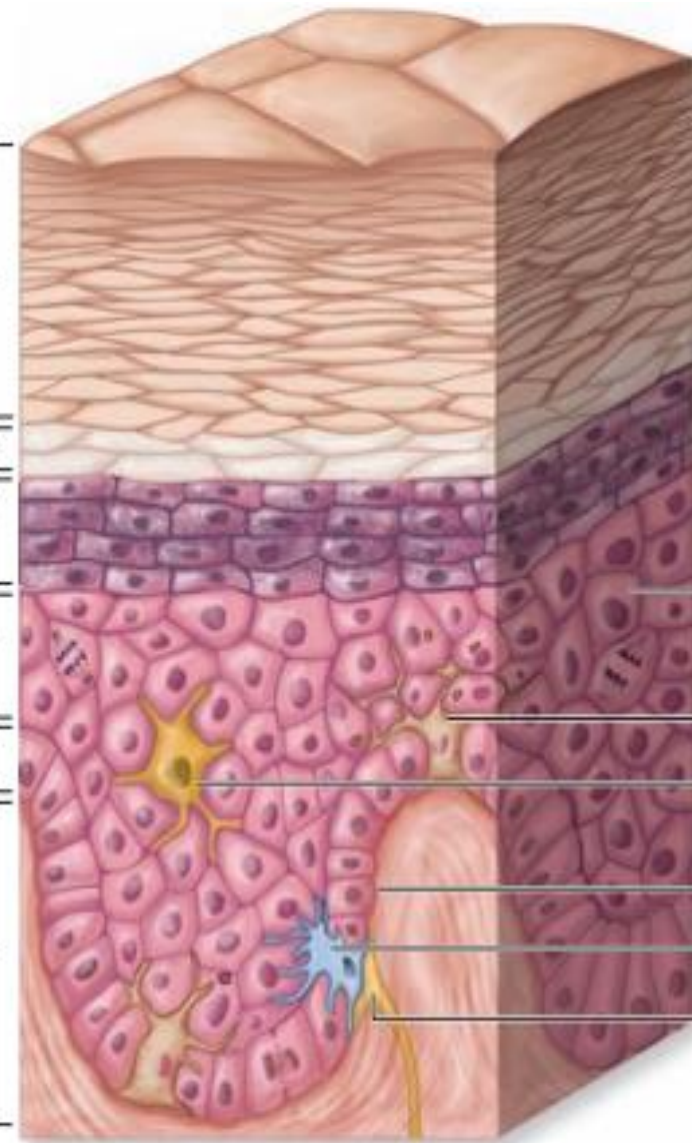
- **Stratum basale (stratum germinativum)**
 - deepest layer
 - Attached to the dermis
 - **Single row of rapidly dividing keratinocytes**
 - Melanocytes and occasionally Merkel cells present
- **Stratum spinosum**
 - Several layers thick
 - Appear spiny in shape
 - Cells contain pre-keratin filaments
 - Melanin granules and Langerhans' cells are present in this layer
- **Stratum granulosum**
 - Three to five layers thick
 - Keratinocytes appear flattened, organelles and nuclei disintegrate; granulated cytoplasm remains; cell membranes thicken

Layers of epidermis,,,

- **Stratum lucidum** (*thick skin only*)
 - Found only in the palms, fingertips, and soles of the feet
 - Appears translucent under the microscope
 - Few rows of clear, flat, dead keratinocytes
- **Stratum Corneum** – outermost layer
 - 15-20 layers of flattened **non-nucleated** cells whose cytoplasm is filled with a keratin (are thus keratinized cells).
 - Protect deeper cells from water loss, the effects of air, chemical, physical, and biological assault
 - These cells are continuously shed at the surface of the stratum corneum.



- Stratum corneum
- Stratum lucidum
- Stratum granulosum
- Stratum spinosum
- Stratum basale
- Dermis

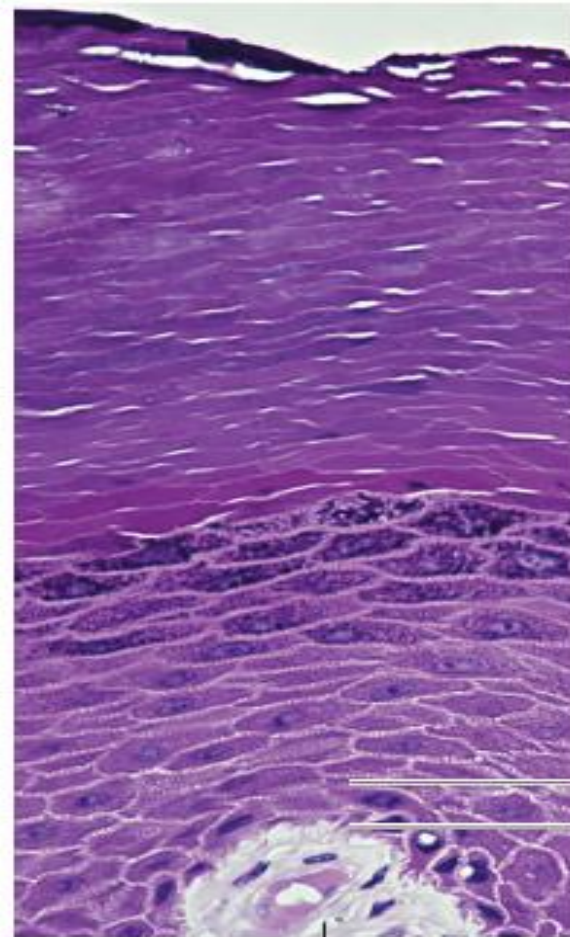


- Dead keratinocytes
- Living keratinocyte
- Melanocyte
- Epidermal dendritic cell
- Basement membrane
- Tactile cell
- Sensory nerve ending

a

b

Layers of epidermis



(a)

Dermis

Stratum corneum

Most superficial layer; 20–30 layers of dead cells represented only by flat membranous sacs filled with keratin. Glycolipids in extracellular space.

Stratum granulosum

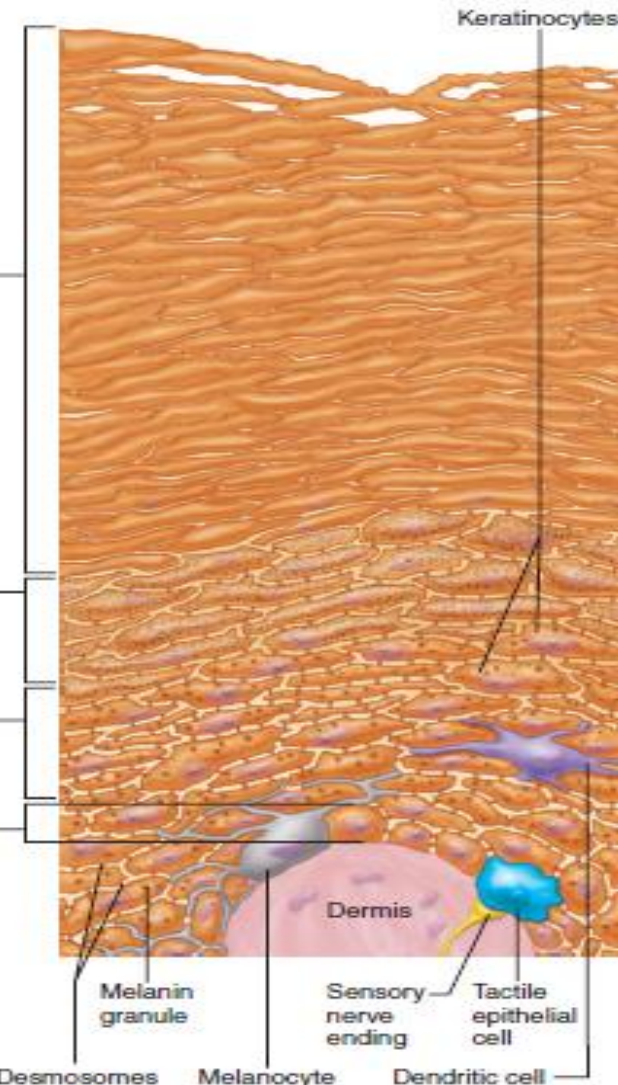
Three to five layers of flattened cells, organelles deteriorating; cytoplasm full of lamellated granules (release lipids) and keratohyaline granules.

Stratum spinosum

Several layers of keratinocytes unified by desmosomes. Cells contain thick bundles of intermediate filaments made of pre-keratin.

Stratum basale

Deepest epidermal layer; one row of actively mitotic stem cells; some newly formed cells become part of the more superficial layers. See occasional melanocytes and dendritic cells.



(b)

Dermis

- The second, deeper part of the skin, the dermis, is composed of a strong dense irregular connective tissue containing **collagen and elastic fibers**.
- This layer supports the epidermis & binds it to the subcutaneous tissue (hypodermis)
- The dermis also has the ability to **stretch and recoil easily**.
- The few cells present in the dermis include **predominantly fibroblasts**, with some **macrophages**, and a **few adipocytes** near its boundary with the subcutaneous layer.
- **Blood vessels, nerves, glands, and hair follicles (epithelial invaginations of the epidermis) are embedded in the dermal layer.**
- **The dermis is essential to the survival of the epidermis.**

Dermis,,,

❑ **Specialized receptors and structures in the dermis are:** Meissner's and Pacinian Corpuscles

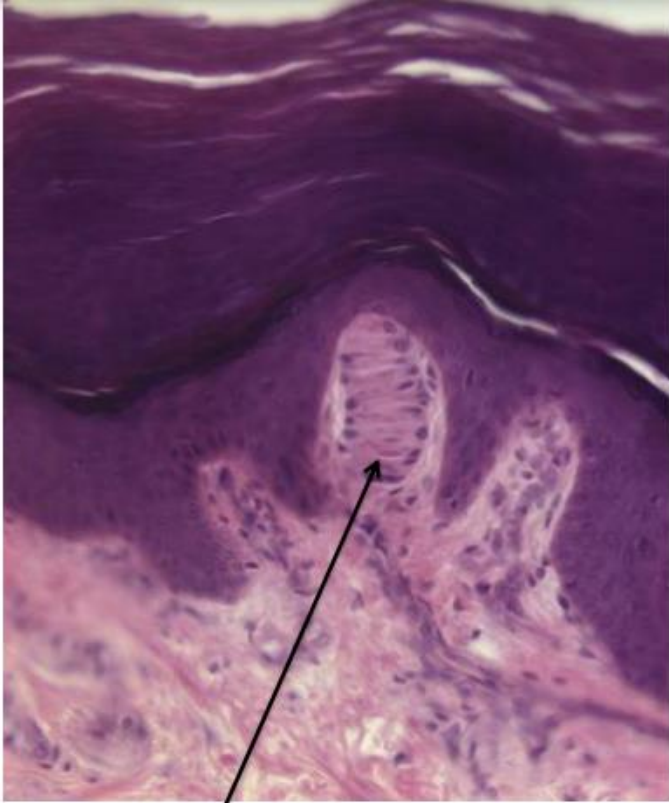
✓ **Meissner's corpuscle •**

- Sensitive to light touch
- Consists of an unmyelinated axon meandering back and forth between flattened Schwann cells.

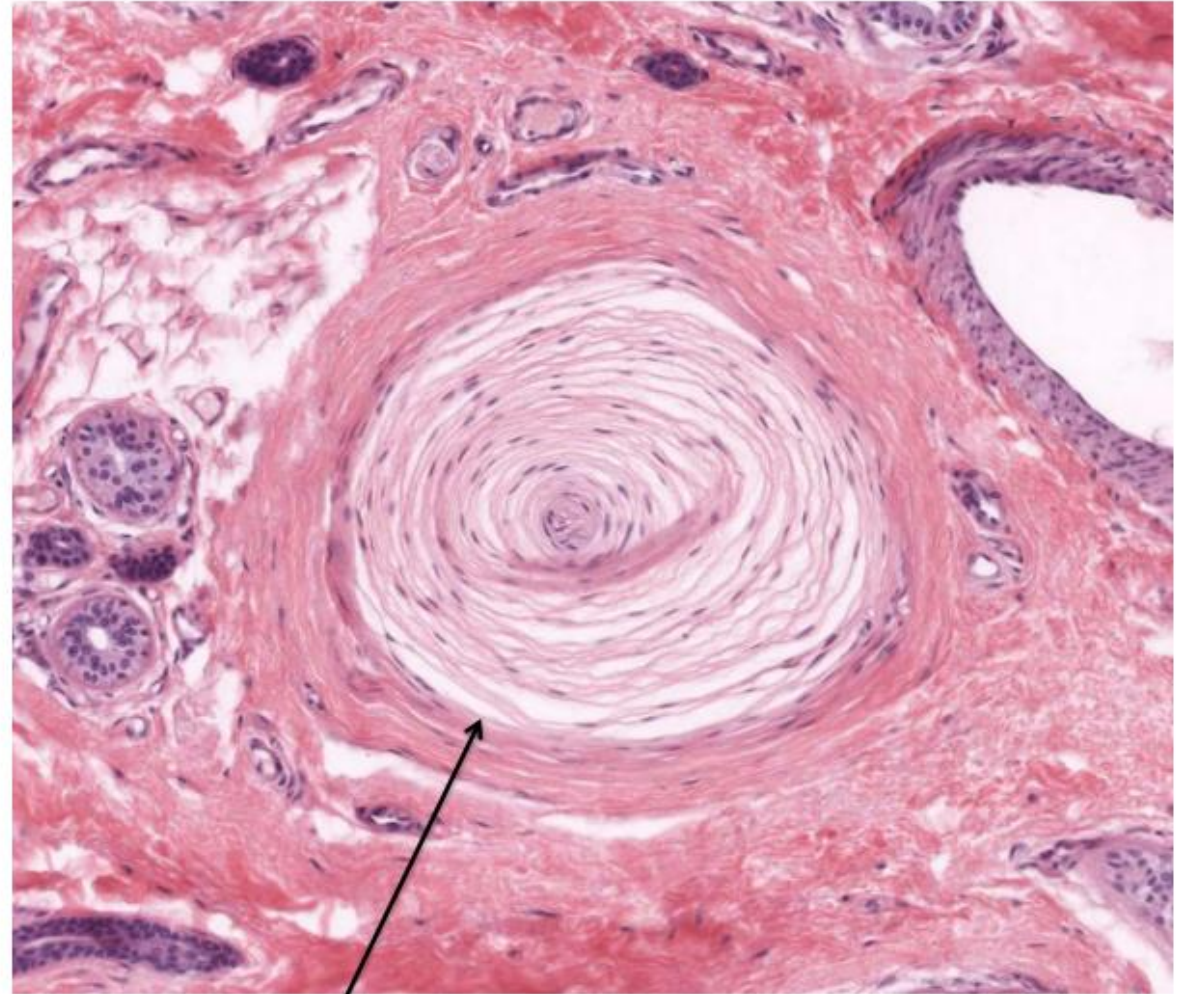
✓ **Pacinian corpuscle •**

- Sensitive to vibration and pressure.
- Consists of unmyelinated nerve terminal surrounded by layers of fibroblasts

Dermis,,,



Meissner's corpuscle: sensitive to light touch

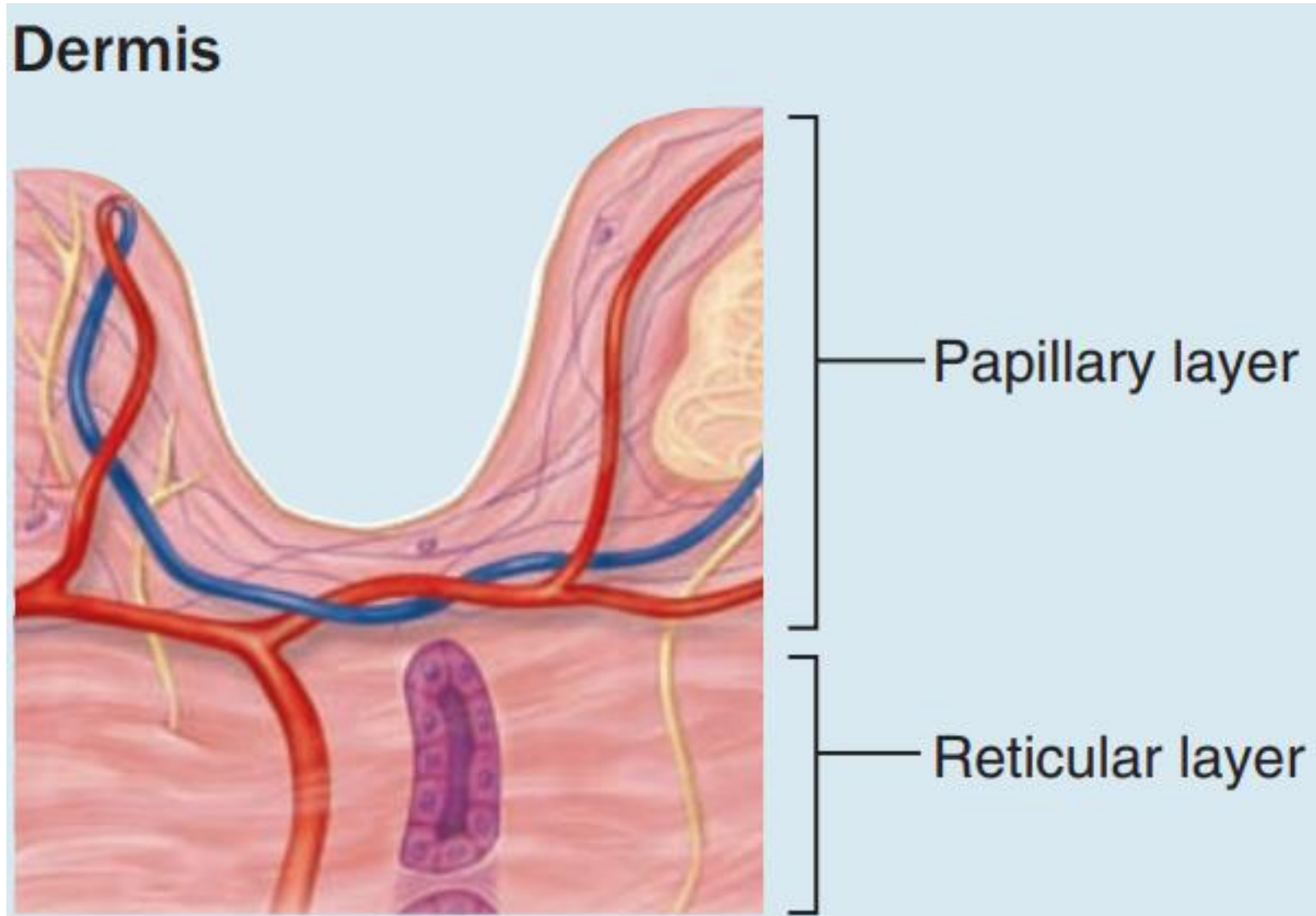


Pacinian corpuscle: sensitive to vibration, pressure

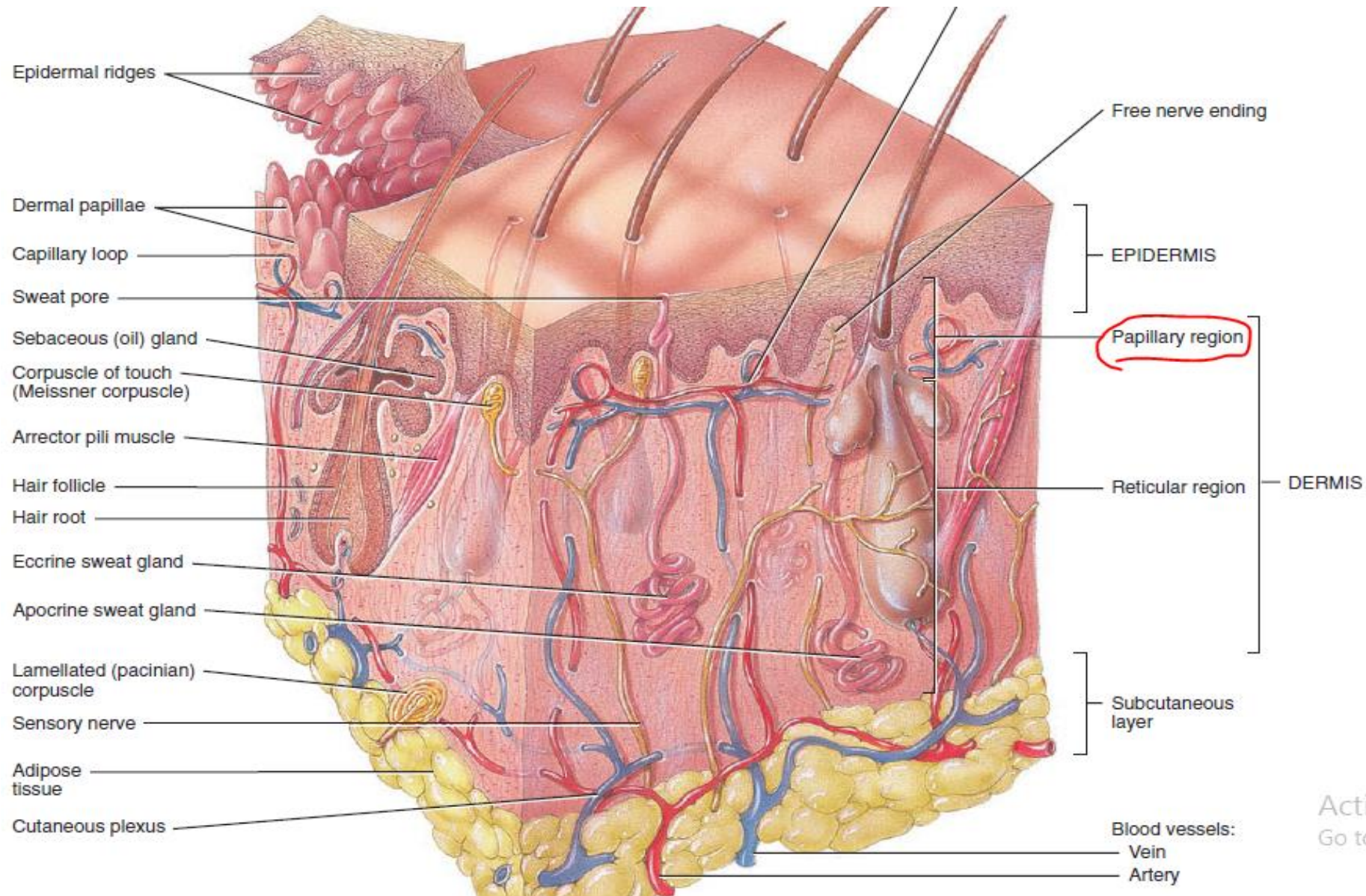
Dermis,,,

- **Based on its tissue structure the dermis has two layers, superficial papillary region and a thick, deeper reticular region.**
- I. **Papillary layer:** The papillary layer forms the superficial layers of dermis and includes the dense connective tissue of the dermal papillae.
 - These papillae are best developed in the thick skin of **the palms and soles.**
 - Each papilla contains a capillary loop (blood vessels).
- I. **Reticular layer:** The reticular layer of the dermis is the deep layer of dermis and consists mainly of thick bundles of collagen fibres.
 - It also contains considerable numbers of elastic fibres.

Dermis,,,



Dermis,,,

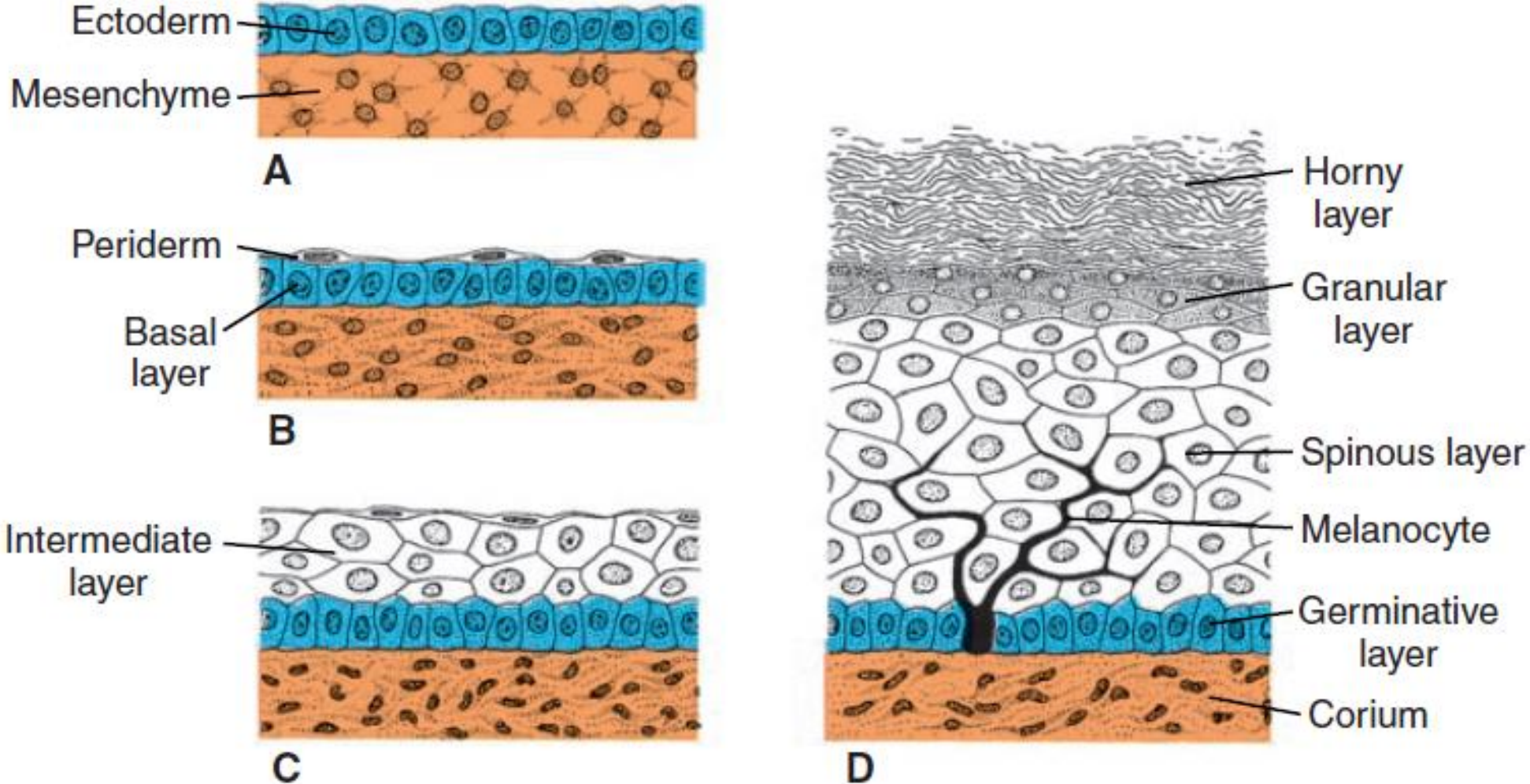


Acti
Go to

Development of the Integumentary system

- The skin develops from two sources:
 - The **epidermis** develops from surface **ectoderm** and
 - The **dermis**, underlining the epidermis, is a deep layer composed of dense, irregularly arranged connective tissue that is derived from **mesenchyme** (*the underlying mesoderm*).
- In addition to the above two sources, the **melanoblasts** (dendritic cells) of epidermis develop from the **neural crest cells**.
- The **appendages of skin** (e.g., sebaceous and sweat glands, hair, and nails) are derived from the epidermis.

Development of the Integumentary system,,



❑ Formation of the skin at various stages of development. A. 5 weeks. B. 7 weeks. C. 4 months. D. Birth.

Development of the Integumentary system,,,

- **Ectodermal** (epidermal)/mesenchymal (dermal) interactions involve mutual inductive mechanisms that are mediated by a conserved set of signaling molecules.
- Skin structures vary from one part of the body to another.
- **For example**, the skin of the eyelids is thin and soft and has fine hairs, whereas the skin of the eyebrows is thick and has coarse hairs.
- The embryonic skin at 4 to 5 weeks consists of a single layer of surface ectoderm overlying the mesoderm

Development of the Epidermis

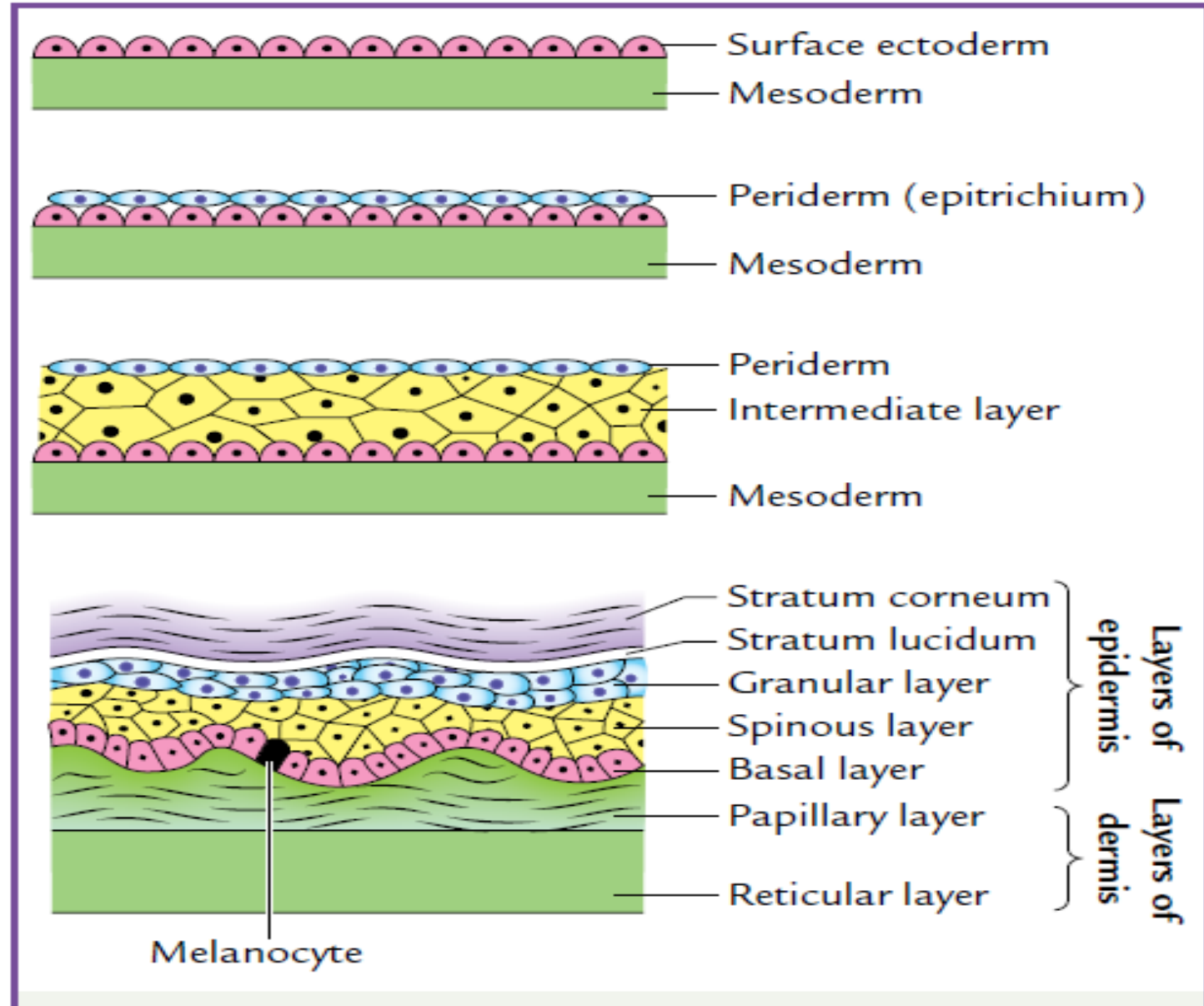
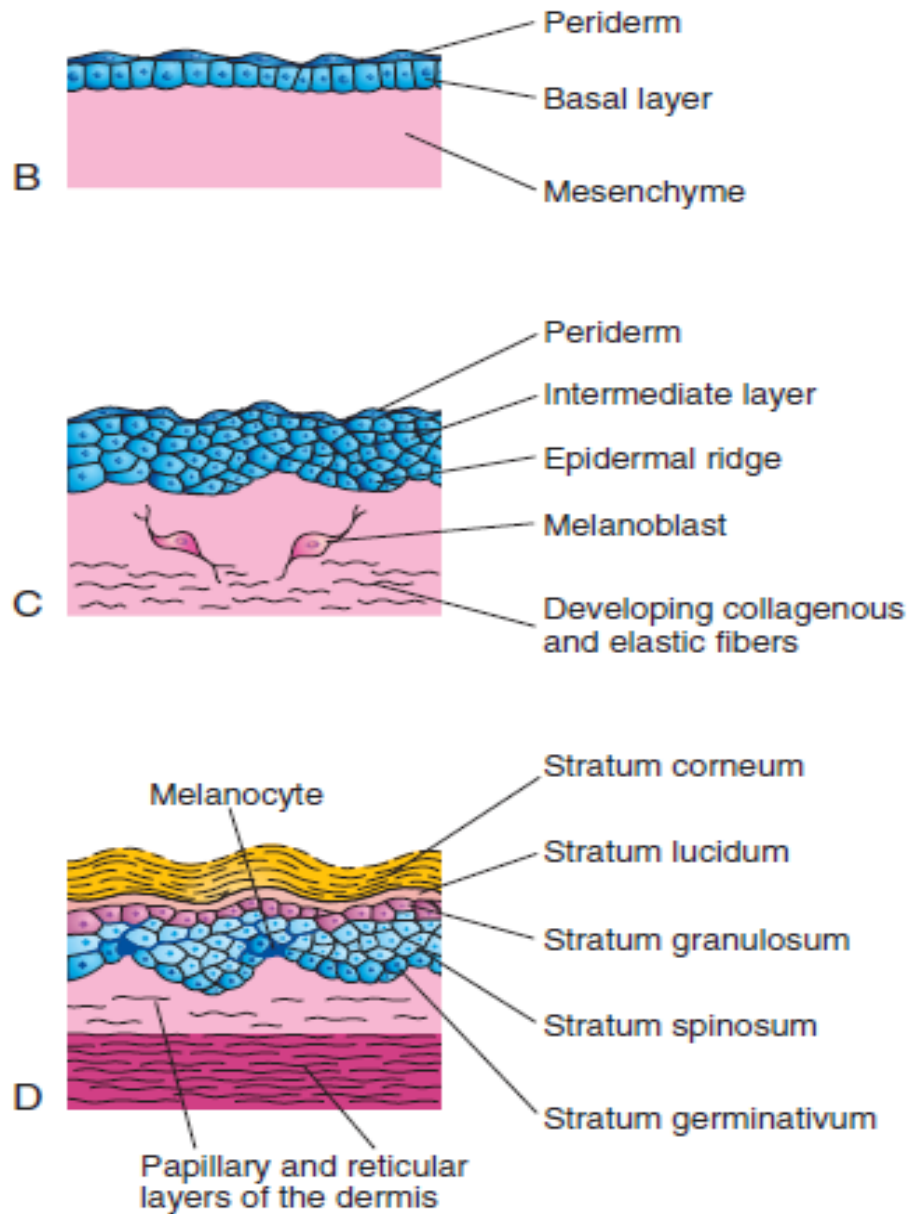
❑ **Initially** the surface ectoderm consists of a single layer of cells.

- **In the second month**, these cells proliferate and form a second layer of flattened cells called **periderm/epitrichium**.

❑ **The epidermis now consists of two layers:** superficial layer of flattened cells (**periderm**) and deep layer of cuboidal cells (**basal layer**).

- With further proliferation of cells in the basal layer, a third **intermediate layer** is formed.
- The **basal layer** functions as a germinative layer and is called **stratum germinativum**.
- **The cells of the basal layer** proliferate and differentiate to form the various layers (strata) of the epidermis.

Development of the Epidermis,,,

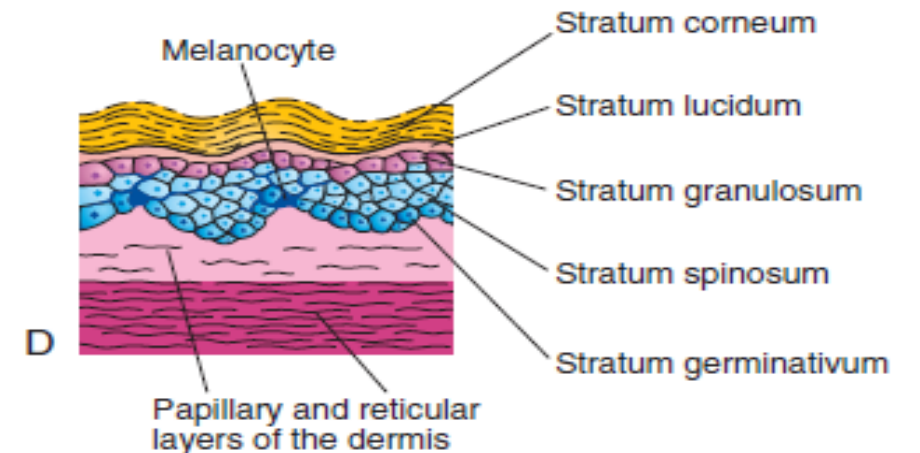
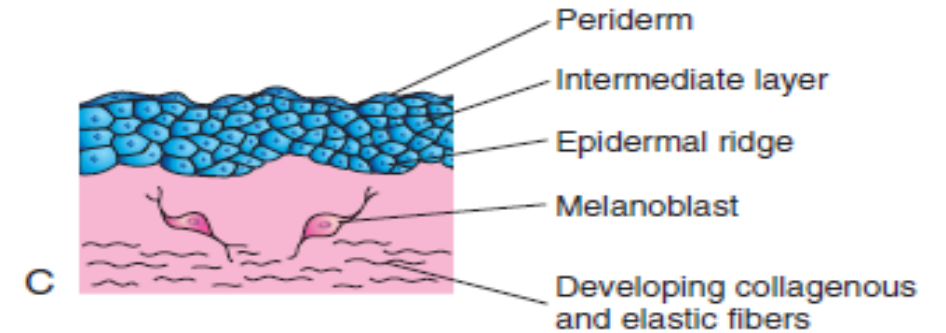
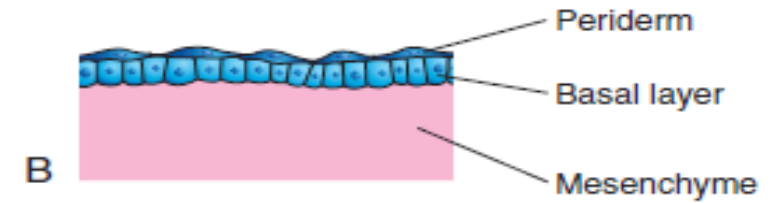


Development of the Epidermis,,

- At the end of the third month, the epidermis consists of four layers.

❖ **From deep to superficial these are:**

- ✓ Basal layer (stratum germinativum)
- ✓ Spinous layer (stratum spinosum)
- ✓ Granular layer (stratum granulosum)
- ✓ Horny layer (stratum corneum).



Development of the Epidermis,,,

❖ **The basal layer** consists of columnar cells.

- They constantly divide mitotically and move superficially to renew the epidermis.
- It usually takes 6–8 weeks for the cells to move from the basal layer to surface of the skin.

❖ **The spinous layer** consists of numerous irregular prickly cells (cells with spine-like processes).

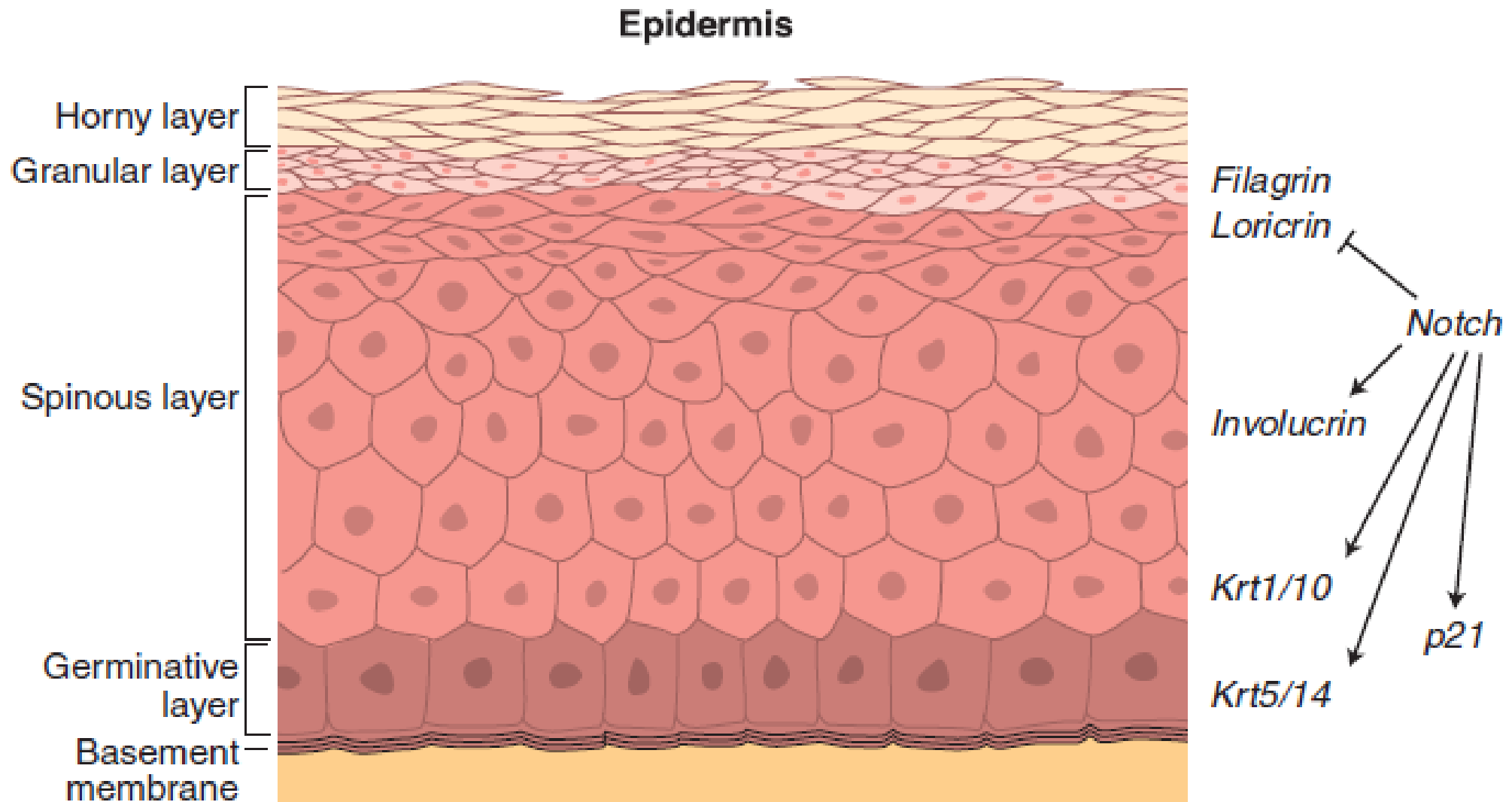
❖ **The granular layer** consists of three or four flattened rows of cells.

- The cytoplasm of the cells of the granular layer contains **keratohyalin granules** *due to the process of keratinization*.

Development of the Epidermis,,

- The cells superficial to granular layer are cells with scanty nuclei, and form a homogeneous layer called **stratum lucidum**. Histologically, this layer appears clear.
- **It exists only in lips and thick skin of the soles and palms.**
- ❖ **The horny layer** consists of several layers of flattened scale-like cells that are continuously shed off as flake-like residues of cells deprived of their nuclei and eventually die.
- The cytoplasm of these cells is filled with *keratin granules*.

Development of the Epidermis,,,



Development of the Epidermis,,,

- During the early part of the 5th month, the intermediate layer is replaced by the three definitive layers of keratinocytes: **the inner stratum spinosum, the middle stratum granulosum, and the outer stratum corneum.**
- This transformation, which involves apoptosis (programmed cell death), begins at the cranial end of the fetus and proceeds caudally.
- The layers of the epidermis represent a maturational series: presumptive keratinocytes are constantly produced by the stratum germinativum.
- **The cells of the stratum germinativum are the only dividing cells of normal epidermis.**

Development of the Epidermis,,,

- The **melanoblasts (dendritic cells)** appear in the basal layer **during the third month.**
- They are derived from **neural crest cells.**
- **N.B.** The periderm disappears when the stratum corneum is formed.
- The cells of the periderm continually undergo **keratinization** (formation of a horny layer) and **desquamation** (shedding of the cuticle in scales), and are replaced by cells arising from the basal layer.
- The exfoliated peridermal cells form part of a white greasy substance **vernix caseosa** that covers the fetal skin.
- Later, the vernix contains **sebum**, a secretion from **sebaceous gland.**

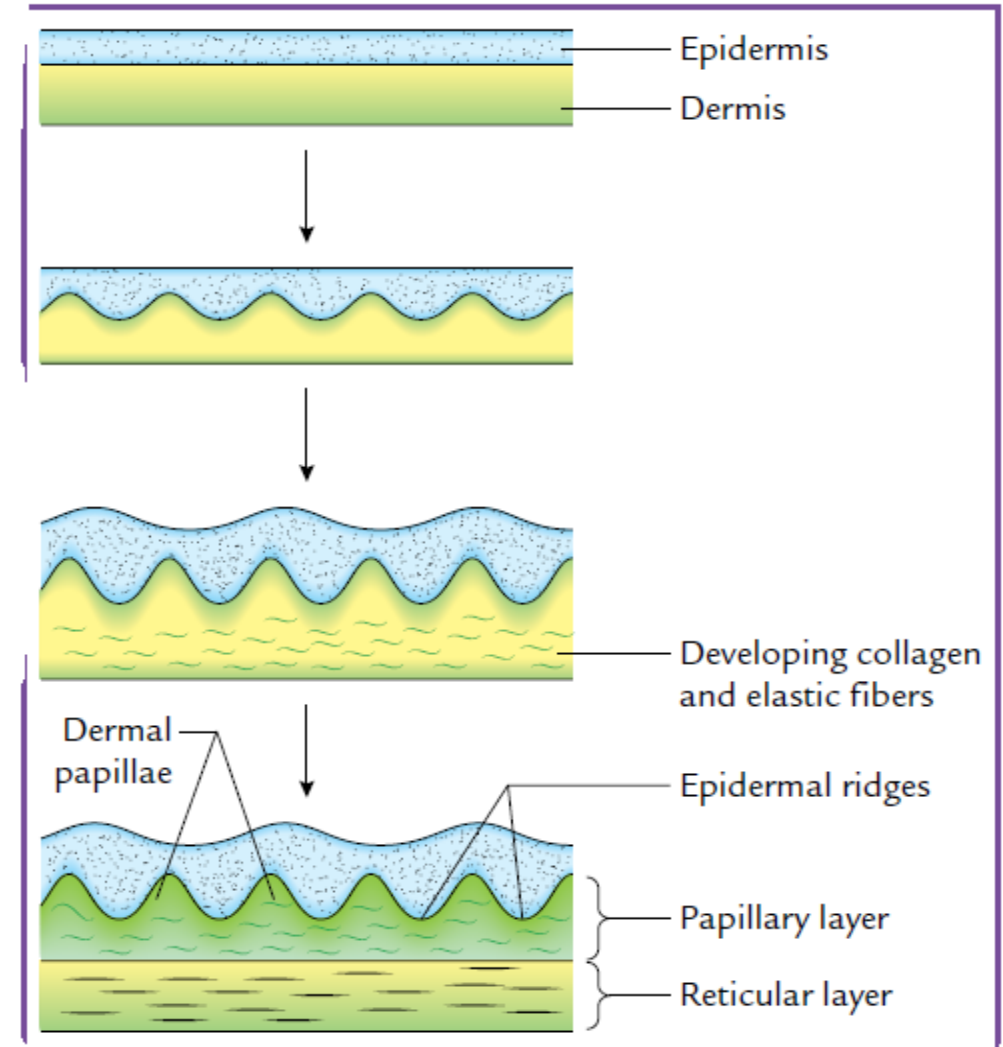
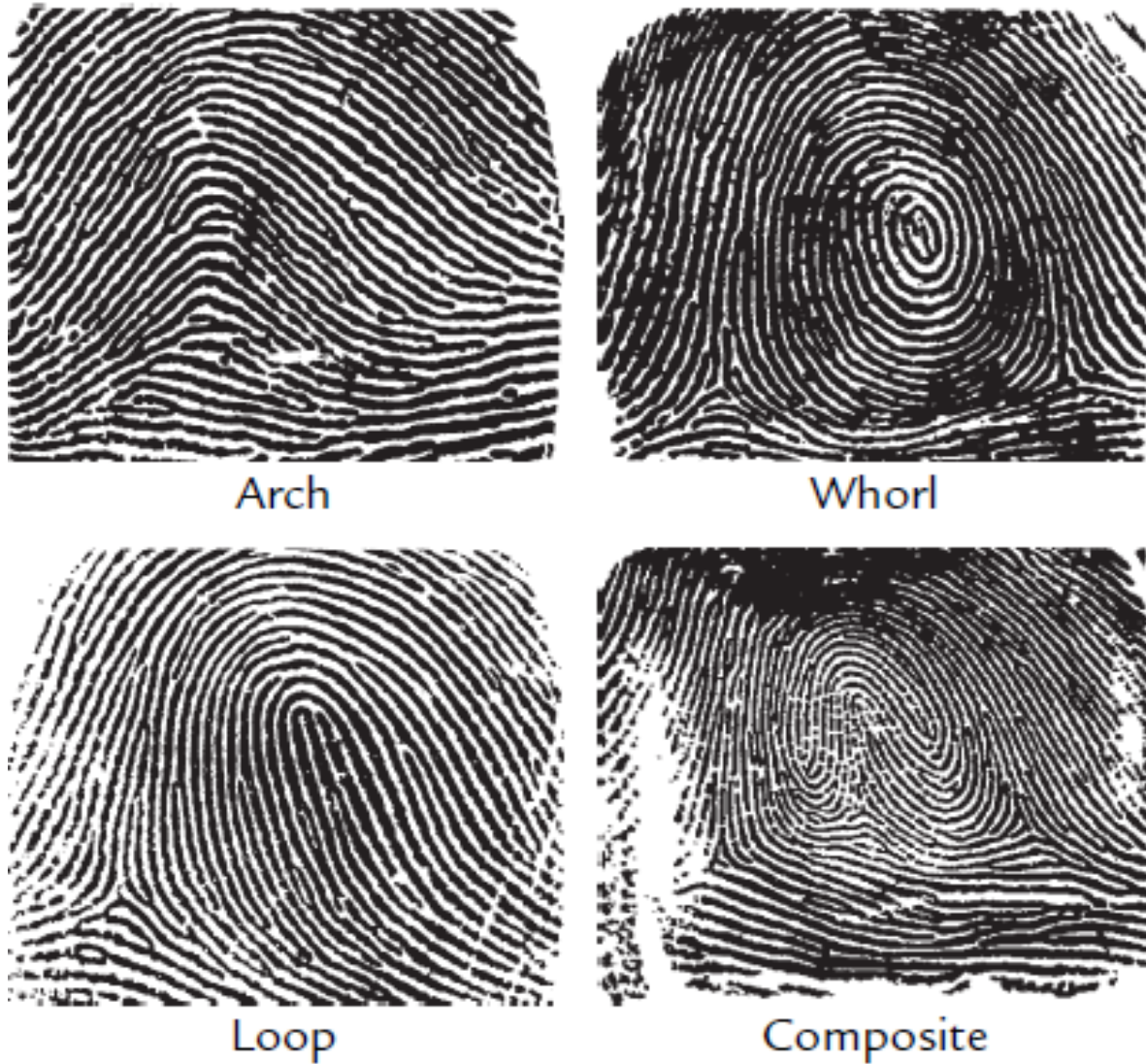
Development of the Epidermis,,,

- **Finally, in the process called cornification,** lytic enzymes are released within the cell, metabolic activity ceases, and enucleation occurs, resulting in the loss of cell contents including the nucleus.
- Consequently, the keratinocytes that enter the stratum corneum are flattened, scalelike, and terminally differentiated keratinocytes, or squames.
- In addition to keratinocytes, the epidermis contains a few types of less abundant cells, including melanocytes, Langerhans cells, and Merkel cells.
- **The pigment cells, or melanocytes,** are differentiate from neural crest cells.
- The density of melanocytes increases during fetal period.

Formation of Epidermal Ridges and Dermal Papillae

- Proliferation of cells in the stratum germinativum forms **epidermal ridges**, which extend into the developing dermis.
- These ridges begin to appear in embryos at 10 weeks and are permanently established by 17 weeks.
- The epidermal ridges produce grooves on the surface of the palms and the soles, including the digits (fingers and toes).
- The type of pattern that develops is determined genetically and constitutes the basis for examining **fingerprints** in criminal investigations and medical genetics.

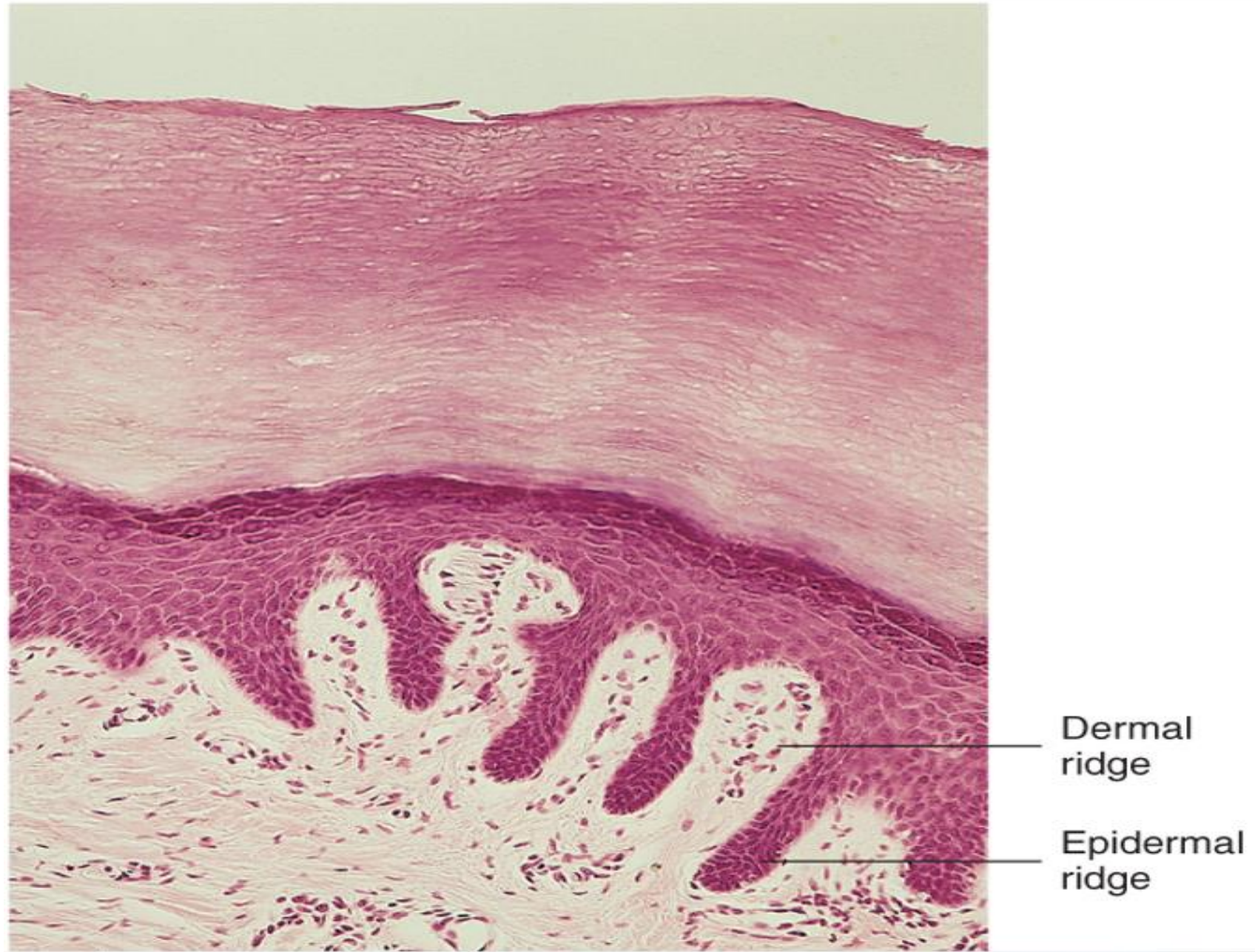
Formation of Epidermal Ridges and Dermal Papillae,,,



Formation of epidermal ridges and dermal papillae.

❑ Four basic types of dermatoglyphic patterns in the digits. 130

Formation of Epidermal Ridges and Dermal Papillae,,



Development of the Dermis

- The dermis, or corium the layer of skin that underlies the epidermis and contains blood vessels, hair follicles, nerve endings, sensory receptors, and so forth is a tissue with a triple embryonic origin.
- The **dermis** develops from mesenchyme lying underneath the surface ectoderm.

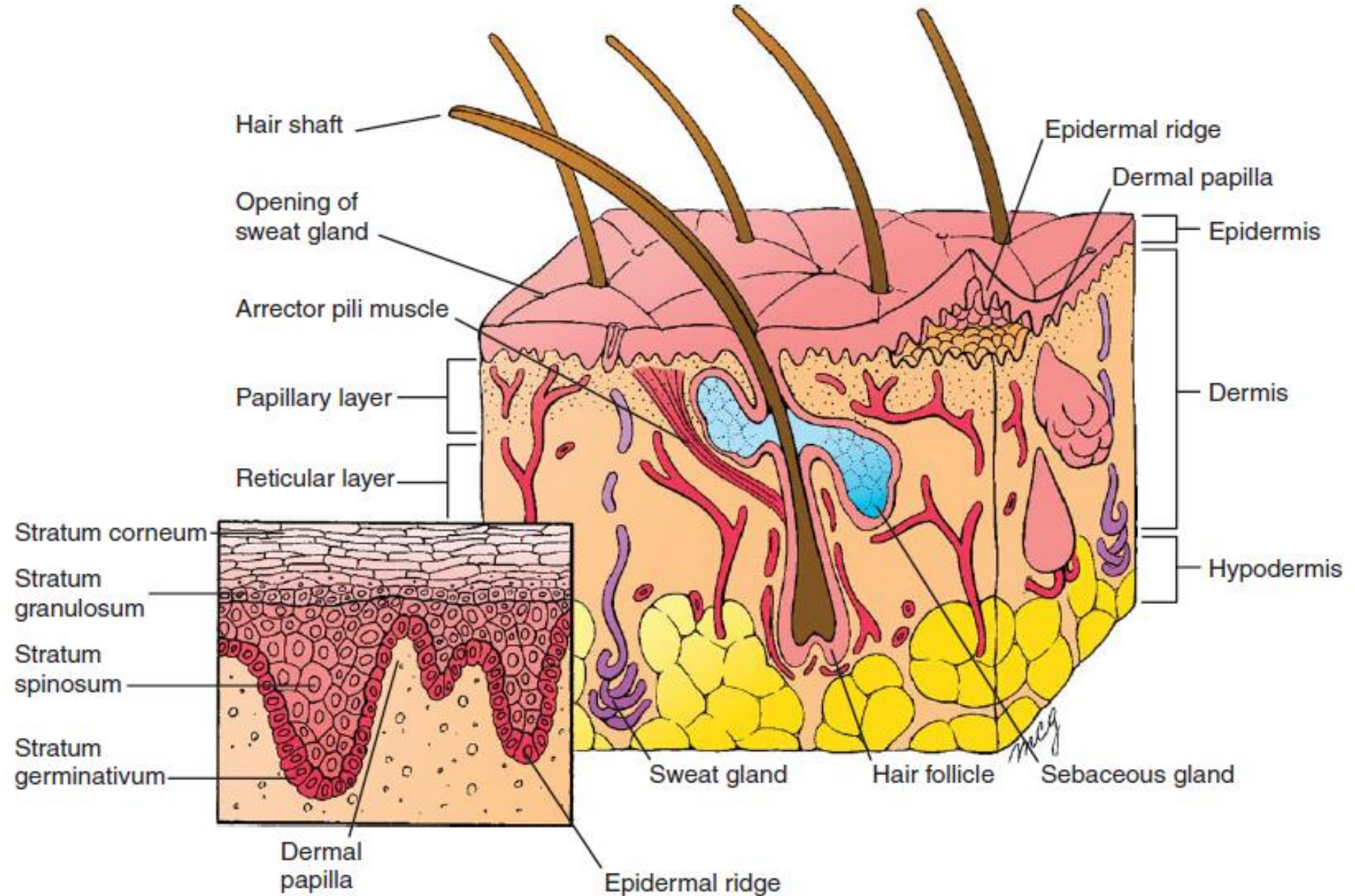
□ **This mesenchyme is derived from three sources:**

- I. Paraxial mesoderm,
- II. Somatopleuric layer of lateral plate mesoderm, and
- III. Neural crest cells.

Development of the Dermis,,,

- The mesenchymal cells differentiate into connective tissue cells that form **connective tissue fibers** (collagen and elastic fibers) and amorphous **ground substance** of the connective tissue.
- During third and fourth months dermis forms many irregular ridges called **dermal papillae**, which project into the epidermis and interdigitate with the **epidermal ridges**.
- The dermis differentiates into two layers:
 - **1.** Superficial papillary layer
 - **2.** Deeper reticular layer.

Development of the Dermis,,,



Development of the Dermis,,,

- In the trunk, the majority of the dermis is derived from the **somatic layer of the lateral plate mesoderm**, but part of it is derived from the dermatomal divisions of the somites.
- ❑ **In contrast, in the head most of the** dermis is derived from neural crest cells, and thus originates from ectoderm.
- During the 3rd month, the outer layer of the developing dermis proliferates to form ridgelike dermal papillae that protrude into the overlying epidermis.
- The intervening protrusions of the epidermis into the dermis are called epidermal ridges.

Development of the Dermis,,

- Most of the dermal papillae of **papillary layer** of dermis contain a small loop of capillary plexus and a sensory receptor (sensory nerve end organ).
- The loops of capillary plexus provide nourishment to the epidermis and to the sensory nerve end organ.
- The **deeper reticular layer** of dermis contains large amount of fatty tissue.
- The blood vessels in the dermis develop initially as endothelium-lined structures that differentiate from the mesenchyme.
- These primordial vessels give rise to the capillaries (**angiogenesis**).
- The angiogenesis of the dermis is completed by the end of first trimester of pregnancy.

Development of Skin Derivatives

- In many regions of the body, the skin gives rise to specialized structures including; **mammary glands, sebaceous glands, hair, sweat glands, teeth, salivary glands, and lacrimal glands.**
- Development of all of these skin derivatives depends on epithelial-mesenchymal interactions.
- All are characterized by the development of **an ectodermal placode**, followed by condensation of cells in the underlying mesenchyme, and then invagination of the epithelium into the underlying dermis.

Development of Glands

- The glands of the skin are derived from the **epidermis and grow into the dermis.**
- Several types of glands are produced by downgrowth of the epidermis.
- ❖ **Three types of glands**—the sebaceous glands, apocrine glands, and sweat glands—are widespread over the body.
- The milk-producing mammary glands represent a specialized type of epidermal gland.

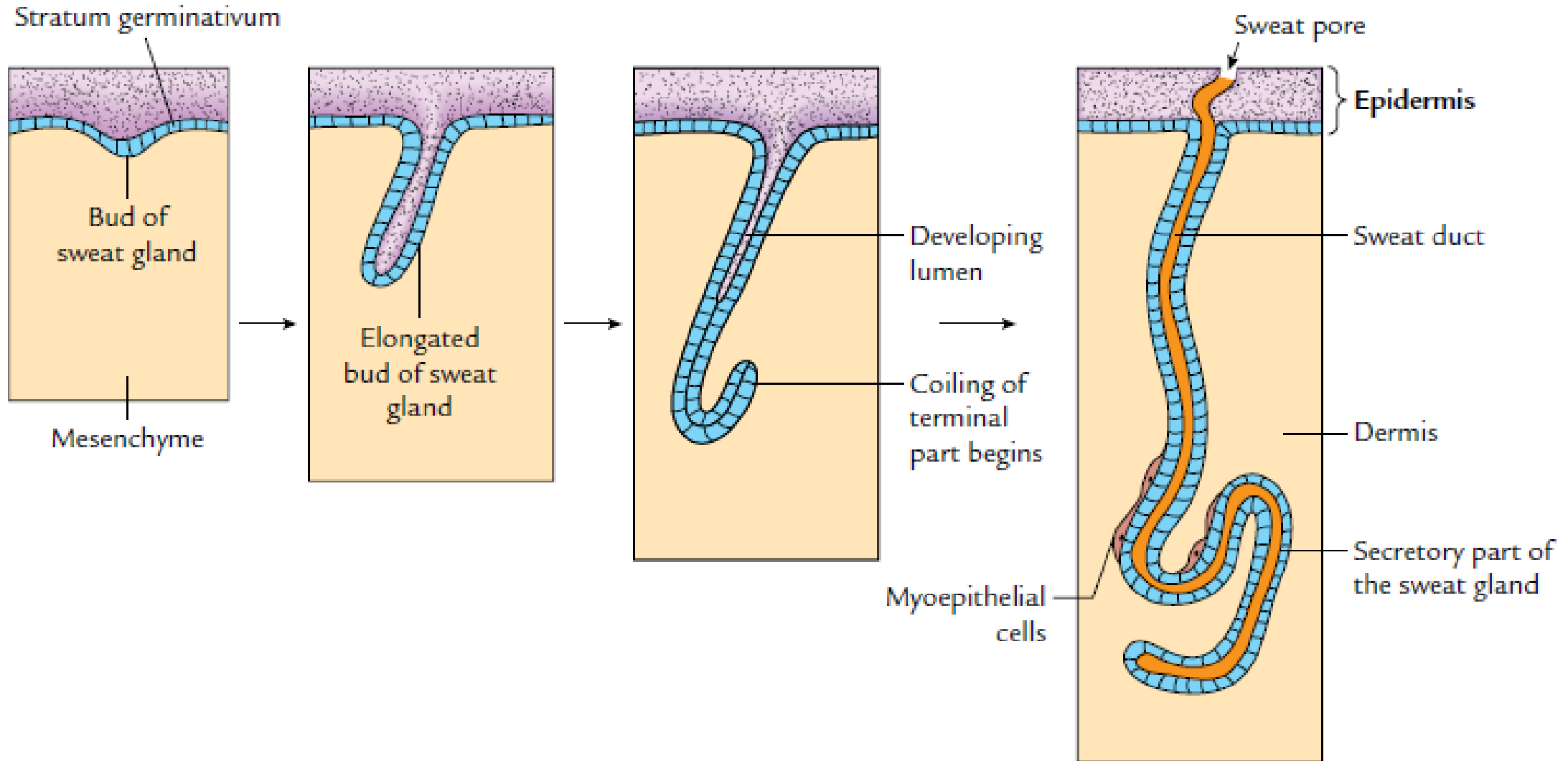
Development of Sebaceous Glands

- The sebaceous gland develops as a bud from the epithelial root sheath of hair follicle.
- Over most of the body, these glands form as diverticula of the hair follicle shafts, budding from the side of the root sheath about four weeks after the hair germ begins to elongate.
- In some areas of hairless skin—such as the glans penis of males and the labia minora of females—sebaceous glands develop as independent down growths of epidermis.
- The bud grows into the dermis tissue and branches to form a small system of ducts ending in expanded secretory acini (alveoli).
- Mature sebaceous glands are present on the face by six months of development

Development of Sweat Glands

- The sweat glands are of two types: **eccrine** and **apocrine**.
- **Eccrine sweat glands** are located in the skin throughout most of the body.
- They develop as epidermal downgrowths (cellular buds) into the underlying mesenchyme.
- As the buds elongate, they form the secretory part of the gland, and the sweat duct.
- The peripheral cells of the secretory part of the gland differentiate into myoepithelial and secretory cells.
- The myoepithelial cells are thought to be specialized smooth muscle cells that assist in expelling sweat from the glands.
- Eccrine sweat glands begin to function shortly after birth

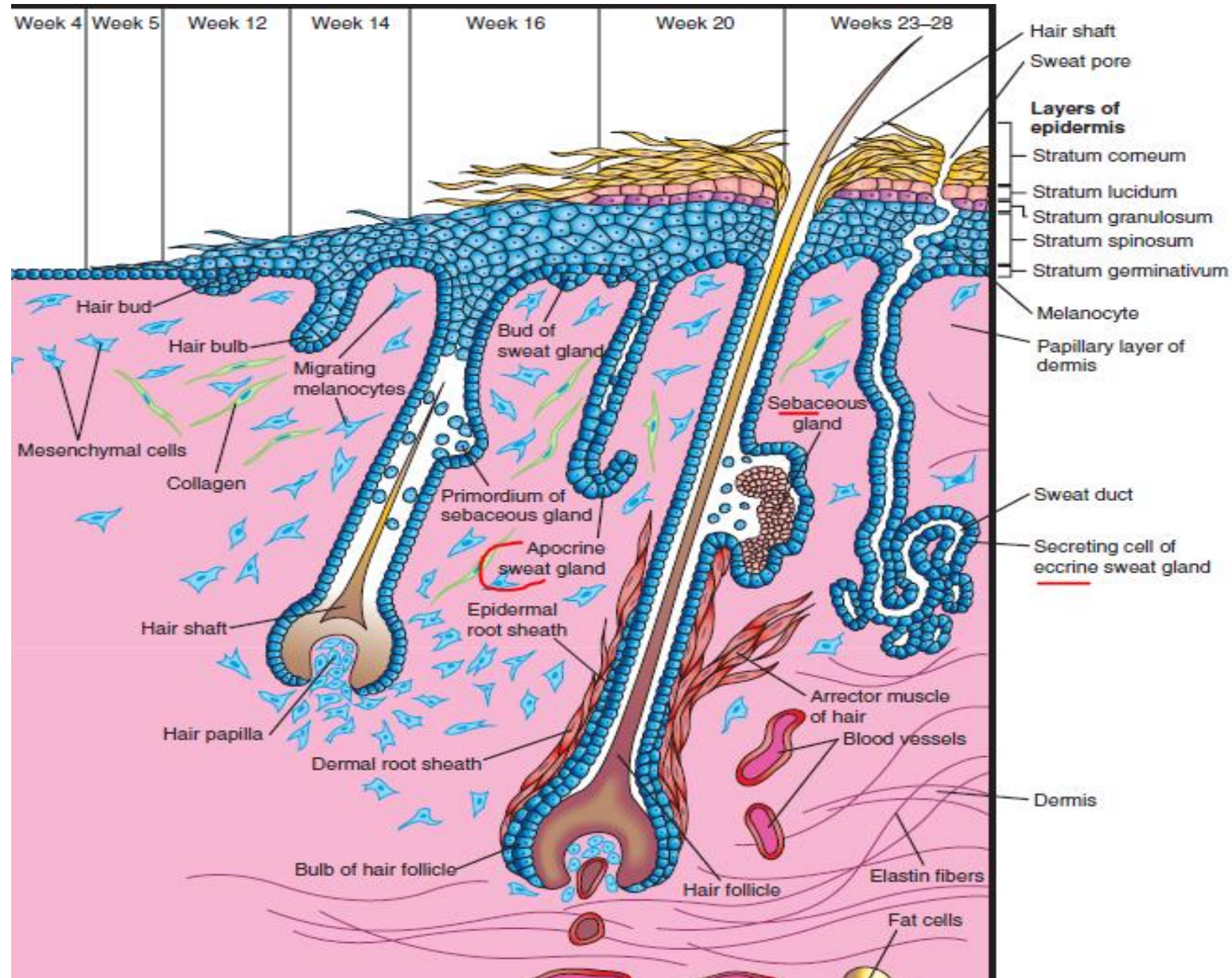
Development of Sweat Glands,,,



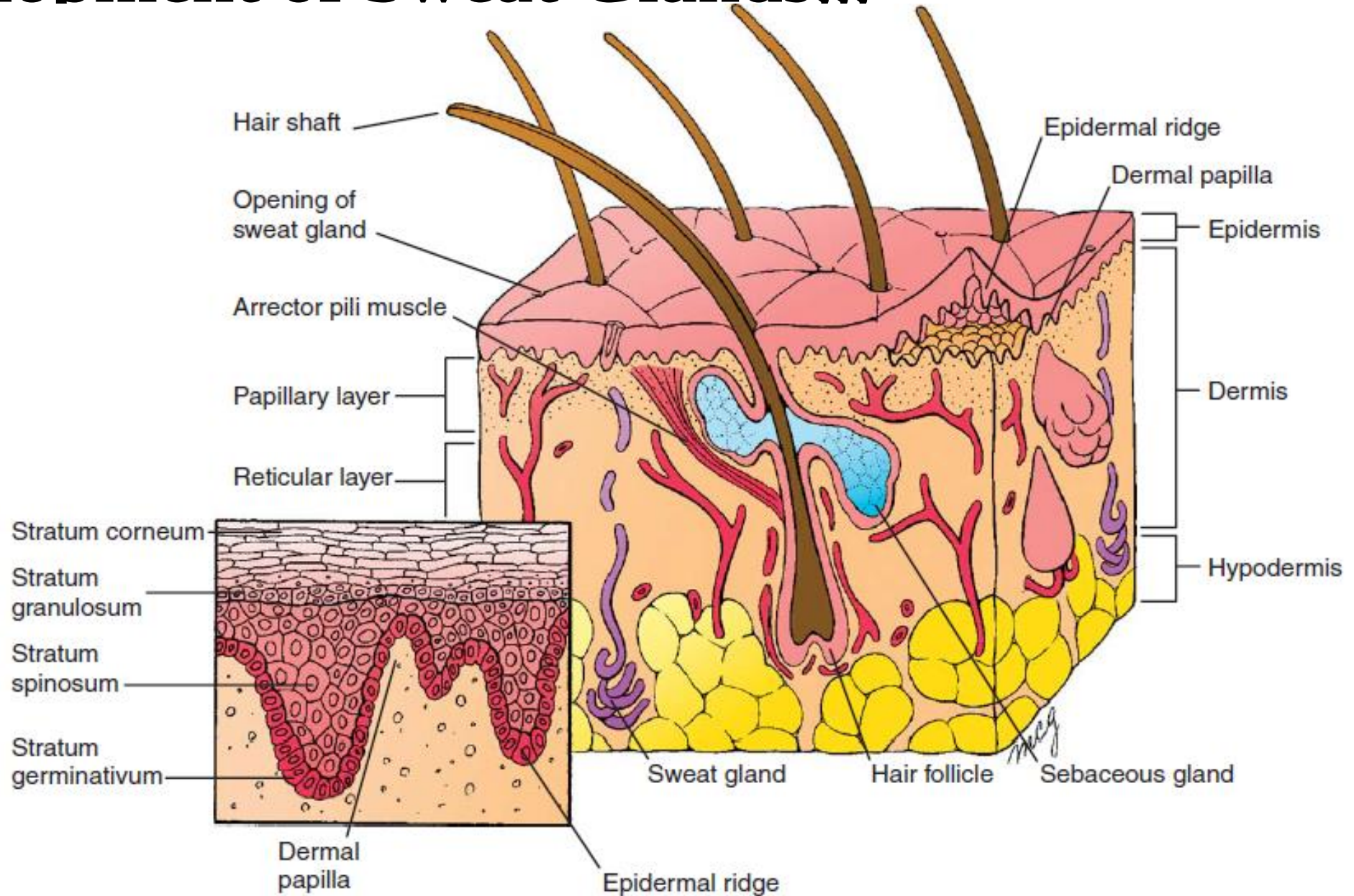
Development of Sweat Glands,,,

- **Apocrine sweat glands** develop from same epidermal buds that form the hair follicles
- The distribution of the large sudoriferous (producing sweat) **apocrine sweat glands** is mostly confined to the axillary, pubic and perineal regions, and the areolae (circular pigmented areas) surrounding the nipples.
- The glands develop from downgrowths of the stratum germinativum of the epidermis.
- As a result, the ducts of these glands open, not onto the skin surface, as do eccrine sweat glands, but into the canals of the hair follicles superficial to the entry of the sebaceous gland ducts.
- Secretion by apocrine sweat glands is influenced by hormones and does not begin until puberty.

Development of Sweat Glands,,

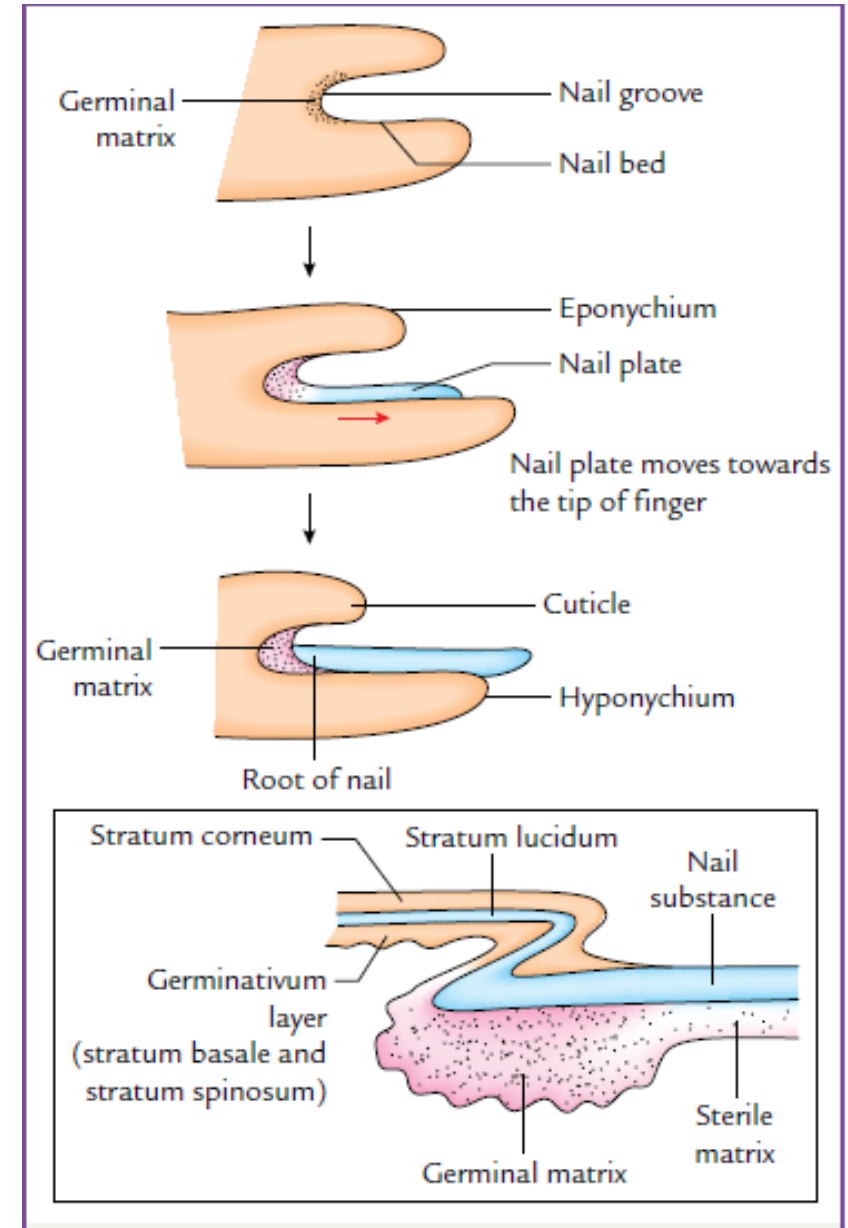


Development of Sweat Glands...



Development of Nails

- The nail develops at tips of digits of toes and fingers approximately at 10 weeks.
- The ectoderm at the tip of each digit thickens to form **primordium of nail**—the **primary nail field**.
- The first indication of development of the nail is the formation of fold of thickened epidermis.
- This fold is ‘U’ shaped and is called **nail fold**.
- The formation of nail fold defines certain structures in the terminal part of the digit-like **nail groove** and **nail bed**.
- The cells of germinal layer at the base of nail groove proliferate to form a thick layer of cells called **germinal matrix**.



- The end!